CS6750: Team Project

Team Alpha

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1 INTRODUCTION

Students constantly need to make decisions about how to allocate their time and attention while managing their academic responsibilities. It requires balancing multiple competing demands from assignments, projects, exams, and personal commitments. There are many tools created for scheduling tasks, but they typically focus on organizing tasks instead of supporting the decision-making process behind prioritization. This causes many students to struggle with deciding what they should do next, instead of simply what to do.

The focus of this project is scheduling and prioritizing academic work, particularly on how students decide the order that the tasks should be completed in. The goal is to design an interface that helps students efficiently decide what to work on based on specific factors and better supports decision-making. Some potential factors are importance, urgency, energy levels, and workload. The system intends to reduce cognitive load in students by assisting them in making informed prioritization decisions and in evaluating tradeoffs.

Dealing with these issues is a common experience among students. As an example, students may have a to-do list filled out with their assignments with different due dates, exams for multiple classes, other ongoing work for all of their current classes, and any personal commitments like appointments, etc. Although these activities and their due dates may be listed, the student may spend a lot of time deciding where to start and the order in which to tackle the tasks. The number of tasks themselves may make some students feel overwhelmed. There are also a number of conflicting priorities. This scenario highlights an existing gap between task organization and efficient decision support.

Considering the human computer interaction perspective, this task explores important questions about decision fatigue, cognitive load, and supporting complex decision-making. An effective interface should present the necessary information to the user and both help them interpret and act on the presented information.

As for student scheduling, this hints to providing transparency, structure, and guidance around the prioritization decisions instead of relying on users to manually do everything. This project explores how an interface designed for students can support them in scheduling and prioritizing their work. Through this process, we intend to identify key challenges in the approaches currently available, then design solutions that reduce the effort needed to make decisions, improve clarity, and increase the user's confidence in the planning process.

2 NEEDFINDING PLAN

To better understand how students currently schedule and prioritize their academic work, along with learning their pain points and key factors, there will be three needfinding activities conducted: surveys, interviews, and heuristic evaluation.

2.1 Survey plan

The surveys will be conducted with approximately 20-25 students from different levels of academic study. The survey is meant to take approximately 8-10 minutes. Participants will be recruited from both classmates and personal networks.

Procedure
The participants will be invited to complete an online survey consisting of multiple choice, Likert-scale, and open ended questions. The questions explore their current scheduling habits, prioritization strategies, and any challenges they have. The questions are designed to discover both behavioral patterns and perceived difficulties of the students with scheduling tools. The full survey questions can be found in *Appendix 16.1: Needfinding Survey Questions*

Goals
The goal of this survey is to provide a broad overview of student behavior and highlight areas for deeper exploration. It aims to identify participants' commonly used tools and strategies for scheduling, key factors that influence prioritization decisions, and recurring pain points and needs.

Potential biases
During the survey process there are some potential biases such as participants overestimating their time management effectiveness (self-report bias), mostly recruiting from classmates may skew to those of a similar academic background

(sampling bias), and participants may interpret the questions differently (interpretation bias).

2.2 Interview plan

The interviews will be conducted with 6 participants, lasting approximately 15-20 minutes. The participants will be recruited from personal networks, specifically those that are familiar with academic scheduling issues. The full interview notes can be found in *Appendix 16.3: Needfinding Interview Notes*.

Procedure

The interviews will follow a semi-structured format to allow for both flexibility and consistency. The interview takes place in several stages starting with understanding the participant's current semester and the tools they use, then the participants move on to describing a recent scenario in which they chose what to work on, then comes the exploration of factors like effort, deadlines, and stress. Following these it moves on to how the participants handle competing priorities, then the roles of mood, motivation, and procrastination. Finally finishing the interview with the perceived gaps and difficulties in their current approaches. These stages are designed to elicit both concrete examples and reflective explanations. This structure allowed us to compare stated strategies and actual behaviors in practice. The full interview questions can be found in *Appendix 16.2: Needfinding Interview Questions*.

Goals

The goal is to understand how students make day-to-day prioritization decisions. More specifically, the goal is to uncover the decision-making process behind prioritization, how students handle uncertainty and tradeoffs, emotional and cognitive factors, and scenarios where current tools fail to effectively support decision-making.

Potential biases

In the interview process there are some potential biases such as participants may not accurately remember past decisions (recall bias), interviewer tone or phrasing may influence responses (interviewer bias), and the participants might present themselves as more organized than they are (social desirability bias).

2.3 Heuristic evaluation plan

The heuristic evaluation will focus on understanding how current tools either support or fail to support decision-making and task prioritization. Due to being the main preferred scheduling tool, from 16 out of the 21 survey participants and mentioned frequently among interviewees, Google Calendar is the chosen tool for the heuristic evaluation.

Procedure

The evaluation will be performed during a simulation of five representative tasks of a realistic student workflow. These tasks are: adding a multiphase project with a distant deadline, deciding between two tasks with equal or similar deadlines, rescheduling plans due to unexpected reasons, estimating and logging time for larger projects, and reviewing and interpreting an entire week's scheduled work. These tasks were formed from patterns directly observed in the earlier needfinding activities. Nielsen's ten usability heuristics will be applied as a framework during the evaluation, although there will be a focus on *visibility of system status, recognition rather than recall, user control and freedom, and flexibility and efficiency of use*. These specific heuristics directly relate to issues observed in the previous needfinding stages. Each usability issue identified will be rated on a scale from 0-4, where: 0-not a usability problem, 1-cosmetic issue that doesn't need to be fixed, 2-low priority minor issue, 3-high priority major issue, and 4-highest priority extreme issue. Each task step will be performed in Google Calendar while documenting any usability violations. Per violation found, the task in which the issue occurred, the heuristic violated, a description of the issue, the location in the interface, and the severity rating will be recorded.

Goals

The heuristic evaluation aims to evaluate how well the interface supports long term planning, decision-making between competing tasks, schedule adaptation flexibility, time estimation and workload management, workload visibility, and cognitive load.

Potential biases

There are several biases that can affect this evaluation such as the evaluation is performed by one person which may limit perspective (single-evaluator bias), the focus of the evaluation only reflects what is chosen from previous needfinding exercises which may highlight certain issues (task selection bias), and by

focusing on Google Calendar issues present in other scheduling tools may not be considered (tool specific bias).

2.4 Bias mitigation across methods

Each individual needfinding activity can introduce potential biases into the project, which is why the activities were intentionally designed to complement and validate each other. Surveys provide a broader view of student behavior and challenges, which is mitigated by the interviews offering deeper contextual insight into their behaviors. Some of the biases introduced by interviews such as recall bias and social desirability bias, are addressed in heuristic evaluation by providing a more system focused and objective perspective. Heuristic evaluation may introduce single-evaluator bias and tool specific bias, which can be mitigated by the fact that the tasks themselves are derived from survey and interview results showing that they reflect real user challenges.

There are some remaining biases, even with all of these mitigating factors in place. The limited participant pool to academic students may reduce the generalizability of the findings. The reliance on simulated interactions or self-reported data without real-world observations may introduce some limitations. With these in mind, the combination of multiple needfinding approaches reduces the effect of any single bias source and gives a more comprehensive understanding of how students prioritize and schedule their work.

3 NEEDFINDING RESULTS

3.1 Survey results

We collected 21 online survey responses from students of different academic levels, although the majority are graduate students. The survey provides a broad overview of student prioritization strategies, scheduling behaviors, and challenges. According to the participants, the majority used Google Calendar as their dominant platform. There were also many respondents that used mostly note taking tools, such as Apple Notes, Google Keep, etc. Smaller portions of the participants mainly used less conventional tools or physical planners. There were a notable number of participants that selected multiple tools, creating fragmented systems instead of relying on a single solution.

As far as usage frequency, the majority of the participants used scheduling tools

daily, usually multiple times a day, highlighting the active engagement of students in planning their work. Although many participants were actively engaged, few reported following their schedules consistently. It was also frequently indicated that they often make changes to their schedules or that they tend to only follow plans when overwhelmed, highlighting the gap between planning and execution.

The respondents also highlighted that the main factor driving prioritization decisions is deadlines for the majority. The other factors were significantly less selected including difficulty, time/energy, importance, and even no prioritization strategy. Overall, the participants also reported having a moderate difficulty with prioritization tasks. This suggests more of a dependence on simple heuristics over a structured decision-making process. From their responses, it also seems that procrastination is strongly linked to uncertainty in decision-making instead of solely a lack of motivation.

According to the responses, the participants reported being reasonably accurate when estimating smaller tasks, but report greater inconsistency for medium and large tasks. Many indicated that they tend to underestimate large or multi-stage projects. The responses also suggest that even though students are actively using these scheduling tools, the tools may not sufficiently support clarity, confidence, or stress reduction in users. Overall the results highlight that the participants struggle most with decision-making, prioritization, and execution. It also suggests the need for systems that reduce the cognitive burden and provides clearer guidance. The full raw survey responses can be found in *Appendix 16.4: Needfinding Survey Data*.

3.2 Interview results

We conducted semi-structured interviews with six students across undergraduate and graduate programs, representing a range of majors and workload structures. Participants balanced varying combinations of coursework, employment, and extracurricular responsibilities, providing a broad view of how students approach day-to-day task prioritization. A structured summary of the key insights derived from these interviews is provided in Table 3.

Across interviews, task prioritization emerged as largely reactive and context-sensitive rather than systematic or plan-based. Participants consistently describe making decisions in response to what feels most urgent in the moment, often

framed as "putting out fires," rather than following a stable prioritization strategy. While deadlines serve as the dominant organizing principle, this reliance is narrowly short-term: imminent due dates drive action, but longer-term tasks are routinely deferred until they become urgent. As a result, participants repeatedly describe cycles of cramming, uneven workload distribution, and stress. Importantly, prioritization is not driven by objective measures such as time required. Instead, decisions are shaped more strongly by perceived cognitive effort and emotional friction. Tasks that feel difficult, unfamiliar, or aversive are systematically avoided, even when recognized as important.

At the same time, prioritization is highly dependent on situational context and internal state. Participants adapt their decisions based on energy level, mood, environment, and available time, often shifting toward lower-effort or more familiar tasks when fatigued or overwhelmed. This further reinforces avoidance patterns and contributes to inconsistent progress across tasks. Existing tracking strategies do little to mitigate this: participants report fragmented use of tools (e.g., calendars, planners, apps) combined with heavy reliance on memory, resulting in unreliable task awareness and increased cognitive load. When multiple tasks compete for attention, users frequently experience uncertainty or choice paralysis, defaulting to heuristics such as urgency, ease, or personal preference. These findings suggest that current approaches to task management fail not because users lack awareness of their responsibilities, but because they lack support for how to decide among competing tasks under real-world constraints.

3.3 Heuristic evaluation results

The heuristic evaluation was performed as planned using Google Calendar and the five representative tasks: adding a multiphase project with a distant deadline, deciding between two tasks with equal or similar deadlines, rescheduling plans due to unexpected reasons, estimating and logging time for larger projects, and reviewing and interpreting an entire week's scheduled work. In total 11 usability violations were found across these tasks, with multiple high or extreme severity issues found. The full heuristic evaluation results can be found in *Appendix 16.5: Needfinding Heuristic Evaluation Data*.

There were several notable limitations identified when attempting longer term, multi-stage work. One is the lack of interface support for either checkpoint creation or intermediate milestones within a task, due to this users would be forced

to create multiple disconnected events instead. This issue resulted in a severity rating of 4 (highest priority, extreme issue) under the user control and freedom heuristic violation. The default system behavior of setting deadlines based on selected dates, suggests the misalignment with real world complex academic planning.

When evaluating two tasks with the same deadlines, the interface offers no prioritization cues, it simply orders them alphabetically. This resulted in a severity rating of 4 under the flexibility and efficiency of use heuristic violation. This hints at another interface limitation due to the lack of support in deciding what to do next, outside of urgency.

The evaluation also revealed that ambiguous interface cues contributed to the usability issues in rescheduling workflows. This is highlighted when, discovered through trial and error, the pencil icon is used for both modifying event details and scheduling resulting in a severity rating of 3 for the match between system and real world heuristic violation. This shows the inefficiency in allowing users to quickly adapt their plans in this system.

The lack of time estimation or tracking functionality in the interface is another notable limitation discovered. Users being unable to log or estimate how long tasks will take resulted in severity rating of 4 for the flexibility and efficiency of use heuristic violation. The need for users to rely on external tools for basic functionality resulted in a severity rating of 3 for the user control and freedom heuristic violation. This apparent fragmentation increases the users' cognitive load and displays their need to mentally manage their workload.

There were several notable visibility issues in the weekly view. Truncated titles could only be viewed if the user opened the item, this resulted in a severity rating of 3 for visibility of system status heuristic violation. The lack of any visual distinction between task workload, urgency, or importance resulted in a severity rating of 4 for the visibility of system status heuristic violation. This is another example of inefficient decision-making and increases in user cognitive load for this interface.

Taking all of these into consideration, Google Calendar fails in both supporting decision-making and prioritization. The findings presented were consistent with patterns observed in both the surveys and interviews among the students.

3.4 Needfinding insights

Insight 1: Students default to deadline-driven prioritization. Survey results showed most participants selected deadlines as their primary factor, with far fewer selecting importance, difficulty, or energy. Interviews reinforced this, with participants describing prioritization as "putting out fires" and focusing on what is immediately due. This suggests students rely on urgency-based heuristics rather than structured prioritization, leading to reactive workflows.

Insight 2: Deciding what to do next is a key point of friction. Survey responses indicated moderate difficulty with prioritization and inconsistent adherence to schedules. Many participants also reported procrastinating due to uncertainty about where to start. Interviews further showed that participants often feel "stuck" or rely on external tools (e.g., AI schedules) to make decisions, suggesting the breakdown is in choosing between tasks, not tracking them.

Insight 3: Current tools do not support context-aware decision making. The heuristic evaluation identified 11 usability violations in Google Calendar, including high-severity issues around lack of prioritization cues and time estimation support. This aligns with interview findings that students prioritize based on effort, mood, and energy, none of which are represented in existing tools. As a result, users must mentally manage tradeoffs, increasing cognitive load.

4 INITIAL BRAINSTORMING PLAN

Our brainstorming process will follow a two-stage approach: individual ideation followed by a group synthesis session. This structure is intended to encourage a wide range of ideas before converging as a team.

Stage 1: Individual brainstorming. Each team member will independently generate ideas based on insights from our needfinding. We will prioritize quantity over quality at this stage, with each member generating no fewer than 20 ideas. Ideas will be documented individually rather than in a shared space to avoid early influence from others.

Stage 2: Group brainstorming. We will then meet as a group (four members) in a synchronous Discord session to review and build on our ideas. This session will be capped at 25 minutes to maintain focus and momentum. The session will be structured as follows:

- Rapid sharing of ideas without critique
- Grouping ideas into themes or categories
- Brief discussion to refine and combine promising directions

Bias considerations and mitigation. We anticipate potential biases such as fixation on early ideas, groupthink, and uneven participation. To mitigate these, we separate individual and group brainstorming, avoid critique during initial sharing, and use structured turn-taking to ensure all voices are heard. We will also briefly revisit less obvious ideas before narrowing. While these steps reduce bias, some influence from group dynamics may still remain.

5 BRAINSTORMING RESULTS

Following the needfinding, each team member conducted an individual brainstorming session, generating a wide range of ideas focused on improving student task prioritization and decision-making. These ideas were then aggregated and discussed in a time-bounded group session, where we identified common themes and converged on a smaller set of promising directions.

Across individual and group brainstorming, several consistent themes emerged. First, many ideas focused on reducing cognitive overhead in decision-making, such as limiting visible tasks, providing "start next" recommendations, or offering lightweight comparison views between tasks. Second, there was a strong emphasis on context-aware prioritization, including adapting recommendations based on time availability, energy level, or external constraints. Third, multiple ideas addressed procrastination and avoidance behaviors, which led to suggestions on interventions like nudges, task reframing, and automatic breakdown of large or repeatedly skipped tasks. Several topics explored improving temporal awareness, including visualizing workload across time and highlighting task clustering or imbalance. Finally, several ideas explored motivation and engagement, including lightweight gamification (like points, streaks, or progression systems) to incentivize follow-through and reduce avoidance.

From this broader design space, we selected three design alternatives to move forward into prototyping. These alternatives were chosen because they address a distinct but complementary breakdown in current task prioritization behavior, while remaining feasible to prototype at medium-fidelity level.

AI-assisted task breakdown. This concept focuses on reducing overwhelm asso-

ciated with large or ambiguous tasks. When a user inputs a complex assignment, like studying for an exam or completing a term project, the system automatically decomposes it into smaller, actionable subtasks with suggested timelines. This direction was selected because multiple brainstorming ideas emphasized the need for scaffolding large tasks and because needfinding showed that users often delay work until tasks become urgent due to unclear starting points.

Weekly cognitive load heatmap. This alternative introduces a visual weekly overview that color-codes days based on projective cognitive load. Users can see when their schedule is overly demanding and rearrange tasks accordingly, like shifting high-effort work away from already heavy days. This concept builds directly on ideas related to workload visualization and clustering alerts, and was selected because it addresses the forward-planning gap identified in needfinding: students struggle to anticipate overload until it is too late.

Gamified "swipe-to-choose" prioritization interface. This concept combines lightweight decision support with ongoing motivation. Users make quick "now vs. later" choices through a swipe-based interface, reducing friction when deciding what to work on. At the same time, the system adds a points-based layer during work sessions: users earn points for completing tasks, with bonuses for streaks, starting early, or tackling avoided tasks. These points can feed into a simple progression system, like growing a garden or advancing a narrative, which can provide a sense of momentum beyond a single task. This direction was selected because it targets both indecision and lack of follow-through, two core issues identified in needfinding.

6 INITIAL PROTOTYPING

After brainstorming, we developed three medium fidelity design alternative prototypes that explore distinct approaches to improve decision-making and task prioritization in scheduling systems. The prototypes did not focus on visual polish, but instead focuses on structure, interaction flow, and usability. Each individual prototype aims to address specific challenges discovered during the needfinding phase.

6.1 Prototype A: AI-assisted breakdown

This prototype presents an AI-assisted task breakdown system that is designed to aid users in transforming larger, more ambiguous tasks into smaller, specified

and actionable subtasks. This addresses challenges including difficulty initiating complex tasks, procrastination from unclear starting points, and over-reliance on reactive decision-making.

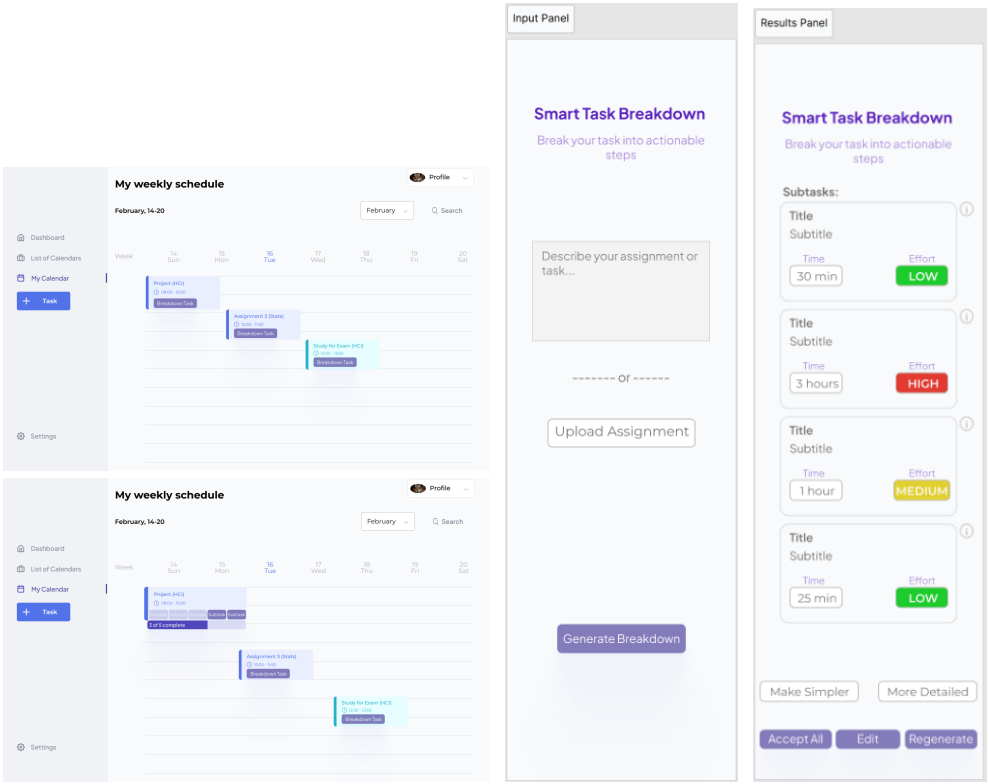


Figure 1—Prototype illustrating the AI-assisted task breakdown workflow. Users begin in a calendar view where a task can be selected for decomposition (top left image), then provide input or upload context to guide the breakdown process (central image). The system generates subtasks with estimated effort and time, which can then be reviewed and adjusted before acceptance (right image). Accepted subtasks are integrated back into the calendar with visible progress tracking (bottom left image). Images created with Figma (Figma, Inc., 2024), using these templates: Kadaner, 2024 & Designer, 2024.

The first screen shows a weekly calendar interface that exists in many scheduling tools. Each task contains a "Breakdown Task" button that initializes the AI-assisted task breakdown process. This introduces a lightweight entry point for additional functionality, that preserves the existing mental model for the user and is grounded in the principle of flexibility and efficiency of use due to this.

The second screen displays the right-side panel that appears after pressing the "Breakdown Task" button. On this panel, the user can either type in a description of the selected task or upload an assignment document. The inclusion of the different input options align with user control and freedom, which allows users to add as much or as little detail about the task as they prefer. The third screen shows the system generated subtask list, after input submission. Each subtask includes a title, subtitle description, estimated completion time, estimated effort level, and an information icon which explains why the specific subtask was generated. This design supports the principle of recognition rather than recall and the explanation improves the visibility of system status and transparency. The fourth screen shows the integration of the subtasks into the main task, displaying the list of subtasks within the tasks and a progress bar of subtasks completed. This supports visibility of system status and reduces the users' cognitive burden. This design transforms the abstract tasks into more manageable units by externalizing task decomposition and providing actionable steps for the user.

6.2 Prototype B: Cognitive heatmap

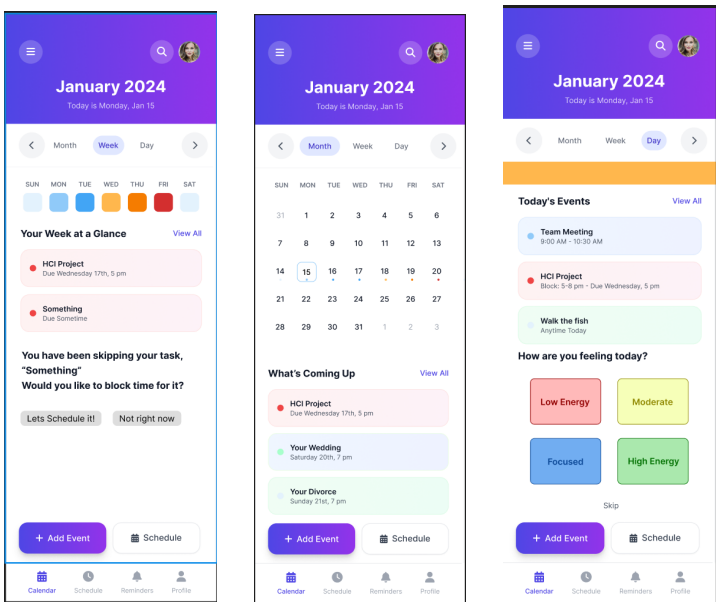


Figure 2—Cognitive heatmap interface across weekly, monthly, and daily views. The weekly view visualizes projected workload and supports time blocking, the monthly view provides a higher-level overview, and the daily view adapts task order based on user energy. The daily banner reflects overall load using the heatmap scale. Images created with Figma (Figma, Inc., 2024), using template Alam, 2024.

This prototype shows a scheduling system that is cognitive load aware, which aims to assist users in understanding and managing their workload over time. This design uses the user's current state to visualize workload intensity, enable context aware task prioritization, and support proactive planning. It addresses concerns discovered during the needfinding stage including users having trouble with long term planning, their lack of awareness concerning workload distribution and their reliance on reactive decision-making.

The first screen displays a weekly calendar containing a cognitive heatmap, which color codes each day depending on the estimated workload intensity. The colors range from tasks that are low workload to those that are extreme workload. This allows the users to quickly identify the underutilized or overloaded days. This screen also shows a summary highlighting important tasks and deadlines, a procrastination prompt that encourages the user to begin tasks earlier, and

a time blocking scheduler. It aligns with both recognition rather than recall and supporting the visibility of system status. The second screen displays a monthly calendar view that uses dots as heatmap indicators to quickly show the user how busy each week is. There is also a section that summarizes the important upcoming events and deadlines. The combination of the high level visualization with summarized task information aligns with the principle of matching the system to the user's mental model of planning over time. The third screen presents an energy based prioritization system that focuses on daily execution. The user is prompted to select their energy level for the day, then the system dynamically reorders the flexible tasks to better fit the user's current capacity. This screen also displays a banner that uses the cognitive heatmap coloring to highlight the workload intensity for that day. Here the flexibility and efficiency of use in adapting to real time context, along with the reduction in user decision fatigue are in alignment. This prototype reframes the static scheduling organization into a more dynamic decision support system.

6.3 Prototype C: Gamification

This prototype uses a gamified approach to task prioritization, designed to increase engagement and reduce decision friction. The focus is to reframe prioritization from a cognitively demanding planning task to an interactive, lightweight process.

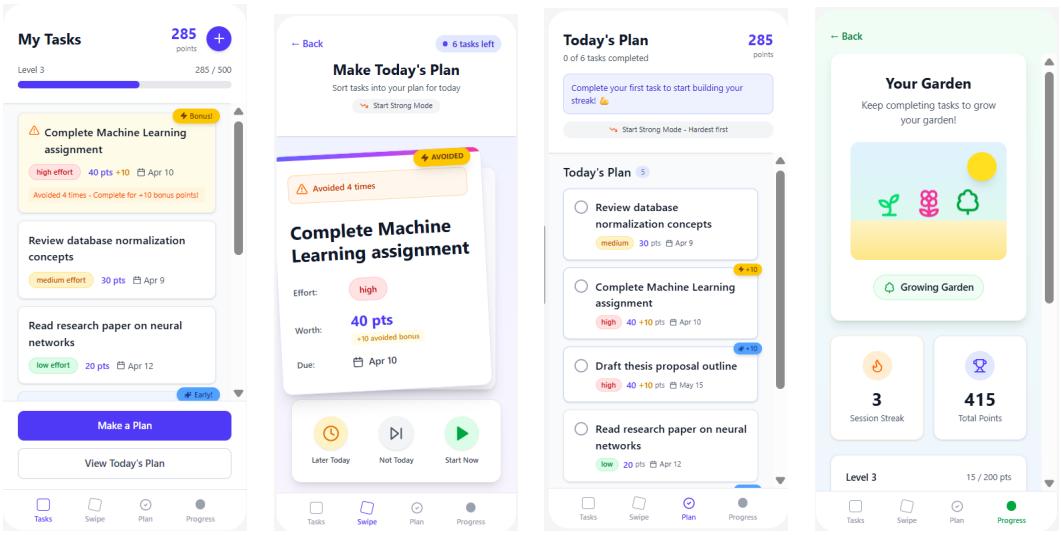


Figure 3—Prototype workflow showing the transition from task management to prioritization, execution, and reward. Tasks with effort-based points and behavioral incentives are sorted via a swipe-based interface into a daily plan, which supports task completion and streak-building. Progress is reflected through accumulated points and a lightweight gamified view. Images created with Figma.ai.

The first screen displays a list of tasks, each including a title, estimated effort, corresponding point value, and either bonus indicators for early completing tasks or previously avoided tasks. Associating the efforts with rewards encourage users to engage with challenging tasks. The second screen uses a swipe-based interaction model in which the user can start a task, put it off for later, or skip it for today. The user can choose to either swipe or use buttons to categorize the tasks that reduces prioritization into smaller, more manageable decisions lightening cognitive load. The third screen shows a structured plan for the day based on the user selections. They're ordered by either urgency or selected prioritization mode, providing clarity and flexibility while reducing uncertainty in transforming the abstract into a more concrete plan. The fourth screen highlights a visual progress representation, like a growing garden, that grows as users earn points and unlock milestones. This addresses the challenge of long term engagement and consistency for users, which was discovered in the needfinding phase. This prototype design encourages consistent task engagement and reduces friction, by making prioritization a rewarding and interactive process. A more detailed view of each screen, along with the prompts use to make them, can be found in

Appendix 16.6.1: Prototype C: Gamification.

7 EVALUATION PLANNING

To evaluate the three initial design alternatives, we will conduct a round of semi-structured interviews with current students. The goal of this evaluation is to compare how effectively each prototype supports decision-making around task prioritization, particularly in situations involving competing demands, uncertainty, and cognitive load.

7.1 Participants and recruitment

We will recruit roughly eight participants, with each team member conducting interviews with approximately two students. Participants will be current students recruited from classmates and personal networks, ensuring familiarity with academic scheduling and prioritization tasks. This aligns with the target user group identified in our needfinding, which focused on students managing multiple academic responsibilities.

7.2 Procedure

Each interview will last approximately 15–20 minutes and will follow a consistent structure. Interviews will begin with a brief set of warm-up questions to establish the participant's current workload, existing strategies for prioritization, and baseline confidence in decision-making.

Participants will then be introduced to the three prototypes. For each prototype, they will be asked to respond to the same scenario involving multiple competing academic tasks (e.g., a near-term deadline, a longer-term project, and an avoided task). This ensures consistency across conditions and allows for direct comparison between prototypes. The order of prototype presentation will be rotated across participants to reduce ordering effects.

During interaction, participants will be encouraged to think aloud as they engage with each prototype. This allows us to capture real-time reasoning, confusion, and decision strategies rather than relying solely on retrospective explanations.

After interacting with each prototype, participants will answer a short set of quantitative and qualitative questions. At the end of the session, participants will complete a cross-prototype comparison and reflect on overall preferences

and gaps across the designs. The full set of interview questions is provided in *Appendix 16.7: Evaluation Plan*.

7.3 Measures

The evaluation includes both quantitative and qualitative measures in order to capture both measurable differences and deeper behavioral insights.

Quantitative measures. For each prototype, participants will provide Likert-scale ratings (1–5) on:

- Ease of deciding what to work on
- Confidence in their decision
- Perceived mental effort
- Likelihood of using the system

Participants will also complete comparative measures, including ranking the three prototypes and selecting which one reduced decision stress the most. These measures allow for direct comparison across designs and support basic statistical analysis.

Qualitative measures. Qualitative data will be collected through think-aloud protocols and open-ended interview questions. These responses will capture:

- How participants interpret and interact with each interface
- What aspects of each design support or hinder decision-making
- How each prototype compares to participants' existing strategies
- Remaining gaps or unmet needs across all designs

This combination of real-time and reflective feedback is particularly important given that prior needfinding showed that decision-making is often intuitive, context-dependent, and difficult for users to fully articulate without prompting.

7.4 Analysis plan

Quantitative data will be analyzed using descriptive statistics, including mean and variance of ratings for each prototype. Because each participant evaluates all three prototypes, we will compare within-subject differences across designs. If appropriate given the sample size, we may conduct simple paired statistical tests (e.g., paired t-tests) to compare ratings between prototypes.

Qualitative data will be analyzed using thematic analysis. Interview notes will be

reviewed and coded to identify recurring patterns in how participants make decisions, where they experience friction, and which design features most strongly influence their choices. Themes will be compared across prototypes to identify which design approaches most effectively support prioritization.

Together, these analyses will allow us to both quantify differences in perceived usability and decision support, and explain why those differences occur.

7.5 Biases and limitations

Several potential sources of bias are present in this evaluation. The participant pool is limited to students within our networks, which may reduce generalizability. Think-aloud protocols may also alter natural behavior, as participants become more reflective than they would be in real usage. Additionally, because the prototypes are medium-fidelity, participant responses may reflect interpretation of the concept rather than fully realistic interaction.

To mitigate these issues, we use a consistent task scenario across prototypes, rotate presentation order, and combine multiple data sources (ratings, rankings, and qualitative feedback). While these steps reduce the impact of individual biases, some limitations remain, particularly related to sample size and prototype fidelity.

8 EVALUATION RESULTS

We performed nine evaluation interviews, with participants from a variety of fields across undergraduate, graduate and professional backgrounds. They were recruited through personal networks, following a semi-structured interview format that included think-aloud walkthroughs of the prototypes. Due to the incomplete results from three of the participants, the cross prototype quantitative analysis will only use the six participants who completed all of the ratings for all three of the prototypes. The qualitative analysis will consider the incomplete findings. The evaluation covered Prototype A: AI-assisted breakdown, Prototype B: Cognitive heatmap, and Prototype C: Gamification. Through all the interviews, the same scenarios were used: one assignment due tomorrow, one large project due next week, and one reading you've been avoiding. After they described what they would work on next using each prototype and why. The raw data from the evaluation interviews can be found in *Appendix 16.8: Evaluation Data*.

8.1 Quantitative Results

There were four Likert-scale questions given for each prototype on topics such as the ease of deciding what to work on, their confidence in their decision, their perceived mental effort, and the likelihood of using the system. We also used Friedman tests for analysis, because we used ordinal ratings from the same participants for all of the prototypes (numiqo Team, 2025).

Table 1—Descriptive Quantitative Results Across Prototypes

Measure	Prototype A	Prototype B	Prototype C
Ease of deciding what to do	3.50	4.50	4.50
Confidence in their decision	4.00	4.33	4.33
Perceived mental effort (lower is better)	2.67	2.42	2.00
Likelihood of using the system	2.50	3.58	3.67
Average ranking (lower is better)	2.33	2.00	1.67
Times ranked most helpful	1	1	4

Prototype C and Prototype B are both tied for the highest mean rating for ease of deciding what to do at 4.5, Prototype A scored lower at 3.5. The confidence in their decision ratings followed a similar pattern with Prototype C and Prototype B at 4.33, and Prototype A lower at 4.0. For perceived mental effort Prototype C scored lowest at 2.0, the Prototype B scored 2.42, and Prototype A scored highest at 2.67. For the likelihood of using the system Prototype C scored the highest at 3.67, then Prototype B scored lower at 3.58, and Prototype A at 2.5.

The Friedman test results did not show statistical significance for any of the four topics. Ease of deciding what to do: $x^2(2)=4.11$, $p=.128$, confidence in their decision: $x^2(2)=2.00$, $p=.368$, perceived mental effort: $x^2(2)=1.63$, $p=.444$, and the likelihood of using the system: $x^2(2)=4.11$, $p=.128$.

The average ranking for each prototype, across participants, in the cross prototype ranking is Prototype C with 1.67, then Prototype B with 2.00, and finally Prototype A with 2.33 as the lowest ranking.

8.2 Qualitative Results

There were several recurring themes discovered during the qualitative process across participants. One recurring theme among participants is that they made

their real world scheduling decisions in a reactive way, focusing mainly on deadlines instead of using a strategy. Hints of this showed in their prototype preferences where they preferred designs that required less active thinking, instead of designs with more setup or interpretation necessary. Further breakdown of qualitative results can be found in Table 5 within *Appendix 16.8.2: Interview Summary*.

Prototype A

This prototype's usefulness for breaking down larger tasks into manageable subtasks were mentioned. Participants also noted that although it made larger tasks more approachable, it didn't actually help them choose between competing tasks. It was deemed as more of a planning aid in a deadline centric process. There were concerns over the way the subtasks fit into the calendar and unease over trusting the AI's capabilities in capturing all of the assignment elements in generating subtasks. The participants desired the ability to manually edit all of the components, place subtasks on different days, clearer visual weighting of subtasks by either importance or effort, and clearer AI transparency.

Prototype B

This prototype assisted participants in thinking beyond the upcoming deadline, including the "what's coming up" summaries and the energy check-ins that helped with managing workloads over time and planning ahead. The acknowledgment of fluctuating energy levels in a visible way during the week was also a feature they liked. The cognitive heatmap itself brought some confusion, more specifically on the color coding definitions not being clear enough. From participant feedback, this prototype could improve with clearer labeling, including text based labels in addition to the colors and more explicit explanations on why tasks are reordered or recommended.

Prototype C

According to the participants, this prototype was by far the most engaging and the easiest to act on. They described the simplified daily planning flow and the swipe based prioritization as helpful in overthinking reduction. They also felt that by only presenting a few tasks at a time, it was easier to commit to a plan. They were able to imagine themselves performing tasks they would normally avoid due to the gamified reward structure. One of the issues, mentioned by several participants, is the desire for more control after swiping, including editing, reordering, and revising decisions later on. Some mentioned wanting clearer explanations of swipe direction definitions and better support for working on

multiple assignments concurrently. Participants enjoyed the low friction decision flow, but still wanted task customization, editing, and more support for flexible, realistic work styles.

8.3 Analysis

The evaluation highlights that the three prototypes addressed a different portion of the decision-making problem. Prototype A was strongest at making large projects less intimidating and task decomposition, but lacked support on deciding what to do next. Prototype B was strongest at helping users think ahead and visualizing their workload, but it lacked clarity in some key interactions and visual language. Prototype C was strongest at helping users commit to actions and reducing decision stress, but it didn't fully support longer term planning and lacked flexibility after decision-making.

Decision support among equally pressing tasks, urgency, and reducing cognitive overhead without heavily front loading work are still priorities for the participants. This evaluation highlighted that users want guidance without losing control. They desire a tool that assists them in understanding what is currently important, what comes next, and why. The results strongly support a combined design direction that integrates task breakdown, progress tracking, energy awareness in planning, and a low friction prioritization flow.

9 SECOND ITERATION PLANNING

9.1 Results and Analysis

The first iteration of the design life cycle confirmed that while current tools support task organization, they offer limited assistance in the decision-making process itself. During the evaluation phase, several insights into users' needs which were not captured during the initial needfinding phase were introduced. Students highlighted that they struggle with prioritization and decision-making, however, the evaluation phase revealed specific requirements around how potential solutions can be designed to support them. It was clear that users need help organizing tasks but they also require real time decision-making assistance in the moment of choosing what to select working on. Multiple participants mentioned that they do not rely on structured prioritization strategies but instead make decisions primarily on deadlines and immediate urgency, often resulting in impulsive behavior. This scenario suggests the need for an application that

actively supports decision-making compared to just presenting a list of tasks. The evaluation phase also revealed that there are other factors which influence task selection, including effort, energy level, mood, and stress. During the initial interviews, we discovered that students often avoid tasks which feel difficult or overwhelming at first regardless of their importance. This situation indicates that procrastination is frequently driven by uncertainty about an assignment or lack of clarity but not through the lack of motivation.

When users are assigned with multiple tasks that share a common deadline, they often experience stalled decision-making due to conflicting priorities. This demonstrated the need for a clear prioritization guidance with explicit recommendations. The evaluation also highlighted that existing task management applications contributed to an overall fragmented work flow and increased cognitive load because students rely on multiple applications to complete tasks. Participants in the study expressed a need for applications that reduce cognitive load by offering a solution that functions as a "one stop shop" without any additional effort to maintain. Lastly, users emphasized the importance of flexibility including their ability to edit their decisions after displaying recommended tasks. This key finding suggests that designs must balance guidance along with user control to be effective. Overall the evaluation refined users' needs towards an application that reduces cognitive load, supports real time prioritization decisions, adapts to human behavior and context.

The Gamification prototype was selected as the foundation for the final prototype due to its strong performance in engagement, usability, and ease of decision-making. Participants responded positively with its seamless interaction flow and its ability to reduce cognitive load while encouraging task completion, indicating potential for improved user retention and sustained engagement. As a result, the final prototype will be built upon the gamification prototype while integrating key features from the two initial prototypes, including task breakdown capabilities from the AI assisted design and the workload visualization from the heatmap design. This combined approach targets to provide prioritization, reduce cognitive load, and increase user engagement.

9.2 Challenges

The next phase of the design life cycle aims to determine whether combining all proposed features into a single application enhances usability or introduces

excessive complexity, potentially turning a simple prioritization problem into a more complicated interface. This question raises concerns about whether the application would effectively guide decision-making or simply function as a feature heavy to do list. Additionally, there is uncertainty around usability and cognitive load, particularly whether users can easily understand and navigate the interface without additional mental effort. Trust in AI driven recommendations is another key question, especially whether students will rely on automated task breakdown and prioritization. Another area for evaluation is the balance between flexibility and structure, specifically the extent of user control before an application becomes overly rigid and less effective. It remains unclear how users would make decisions when presented with multiple overlapping features, and how they will prioritize when choosing which task to work on.

9.3 Think Aloud Study

In general, these questions demonstrate the need for further investigation into the interface complexity, flexibility, trust, and decision-making behavior. To address these questions, a think aloud study will be conducted in the next phase of the design life cycle. The goal of the next iteration is to understand how users interact with a feature rich decision-making application and to assess whether it is effective and efficient in supporting task prioritization or whether it is overly complex in situations that involve cognitive load, conflicting decisions, and uncertainty.

9.4 Participants

We will recruit a total of five participants with each team member conducting one interview with a student. Participants will be current students recruited from personal and academic networks, ensuring familiarity with scheduling and prioritization of tasks in academic settings. This aligns with the target user group identified in the initial needfinding phase, which focused on students managing multiple academic activities.

9.5 Procedure

Each think aloud interview session will last approximately an hour. Participants will begin by answering a set of brief questions about their academic workload and prioritization strategies, followed by an interaction with a single prototype integrating multiple features (e.g. prioritization, breakdown, visualization, en-

gagement, and interactive tools). Participants will engage in scenarios including conflicting academic tasks with the same deadline, urgent vs. important tradeoff tasks, high workload, procrastination/low energy, interrupted workflow scenario, partially completed work, and a vague task. These scenarios were designed to reflect the real world variability in student workload which allows us to evaluate decision-making under different contexts and pressures.

9.6 Measures

Quantitative: Ratings (1-5) to measure the ease of decision-making, cognitive load, trust in recommendations, clarity, and likelihood of using the features presented. (Refer to Appendix 16.8.2 for questions and results) Qualitative: Think aloud data which will capture how users interpret features, what things influence their decisions, usefulness of features, trust in automation, and desired level of flexibility. (Refer to Appendix 16.8 for interview transcript)

This Think-Aloud study will evaluate whether an application with multiple integrated features supports decision-making effectively or introduces unnecessary complexity, and how users balance guidance, control, and interactions when prioritizing tasks.

10 FINAL PROTOTYPE

The final prototype combines elements from all three initial design directions into a single system that supports the full workflow of student task management: collecting tasks, planning them, deciding what to work on, and following through. Instead of separating scheduling and prioritization, the system treats them as part of the same continuous process. This directly addresses the main issue identified in needfinding: students already track their work, but struggle with deciding what to do next and acting on it.

Rather than optimizing for perfect planning, the system is intentionally designed around how students actually behave. Our needfinding and prototype evaluations found students are frequently replanning, shifting priorities, and making context-dependent decisions. This framing guided many of the final design decisions for our prototype. A link to a live version of the final prototype can be found on Figma.ai:
https://www.figma.com/make/OdlGx5BvUulA3aHjpwNHEj/Student-Scheduling-Tool-Design?t=hnWx4gljLHHiC42r-20&fullscreen=1

10.1 Weekly view and system overview

As shown in Figure 17, the primary entry point is a weekly view that integrates events, deadlines, and study blocks into a single interface. This view keeps scheduling and task management in the same space, reducing the need to switch between tools.

Users can see their commitments and available time at a glance, while also accessing both the task pool and Focus Mode. This design reduces reliance on memory by making workload and structure visible directly in the interface.

In contrast to traditional calendar tools, this view is not only for displaying events but also for supporting decision-making. By showing both fixed commitments and flexible work time together, the interface helps users reason about tradeoffs earlier rather than reacting at the last minute.

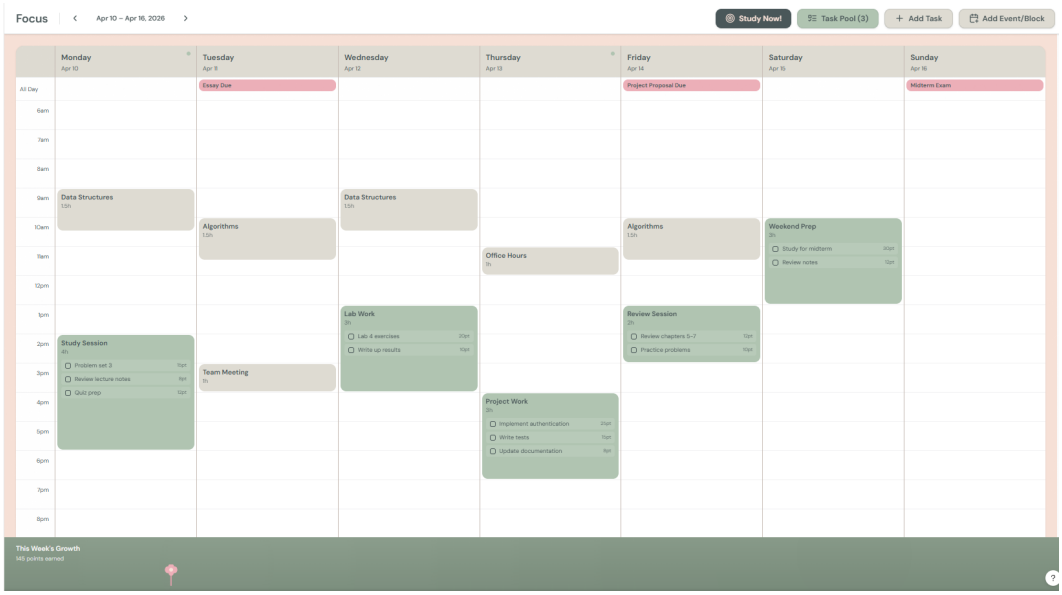


Figure 4—Weekly view integrating events, deadlines, study blocks, and entry points to task management and Focus Mode. Image created with Figma.ai.

10.2 Task assignment and planning

Figure 21 shows how tasks are incorporated into the schedule. Users can assign tasks from the task pool into study blocks across the week. These blocks are designed to be flexible rather than rigid; instead of committing to an exact plan,

users create a rough structure for when work will happen. This directly responds to earlier findings that strict schedules often break down and get abandoned.

By allowing tasks to be easily added, moved, or removed, the system supports planning without over-constraining the user. Each task also includes an effort indicator, which allows users to see how much work is being placed into a given block as they assign tasks. This supports a lightweight version of the cognitive heatmap concept by helping users distribute effort more evenly across their week rather than unintentionally overloading specific sessions.

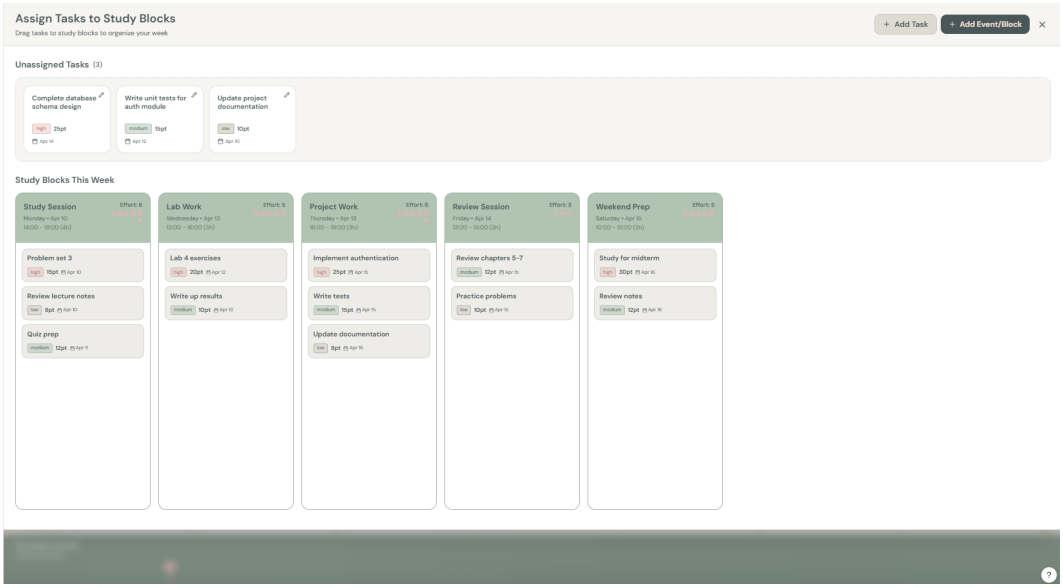


Figure 5—Task assignment into study blocks. Users drag tasks from the pool into flexible work sessions across the week. Image created with Figma.ai.

10.3 Study block interaction

Once tasks are assigned, users can view them at the study block level. As shown in Figure 22, each block displays its associated tasks, progress, and an entry point into Focus Mode.

This layer connects planning to execution. Users are not just scheduling time, but can see exactly what work is tied to that time. At the same time, the structure remains flexible, allowing adjustments without needing to rebuild the entire schedule.

This also introduces a clearer mental model of "what this session is for," which helps reduce the ambiguity that often leads to procrastination. Instead of starting a block and then deciding what to do, users begin with a defined set of options.



Figure 6—Study block detail views showing assigned tasks, progress, and direct entry into Focus Mode. Image created with Figma.ai.

10.4 Focus mode and task prioritization

The core decision-making interaction happens within Focus Mode, shown in Figure 24. Users can enter Focus Mode either from a study block or from the weekly view.

Inside Focus Mode, tasks are presented one at a time and sorted into categories including *do now*, *later this session*, or *not today* using swipe or button interactions.

This reframes prioritization as a sequence of small decisions rather than a single complex one. Instead of comparing multiple tasks at once, users only evaluate one task at a time. This reduces cognitive load and directly addresses patterns observed in interviews, where user often felt stuck or defaulted to urgency-based choices.

The one-at-a-time interaction also helps interrupt avoidance behaviors. Tasks that might otherwise be skipped in a list view still need to be acknowledged and

categorized, which creates a low-friction moment for re-engagement. The system also flags items that have been 'carried over' or skipped in the past, allowing the user to prioritize those tasks more easily.

The structured categories provide guidance without removing control, helping users move forward without overthinking the decision.

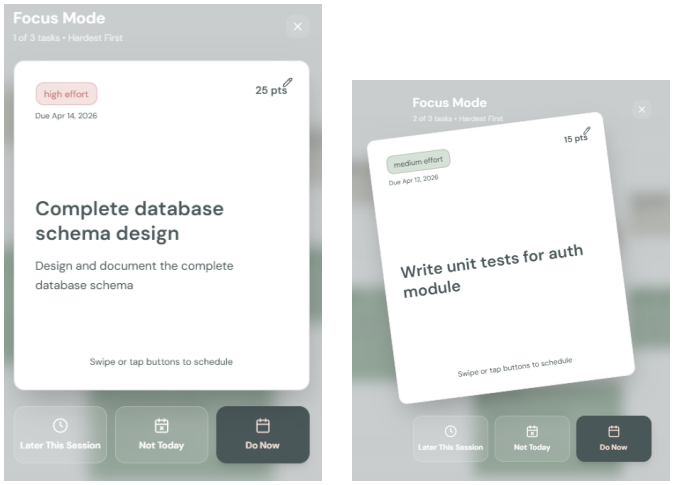


Figure 7—Focus Mode prioritization. Tasks are evaluated one at a time and sorted into do now, later, or not today using swipe or button interactions. Images created with Figma.ai.

10.5 Execution and progress feedback

After prioritization, users transition into execution within the same session. The interface maintains a clear sense of what is being worked on while still showing what comes next.

To support follow-through, the system includes a lightweight points system tied to task completion. As shown in Figure 26, these points contribute to a garden-based progress view that visualizes accumulated effort over time.

This feature builds on the earlier gamification concept but is intentionally minimal. The goal is to provide feedback and reinforce progress without distracting from the work itself. By tying progress to completed effort rather than just finished tasks, the system also helps make incremental work feel visible. This is particularly important for larger or ongoing tasks, which were frequently described as difficult to track and easy to delay.

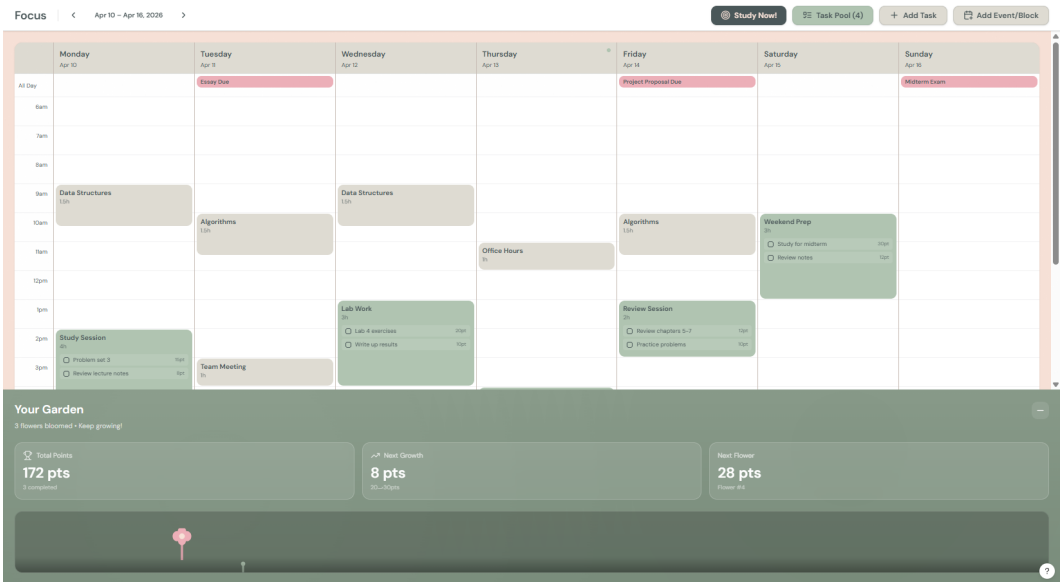


Figure 8—Garden progress view visualizing accumulated points and weekly progress. Image created with Figma.ai.

10.6 Supporting features and design integration

In addition the the core workflow, the system incorporates elements of AI-assisted task breakdown. When users add or edit large or unclear tasks, they have the option to break the task into smaller subtasks to better enable time management. The system also incorporates an AI element that can suggest these smaller subtasks and estimated effort levels. These suggestions are editable, maintaining user control while reducing the difficulty of getting started. This directly addresses a consistent issue from needfinding, where users delay tasks because they lack a clear entry point.

The final design represents a combination of all three initial prototypes. It retains task breakdown from the AI-assisted design, workload awareness from the heatmap concept, and prioritization plus motivation from the gamification approach.

This reflects evaluation findings that no single prototype fully addressed both decision-making and follow-through. The integrated system instead supports the full process, from planning to execution, while keeping interactions simple and adaptable. Overall, the design focuses on reducing cognitive load and helping users move more consistently from intention to action.

11 VIDEO PROTOTYPE

To complement the static representation of the final prototype, we created a video prototype demonstrating a full interaction flow through the system. The video presents a continuous, first-person walkthrough of the interface, focusing on how a student moves from an unstructured set of tasks to a prioritized, actionable plan. The full video prototype can be accessed at the following link: https://drive.google.com/file/d/1Wr2L1-AXlNqDEY2QVxM7HdDMLhkLKr5X/view?usp=drive_link

The walkthrough begins in the weekly overview, where the user can view scheduled study blocks and access the task pool. From there, the video demonstrates task creation and AI-assisted breakdown, emphasizing how larger or ambiguous tasks are decomposed into smaller, actionable components. This directly addresses the needfinding insight that students struggle to initiate work on large tasks due to unclear starting points.

The video then transitions into task assignment, showing how users move tasks from the task pool into specific study blocks. This portion highlights system feedback by giving due date warnings when moving tasks across time. It also reinforces the design goal of supporting forward-looking planning rather than reactive decision-making.

The core interaction is demonstrated through Focus Mode. The user enters Focus Mode from a study block and selects a prioritization strategy. Tasks are then surfaced as individual decision cards, which the user sorts into *start now*, *later this session*, and *not today* categories using swipe or button-based interactions. This interaction is intentionally lightweight and fast, reducing cognitive overhead at the moment of decision. The video also shows how unassigned tasks from the task pool can be incorporated into this process when entering Focus Mode from a study block, while entry from the main weekly view restricts prioritization to unassigned tasks only.

Finally, the walkthrough shows the outcome of this interaction: a reorganized study block with a clear, ordered set of tasks. This end state highlights the system's role in bridging the gap between deciding what to do and actually executing work, which was a primary failure point identified in both interviews and heuristic evaluation.

12 FINAL EVALUATION PLANNING

In this final evaluation, the focus will be on assessing the effectiveness of the integrated final prototype. It will be assessed on how well this newly created system supports decision-making from planning to execution. The plan is to recruit 20-25 participants from both classmates and personal networks to take an online survey.

12.1 Evaluation Procedure

The evaluation will be based on the video prototype by using a structured survey, taking about 8-10 minutes per person. Participants will first watch the video prototype, displaying the functions and features of the integrated prototype. They will first be asked to imagine themselves in scenarios (for example, various urgency levels of multiple assignments with competing deadlines) and complete survey questions consisting of both quantitative ratings and open ended questions. This consistency in approach is to ensure all participants evaluate the same interaction flow, while capturing their subjective feedback and interpretation.

12.2 Measures

Both quantitative and qualitative measures will be taken to asses decision support effectiveness, usability, and the user's perception. The quantitative measures will consist of 1-5 rating Likert scale questions covering the ease of deciding what to work on next, confidence in decision making, perceived mental effort, the clarity of system recommendations, and the likelihood of the user using the system in real life. These are directly related to challenges discovered during both the needfinding phase and the first evaluation. The qualitative measures will focus on open ended questions covering the topics of the most helpful or intuitive aspects of the system, the confusing or unclear aspects of the system, the system's effect on their prioritization approach, and suggested improvements or missing features. These questions would give deeper insights into the user experience.

12.3 Analysis Plan

Descriptive statistics will be used to evaluate the quantitative responses, including both the mean and standard deviation per measure. This will help highlight overall trends in the effectiveness and usability of the system. Moving from multiple prototypes to one has limited the options for inferential statistic testing,

but there will be an examination of response distributions to showcase patterns in user perception and areas of concern. The qualitative data will be analyzed via thematic analysis, by identifying recurring themes present across responses relating to cognitive load, decision support, flexibility, clarity, and user trust in the system. The results will be compared against earlier findings to determine whether the integrated design successfully addresses the previously identified issues.

12.4 Biases and Limitations

Some of the limitations present in this evaluation is the limited generalizability due to recruiting from classmates and our personal networks. There is also the fact that we are using the video prototype instead of having the participants freely participate with an interactive system directly, which may have some influence on participant responses. The self-reported nature of surveys may introduce some bias from the user, including them overestimating preferences or usability. This is mitigated by using both quantitative and qualitative feedback, giving a more comprehensive understanding of the user's experience. These limitations, although present, don't prevent the evaluation from providing a structured assessment of the final prototype's effectiveness and showcasing areas where improvement is needed. The full survey questions can be found in *Appendix 16.10: Final Evaluation Survey Questions*.

13 FINAL EVALUATION RESULTS

The final evaluation was conducted as planned using an online survey with participants recruited from classmates and personal networks. A total of twenty participants completed the online survey after viewing the video prototype demonstrating the functionality and the interactive flow. One response was identified as invalid based on an internal inconsistency: all ten quantitative ratings were scored as 1 (Strongly Disagree), while the corresponding qualitative responses were uniformly positive and generic (e.g., "Everything was clear and easy to understand," "Solid prototype"), suggesting the response was not completed in good faith. The analysis that follows is based on the remaining 19 valid responses. The raw survey data can be found in Appendix 16.11: Final Evaluation Survey Data.

13.1 Participants

A total of 19 valid respondents were included in the final evaluation. Of these participants, 18 identified as graduate students and 1 as a high school student or lower, reflecting the primarily graduate-student population recruited from academic networks and aligned with the target user group. The sample participant pool is consistent with the population identified during the earlier needfinding phase, which focused on students managing multiple academic responsibilities and competing deadlines.

Participants reported varying planning frequencies: 11 indicated planning their academic work on a weekly basis, 4 monthly, 2 daily, and 2 rarely. This distribution captures a range of planning behaviors providing perspectives from users with different approaches to task management and prioritization.

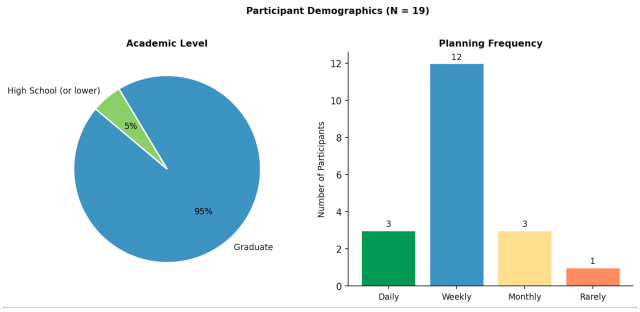


Figure 9—Demographic composition of evaluation participants, including academic level and reported planning frequency.

13.2 Quantitative Results

Participants responded to the ten Likert scale items (1-5) following the review of the video prototype. As mentioned in the analysis plan, descriptive statistics were calculated for each measure. Since the evaluation centered on a single integrated prototype rather than comparing multiple designs, inferential statistical tests were not conducted. Analysis focused on mean ratings, standard deviations, and distributional patterns to identify overall trends in usability, decision support, and potential areas for improvement.

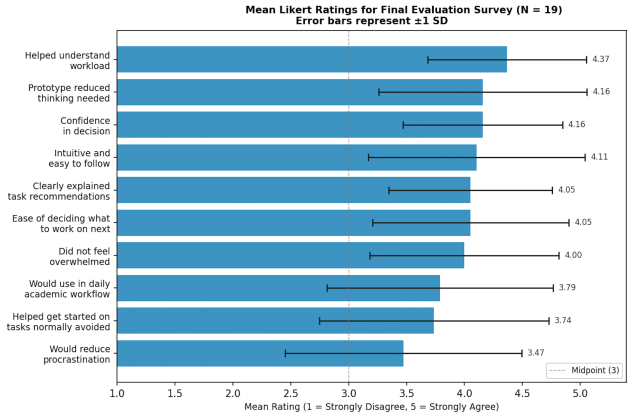


Figure 10—Mean Likert Ratings for Final Evaluation Survey. Average participant ratings across nineteen survey measures, providing an overview of perceptions of usability and decision support effectiveness.

Descriptive Statistics for Final Evaluation Survey (N = 19)
Amber rows indicate the two lowest-rated items.

Survey Item	Mean	SD	Min	Max
Ease of deciding what to work on next	4.05	0.85	2	5
Confidence in decision	4.16	0.69	3	5
Prototype reduced thinking needed	4.16	0.90	2	5
Clearly explained task recommendations	4.05	0.71	3	5
Helped understand workload	4.37	0.68	3	5
Would use in daily academic workflow	3.79	0.98	2	5
Prototype felt intuitive and easy to follow	4.11	0.94	2	5
Did not feel overwhelmed	4.00	0.82	2	5
Helped get started on tasks normally avoided	3.74	0.99	1	5
Would reduce procrastination	3.47	1.02	1	5

Table 2—Descriptive Statistics for Final Evaluation Survey. Mean and standard deviation values for each survey measure, summarizing usability, decision making, task support and user variability.

Table 2, presents descriptive statistics for the final evaluation survey (N=19). Overall, the participants of this survey rated the prototype positively across usability, decision support, and task management measures. The highest-rated item was "The prototype helped me understand my workload" (M = 4.37, SD = 0.68), indicating that the integrated design was particularly effective at externalizing and communicating workload structure. Confidence in decision-making (M = 4.16, SD = 0.69) and reduced cognitive effort in task selection (M = 4.16, SD = 0.90) also scored highly, suggesting that the system's approach to surfacing tasks and guiding prioritization was effective. Clarity of task recommendations (M = 4.05, SD = 0.70) and ease of deciding what to work on next (M = 4.05, SD = 0.85) were similarly strong, with relatively low variability across participants

who found the prioritization logic both accessible and useful.

Usability ratings were favorable with participants reporting that the prototype felt intuitive and easy to follow (M = 4.11, SD = 0.94) and did not feel overwhelming to use (M = 4.00, SD = 0.82). Together, these findings reflect favorably on the usability of the final design. The slight differences in standard deviations for these measures, relative to the workload and decision confidence measures suggest modest variation in how participants perceived and navigated the interface's complexity.

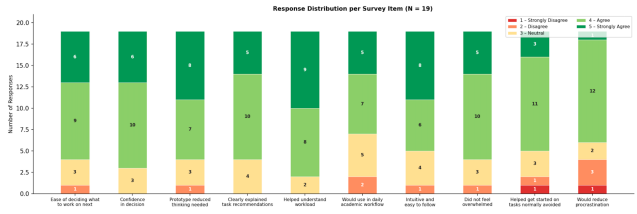


Figure 11—Shows the distribution of survey responses across usability and impact statements, including decision making ease, confidence, intuitiveness, task initiation, and procrastination reduction.

The daily workflow adoption intent (M = 3.79, SD = 0.98) was moderate, with greater dispersion than the core decision-making items. This distribution suggests that while the system resonated strongly with some participants, others with established scheduling planning practices were less certain it would displace their existing approaches. The two lowest-rated measures related to the prototype's ability to support initiation of task participants normally would avoid (M = 3.74, SD = 0.99) and its perceived effectiveness in reducing procrastination (M =3.47, SD = 1.02). The comparatively low mean ratings coupled with higher standard deviations indicate greater variability in participant responses and suggest that the prototype's impact on avoidance related behaviors might be inconsistent across users. These findings suggest room for refinement in strengthening the application's support for behavioral follow.

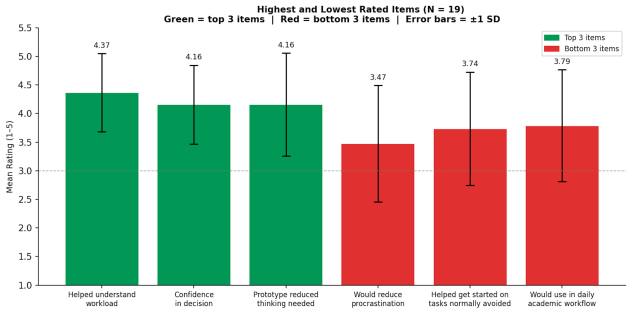


Figure 12—Highest and Lowest Rated Items. Comparison of highest and lowest scoring survey measures, highlighting the strongest and weakest rated aspects of the prototype access usability, decision support, and behavioral outcomes.

13.3 Qualitative Results

The qualitative data from the seven open-ended questions was analyzed using thematic analysis. Responses were reviewed across all questions to identify recurring patterns in what participants valued, points of friction, and anticipated use of the system.

Task organization and prioritization features emerged as the most valued aspects of the final prototype. Participants most frequently highlighted task and sub task creation, prioritization recommendations, and effort-level indicators. Several survey participants highlighted the ability to break tasks down into sub tasks automatically as particularly useful for initiating complex work. Others emphasized the task recommendation feature as the most valuable aspect of the system, describing it as reducing the mental overhead of deciding where to begin. Effort level indicators were consistently appreciated across multiple participants for making task demands more explicit without requiring users to retain information mentally. Overall, these features were perceived as directly addressing decision fatigue and planning friction identified in earlier needfinding.

Gamification elements received mixed feedback often in context dependent manner. The points system and effort-based rewards were described as motivating by several participants, particularly for distinguishing difficulty of tasks and encouraging engagement with less desirable tasks (that would otherwise be avoided). However, the garden visualization was frequently described as unclear or limited in its usefulness, with some participants indicating that they would not person-

ally engage with it even if they recognized its potential appeal to others. The point allocation logic was also a mild source of confusion for some respondents who were uncertain about how points were earned or calculated. This pattern is consistent with feedback from the first evaluation round, where Gamification elements were broadly engaging but not universally compelling.

Focus Mode and study blocks were the most commonly misunderstood features. These two features were frequently mentioned in responses related to confusion or usability challenges. Several participants expressed uncertainty around how study blocks related to Focus Mode, as well as differences between entering Focus mode from a study block versus from the main weekly view. The workflow of assigning tasks to study blocks before execution also required clarification for some users. While participants generally responded positively once the concepts were understood, the findings suggest that these interaction flows are not immediately intuitive and would benefit from improved onboarding, labeling, or guidance.

A majority of participants indicated the system would change how they approach task selection. Approximately 14 of 18 valid respondents reported that the prototype would influence their decision making process, citing improved visibility into workload, reduced decision fatigue due to automatic task surfacing, and the incorporation of effort level context in prioritization. The remaining participants generally referenced established workflows they were already satisfied with, or a preference working under pressure rather than structured planning.

Desired improvements primarily focused on integration and increased user control. The most commonly requested enhancements included integration with external tools such as Canvas or Google Calendar, the addition of a Pomodoro or focus timer, and the ability to customize priority weights or manually adjust automated system recommendations. A smaller number of participants suggested support for recurring or template tasks. These suggestions are consistent with feedback from the first evaluation round, reinforcing the importance of balancing user control alongside system-generated guidance.

Lastly, intended use cases aligned closely with the original design context. Participants most commonly described using the prototype at the start of each week for planning or during periods of heavy workload when prioritization becomes the primary obstacle. Several participants specifically highlighted the applica-

tion's usefulness in scenarios involving multiple competing deadlines of similar urgency, directly reflecting the problem space originally identified in the initial design phase.

13.4 Analysis

The final evaluation results suggest that the integrated prototype successfully addressed several core gaps identified during earlier needfinding and first-round evaluation. The highest-rated measures including workload understanding, decision confidence, and reduced cognitive effort, correspond directly to the issues most frequently identified in needfinding, where students reported difficulty in prioritizing competing tasks and often experienced hesitation or reactive decision making when choosing what to work on. Compared to the initial evaluation where no single prototype fully supported both decision making and translating planning into execution. The final prototype with the integrated design demonstrates clear improvement across these dimensions and was seen by participants as addressing the intended problem.

However, some limitations remain. The lower ratings for procrastination reduction and support for initiating avoided tasks suggest that the prototype does not fully address the behavioral and avoidance related challenges identified earlier, where procrastination was often linked to uncertainty and cognitive load rather than the lack of motivation. This may indicate that the current approach to highlighting overdue or avoided tasks is not sufficiently different from general task recommendations to influence user behavior consistently. In addition, the greater variation in adoption intent suggests the application may be better suited to users who already follow structured planning routines rather than those with more flexible or reactive habits.

Usability feedback around Focus Mode and study blocks represent an important gap, given these features are central to the system's bridging planning and execution. Participant responses suggest that while the interaction concepts are viable, they require clearer communication and explanation. The integration requests raised by multiple participants highlight a remaining point of friction: the system's usefulness is partially constrained by the effort required to manually enter and maintain task data, which participants in both the initial and final evaluations identified as a barrier to sustained use.

Overall, the final evaluation indicates that the integrated prototype effectively

supports early stage planning and task prioritization. Key design elements such as effort visibility, task recommendations, and structured daily planning were well received and validated by participants. Future improvements should focus on strengthening support for execution of intended actions, simplifying onboarding for more complex features/interactive flows, and reducing data entry burden through integration with existing academic tools.

14 INDIVIDUAL REFLECTIONS

Hasan Shaikh

In the need-finding, I did the Heuristic Evaluation plan and results. My idea in the brainstorming was the cognitive load heatmap, I also made the initial prototype adjacent to this idea along with other discussed features (Prototype B). For the evaluation plan, I did one interview (the rest of my team did two each minimum). For the key insights of these interviews, I aided in discussion only. For the final evaluation, I made all the figures and created a simple rough draft of the analysis. For the final prototype, I only helped in initial ideation and schematics planning. The design was then prompt engineered by teammates in Figma.

Sophia helped keep the project ahead of schedule, did interviews at several stages of the process, made Prototype C, largely prompt engineered the final design, did the video, did some writing. Carline did most of the writing and data organization, made Prototype A, helped refine aspects of the final design, and some interviews. Shivani did the surveys for the initial need-finding, some writing, the final draft of the final evaluation/analysis, and some interviews.

The project progressed well, my teammates were always very ahead of schedule and expected me to finish tasks well before checkpoints. I think communication could have been better and overall expectations should have been set firmly from the start. There seemed to be a miscommunication between team members that wanted to get ahead of schedule and team members that seemed to function better on procrastination. The rushed deadlines plus miscommunication caused a lot of unneeded tension that should have been addressed earlier in the semester. I wish the groups were mandatorily created by force earlier into the semester to allow us the time to get acquainted with one another without the stress of starting the project immediately. Perhaps a mandatory team contract with an

accountability clause a week before the project started would have helped us get acquainted while setting expectations. Our team was constantly in a rush to finish things and unlike my other classes, I do not feel like I got to connect with my teammates personally although I tried my best to make light-hearted conversation during our meetings and communicate my workflow. Granted, I missed many Saturday morning meetings due to my work the night before always going far past midnight so perhaps I contributed to this problem. Overall, I hope I had a net positive effect on the team atmosphere and contribution though.

Sophia Greathouse

I contributed across all stages of the project, with most of my work focused on coordination, synthesis, and final deliverables. I handled scheduling and communication, organized and ran meetings, and made sure we were staying on track across sections.

During needfinding, I wrote and conducted the interview activity and later helped synthesize results across the survey, interviews, and heuristic evaluation.

For design, I developed Prototype C (gamification) and then built the final prototype by combining ideas across all three directions. I also wrote the final prototype section. This was originally assigned to another team member, but I took it on when it was not submitted in the shared document on time.

I also picked up additional writing beyond what I was originally assigned, including the brainstorming results and evaluation planning sections, so we could stay on schedule. I designed the evaluation instruments and helped structure both the evaluation plan and results. A lot of my work in the later stages was pulling together themes and making sure everything connected back to our findings.

For the final deliverables, I created the video prototype, including the storyboard and walkthrough (with help from teammates on the call), and wrote the video section. I also completed the final readthrough of the paper, making edits for grammar, spelling, and consistency, and rewriting portions as needed to better align with assignment requirements.

Team contributions

Carline was very consistent and reliable throughout the project. She attended every meeting, developed Prototype A (AI-assisted breakdown), contributed a large portion of the writing and references, and supported evaluation planning and final deliverables. She also stepped in to complete the initial prototyping section when timelines became constrained and handled submission.

Hasan developed Prototype B (cognitive heatmap) and completed the heuristic evaluation. He contributed to design iteration and parts of the evaluation and final review, and attended most meetings. He contributed to some of the writing, with his contributions strongest on the design side. He also completed the final analysis and co-wrote the final evaluation results section with Shivani.

Shivani completed the needfinding survey and was assigned several writing sections, including second iteration planning and the final prototype writeup. However, her meeting attendance was inconsistent, and some of her assigned sections were not completed in the shared document or were not accessible to the team prior to deadlines. This made it difficult to assess progress and determine what work still needed to be completed. As a result, Carline and I divided and completed those sections to ensure the project stayed on schedule. She also co-wrote the final evaluation results section with Hasan.

Reflection on project process

Overall, the project turned out strong, especially in terms of the design process and final prototype. One thing that worked well was keeping the focus on decision-making instead of just task organization. That helped keep the project aligned with the original problem we were trying to solve.

At the same time, the process required some adjustment, especially in the later stages. When work was not consistently completed in a shared and visible location, it became harder to track progress and required redistribution of tasks to stay on schedule. We were able to adapt and still produce a cohesive final result, but it did add extra coordination and workload.

If I were to do this again, I would set clearer expectations earlier around working in a shared location and add more structured checkpoints so we can catch gaps sooner. I also would plan more explicitly for how to redistribute work if needed, rather than figuring that out in real time.

Overall, this project reinforced how important visibility and coordination are in team-based work. Even with some uneven participation, we were able to finish strong by staying flexible and making sure everything came together at the end.

Carline Durmus

During this project, I wrote the introduction and needfinding plan sections, excluding the interview subsection. I also completed the interview results and heuristic evaluation results subsections of the needfinding results section. I assisted in writing the initial prototyping section and created prototype A (AI-assisted task breakdown) within the initial prototyping section. For this prototype, I used Figma to create it and completed both the evaluation results and the final evaluation planning section. In the design process, I participated in both the individual and group brainstorming sessions, conducted a portion of some of the evaluation interviews, and both created and wrote the questions for the evaluation survey.

Throughout the process, I attended every scheduled group meeting and added many different pieces of supplemental information to the appendix. To the appendix, I added the needfinding survey questions, the needfinding interview questions, the needfinding interview notes, the needfinding survey results data, the needfinding heuristic evaluation results table and walkthrough text, the raw data from the evaluation interviews, and the final evaluation survey questions. These are my detailed contributions to the project.

Group Member Contributions

Sophia greatly contributed to the success of this project by organizing the group meetings, taking notes, and posting them after every group meeting. She also created and conducted the needfinding interviews, participated in individual brainstorming, group brainstorming sessions, and conducted a portion of the evaluations interviews. She also completed writing many sections of the project, including the interview subsection of the needfinding plan section, the interview results subsection of the needfinding results section, the initial brainstorming plan section, the brainstorming results section. She assisted in writing the initial prototype section and completed prototype C (gamification) using Figma.ai for the initial prototype. She completed the evaluation planning section, the final prototype section, and the video prototype with the corresponding section. Sophia also added many supplemental materials to the appendix.

Hasan performed the needfinding heuristic evaluation and completed prototype B (heatmap) within the initial prototype using Figma. He also participated in

the group brainstorming session and performed his individual brainstorming beforehand. He conducted a portion of the evaluation interviews, assisted in producing the final prototype and assisted with the final evaluation results section. Shivani created and conducted the needfinding survey and conducted a portion of the evaluation interviews. As far as writing, she completed the second iteration planning section and assisted in the final evaluation results section.

Reflection

Overall, the project progressed well. We were able to stay on top of the content deadlines that we set for ourselves and we were able to get everything completed in an orderly manner. There were some communication issues that could have been better, due to varying schedules and work habits, but we were able to successfully complete the project without undue stress on a single person. There is nothing that I would've wanted to know before the start of the project, but it would've been better if we had more time to acclimate towards one another before the deadlines started coming.

Shivani Gurrapu

Individual Contributions

I helped bring the group members together by forming and initiating the team at the beginning of the project. I started the initial conversations and set up a group channel to facilitate efficient communication. I helped guide the initial direction for the final project by sharing ideas, identifying common interests, and organizing the topic selection process to finalize the focus of the project. Throughout the project, I completed my assigned work on time and met my deliverables as committed. While I was unable to attend two meetings due to conflicting work, I remained engaged by contributing to team discussions and delivered tasks on time. For this paper, I contributed by attending team group meetings, conducting think aloud interviews, creating needfinding surveys, writing the evaluation, second iteration planning, and final evaluation sections.

Team Contributions

Sophia set up group meetings, conducted interviews, wrote some of the sections in the paper, generated one of the initial prototypes, helped combine research findings, video prototype, and worked with Hasan to build the final prototype.

Carline contributed across writing, design, and evaluation. She wrote the introduction, needfinding sections, created prototype A, and worked on the evaluation results. She participated in brainstorming and interviews, added materials to the appendix, and delivered tasks on time.

Hasan mainly worked on writing the design, evaluation, and final evaluation sections. He created prototype B with cognitive heatmap and completed the heuristic evaluation. He joined the weekly meetings, helped improve the design and took part in evaluation interviews.

Final Reflection

Overall, the project has progressed well. There were some miscommunications around check-in deadlines and expectations. I worked mostly based on the deadlines communicated in class and consistently submitted my work on time while balancing other priorities in the OMSCS program. The process could've improved if the team had agreed on clear expectations for check-in deadlines and if groups were formed with similar schedules in mind.

At certain points in this project, there was a focus for some group members on submitting sections quickly rather than taking time to refine ideas and discuss design decisions in depth. I also think it would've helped if teams were formed with consideration on working styles and schedules, especially in an asynchronous class environment. Going into the project, I wish I had better understood how differently members might approach the check in deadlines.

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16 APPENDICES

16.1 Appendix 1: Needfinding Survey Questions

1. What is your current level of study?
   - Middle School
   - High School
   - Undergraduate
   - Graduate
   - Other
2. What applications do you currently use to manage your schedule and prioritize tasks? (Select all that apply)
   - Google Calendar
   - Notion
   - Notes app (e.g., Apple Notes, Google Keep)
   - Physical Planner/Journal
   - Other
3. How often do you use your primary scheduling/task management tools?
   - Daily (Multiple times a day)
   - Weekly (Several times a week)
   - Occasionally (A few times a month)
   - Rarely (Less than once a month)
4. In general, how often do you successfully follow the schedule or plan you create?
   - Daily (I stick to it most of the time)
   - Weekly (I try to follow it, but often make changes)
   - Occasionally (I follow it when tasks become overwhelming)
   - Rarely (I often abandon my schedule quickly)
5. Which factors primarily drive how you prioritize your work/tasks? (Select the most applicable factor)

- Deadline (Most urgent tasks first)
- Importance (Tasks that have the biggest impact)
- Difficulty (Starting with the easiest or hardest task)
- Time/Energy (Prioritizing tasks based on my current energy level)
- I generally do not prioritize anything (I work on things as they come up)

6. On a scale of 1-5(Very Hard - Very Easy), how easy is it for you to prioritize tasks effectively?
7. What major challenges do you face when managing your tasks and schedule? (Please describe briefly)
8. Do you ever procrastinate because you are unsure of what task to start next?
   - Yes, frequently
   - No, never
   - Sometimes
9. What features would be beneficial in a tool designed to manage and prioritize your academic work? (Select all that apply)
   - Task prioritization suggestions
   - Smart reminders and alerts based on location or time
   - Automatic scheduling of tasks based on estimated time and availability
   - Visual timelines/Gantt charts of deadlines and project milestones
   - Integration with other academic platforms (e.g., Canvas, Blackboard)
   - Progress tracking and completion analysis
   - AI suggestions for breaking down large projects
10. How appealing would you find an app that analyzes your activity data and provides personalized recommendations for your next steps (e.g., suggesting a 30-minute break, or starting a medium-difficulty task)?
    - Very appealing
    - Slightly appealing
    - Neutral
    - Not appealing
11. When you are planning your week, how do you typically estimate the time required for a task?
    - Small Tasks (less than 1 hour)
      - Overestimate significantly
      - Slightly overestimate
      - Accurate estimate
      - Slightly underestimate

        - Underestimate significantly
    - Medium Tasks (1-3 hours)
        - Overestimate significantly
        - Slightly overestimate
        - Accurate estimate
        - Slightly underestimate
        - Underestimate significantly
    - Large Projects (3+ hours or multi-stage)
        - Overestimate significantly
        - Slightly overestimate
        - Accurate estimate
        - Slightly underestimate
        - Underestimate significantly
12. How would you rate the effectiveness of your current task management system in reducing stress? (1-5)
13. On a scale of 1-5 (Not prepared at all - Fully prepared), how prepared do you feel at the start of a typical week regarding your academic workload?

16.2 Appendix 2: Needfinding Interview Questions

16.2.1 *Introduction/warm-up*

- Can you tell me a bit about your current semester? What kinds of responsibilities are you balancing, in and out of class?
- What tools or methods do you currently use to keep track of your tasks or assignments?

16.2.2 *Recent scenario*

- Can you walk me through a recent time, like in the past few days, when you had multiple things to do and had to decide what to work on?
- What were the options you were choosing between?
- What did you end up choosing?
- What made you choose that?
- Did you consider anything else before deciding?
- Did you feel confident in that decision?

16.2.3 *Decision process*

- When you're deciding what to work on, what factors usually matter the most to you?
- How much do deadlines matter?
- How much does how long something will take matter?
- Do you think about how hard or stressful something feels?
- Do you have any kind of system or mental process for prioritizing, or is it more intuitive?

16.2.4 *Tradeoffs & conflict*

- What happens when two tasks feel equally urgent or important?
- How do you break that tie?
- Do you ever feel stuck deciding?
- Have you ever chosen something and later felt like it wasn't the right choice?

16.2.5 *Emotion & avoidance*

- Are there tasks you tend to avoid or put off? Why?
- What makes a task feel avoidable?
- What do you usually do instead?
- How does your mood or energy affect what you choose to work on? For example, if you're tired or overwhelmed, does that change your decisions?

16.2.6 *Reflection & gaps*

- Do you feel like your current way of deciding what to work on works well for you?
- What's the hardest part about deciding what to work on?

16.3 Appendix 3: Needfinding Interview Notes

16.3.1 *Introduction/Warm-up*

Can you tell me a bit about your current semester? What kinds of responsibilities are you balancing, in and out of class?

- **Ashton:** Working 33 hrs a week, taking 3 classes. 2 of the classes are psychology classes, 1 is developmental psych; that one is equal parts science and humanities. Social psych is mostly humanities—it's my easy class because I have an

affinity for writing. I give the most attention to stats because it's math. Every week for my psych classes I have a chapter to read and take notes—each chapter is 25–30 pages long. I have two assignments per chapter and one quiz per chapter for both of those. I'd say a quiz on 3 chapters every 3 weeks for both of those. Stats is a chapter per week along with a 40-question homework assignment, a quiz, and a discussion board every two weeks. In a perfect world I prioritize stats the most because it takes the most mental energy. In the beginning of the semester I got good at working ahead; now I'm doing readings the day they're due. That takes 3–4 hours per class with notes, so the days that I'm not working I do school the entire day, and I do school before work.

- **Austin:** Currently full time—9 hours in school. I also do research assistant work, and that typically is in office 2 days a week for 2 hours a session and work from home in my free time. I also work a job, typically Saturday and Sunday all day.
- **Annalise:** I'm taking one class that has a heavy and inconsistent workload (assignment types change throughout the semester). I typically spend over 15 hours a week on that course alone, and often much more. I also have a full-time job during the day that usually requires me to work in-person and not have access to regular network systems, and I have a job nights and weekends at a bar.
- **Gabriel:** I am currently a 3rd year in art school, a full-time art student studying graphic design. I'm balancing 4 full-time classes. In order to make time to afford my lifestyle and schooling, I have scheduled all four of my classes to be on two days. On school days, Mondays and Wednesdays, I pull up to school at 8 a.m. and stay until 12:30, then I take my graphic design class from home. I have a 2-hour break that I use to do schoolwork before my queer literature class. Then I go into my night class at 7 p.m., which varies in length between 15 minutes and 3.5 hours. I also work 7 shifts a week across two jobs. I work at Polaris Mall selling jewelry at a store called Swarovski (4 shifts a week) and I work 3 8-hour shifts at a bar called Cobra—the best bar in Columbus—from 3:30 p.m. to midnight or 7:45 p.m. to 3:30 a.m. I reserve one evening a week to socialize where I can, and try to keep in touch with family throughout the week.
- **Callum:** Right now has been the busiest semester I've had of school so far. We picked up extra classes, so it's a relatively high credit count; I'm also trying to get to the end of my one-year-long class and its group project. I'm also doing

applications for summer jobs. There's a lot of reading for each class every day—roughly 200 pages or more a week, and taking notes and discussing. This is on top of two extracurricular writing things and a competition.

- **John:** In class is more straightforward, I'm taking two intro-level asynchronous classes online. The workload is fairly rep-focused, a surface-level understanding survey. The assignment load is pretty light, one or two topics per week, roughly 2–6 hours of my time spent on those. But I really haven't been watching the lectures, I've been vibing through the homeworks and Googling where needed. Outside of school, I have a 9–5 that has exploded in terms of workload. I'm in a few different projects, moving into management, working on proposals, and more—usually 40 hours but sometimes up to 55. Outside of that I have hobbies, spending time with friends, household chores.

What tools or methods do you currently use to keep track of your tasks or assignments?

- **Ashton:** The calendar on Canvas is good because I can scroll ahead and it shows what is due when, and lists the type of assignment that's due. If I log on I can see that stats has three things due today, so I know that ahead of time. Last week I had fewer assignments for my psych class, and I knew because of the calendar that it was a stats-heavy week.
- **Austin:** Usually just trying to keep with the syllabus and weekly module setups. I typically try to put out the biggest fire in front of me; whatever is most pressing in the moment I try to focus on.
- **Annalise:** Nothing consistent. I usually use a planner, either physical or on my tablet, that I fill out at the beginning of the semester with all of the due dates from the syllabus. I use that for the first few weeks, but I haven't been good at being consistent with it. In reality I just go to all of my class pages (this semester only one) and see what's due next and try to keep mental note of longer-term deadlines.
- **Gabriel:** Actually this is the thing that I bust out not very often at all, but there is this app called "Structured." If I have a crazy week, I use it as an hour-by-hour planner with reminders. It's helpful because it gives countdowns before the next obligation, and it's pink and I like pink. Usually I'm checking between photos of my schedules or my schedules app, but mostly it's logged in my brain. If I need to do something unusual I'll "hey Siri" it into my reminders app.

- **Callum:** I'm generally pretty analog, so I'm relying on the syllabus for what the assignment itself is. Outside of the textbook, if there's an additional thing I need to read, I go back to the syllabus through Canvas. Otherwise, Outlook Calendar is where my due dates, timings, and things like that go, and where I have things due.
- **John:** I put everything in Notion.

16.3.2 *Recent Scenario*

Can you walk me through a recent time when you had multiple things to do and had to decide what to work on?

- **Ashton:** I made a risky move but it paid off. I had the discussion board, the homework, and the quiz of stats due tonight. Yesterday I had the chapter for developmental psych and the quiz due. In the past I prioritized whatever is due first, but this week I did stats first because I knew it was a stats-heavy week and it ended up working really well. I did much better on the assignments because I scattered the topics. Cramming psych was tough though because I worked that day as well. But because I'm better at it it's less hard than cramming stats.
- **Austin:** Currently I have a big data project due on Friday, a problem set for economics due on Tuesday, and a data assignment for economics due the Tuesday after that. The organization comes from what groups I'm meeting with and when.
- **Annalise:** I had a quiz due that I had to study for, as well as a project to finish. I honestly put all of the tasks into an AI prompt and had it make a checklist schedule for me. I asked it to kind of mix up the work so I wouldn't get bored doing the same topic for too long.
- **Gabriel:** Almost all of my classes have me do all of the work in class; there's very little to do outside of class unless you need to catch up. But my queer literature class we have to do all of the work outside of class and discuss inside class using the Socratic method. We often have 5 pieces of literature to read in between lectures. I can usually fit it into my break within classes, but I have to cram it in right before class, which may not be great for retention but I am prepared for class. I also find that drunk Gabe takes better care of me than sober Gabe, and sometimes I'll come home and get tasks done late at night. One upside of working late is that I have many hours late into the night where I am productive.
- **Callum:** We had a big final paper due yesterday, so I was in the final stages of

that and finishing getting with my group project partner and getting the work done while also having to do normal coursework. Prioritization comes to what comes next and in a hierarchy of the most important thing—so an imminent due date is more important than a reading assignment, and needing a time with my partner's schedule for the group project is also a factor.

- **John:** I've been continuously doing that over the past few weeks, as I'm getting reprioritized from above at work and they're pushing pretty aggressive timelines. I've had to communicate a lot about "if you need this by X date, what can drop in priority?" I'd say the biggest sacrifice in terms of things that didn't get done is homework; there are times that I just didn't allocate the time to do it. If it's only worth 1% of my grade I won't prioritize it over my actual career.

What were the options you were choosing between?

- **Ashton:** Developmental psych (both a chapter and a quiz) and stats.
- **Austin:**
- **Annalise:** I put in all of the subtasks for my quiz and project, so I could break it out into individual chapters (reading my notes for chapter 3.1, reading other students' notes for chapter 3.1, making flashcards for 3.1, etc.).
- **Gabriel:** Often it's one larger piece and a collection of shorter pieces. So maybe a short story and a collection of poems.
- **Callum:**
- **John:** Work tasks and school tasks, chores. A lot of things are dropping off.

What did you end up choosing?

- **Ashton:** Stats.
- **Austin:** Whichever group I'm meeting with next, they're expecting results when we meet.
- **Annalise:** I just went by the schedule the AI made for me. I'd mix it up a bit but mostly stuck to the plan.
- **Gabriel:** I often have to go by what can be skimmed the easiest. I can skim a short story but the poems have to be more deeply analyzed.
- **Callum:**
- **John:** Work tasks, prioritizing career. Order of priority is generally work task, mental health/time balance, school tasks, then home chores, etc.

What made you choose that?

- **Ashton:** I knew it was a stats-heavy week.
- **Austin:**
- **Annalise:** I had a hard time getting started and offloading the decision-making was enough momentum for me to get work done.
- **Gabriel:**
- **Callum:** The specifically due item for a grade comes first over a reading assignment for the next day's class, because that's not directly for a grade and can be made up in certain ways.
- **John:**

Did you consider anything else before deciding?

- **Ashton:**
- **Austin:** That's the main one, because people are relying on my work at that point of time.
- **Annalise:** No.
- **Gabriel:** No—sometimes she'll send out emails about which texts we'll discuss in class, so I'll focus on those. But if that's not an option, then I have to go with what I can do most easily.
- **Callum:** Schedules of my classmates for group projects.
- **John:** The organization shifts week to week—mental health will upgrade in priority if I've had a low mental health week, where work stays at a static priority level. If I haven't gone outside all week, I'll prioritize that the next week.

Did you feel confident in that decision?

- **Ashton:** Cramming psych was tough though because I worked that day as well. But because I'm better at it it's less hard than cramming stats.
- **Austin:** No, I wouldn't say I feel confident in it. I'd like to adjust it but I can't commit to doing that yet. It makes everything feel more difficult and hectic all the time.
- **Annalise:** Mostly yes, although in some cases it was unrealistic and heavily favored better studying at the front of the task list and cramming towards the end.
- **Gabriel:** I'm never 100% confident because the professor will sometimes throw a curveball and discuss something I'm not anticipating. But I'm usually pretty good at focusing on what I know will be discussed.

- **Callum:** I think so, I think I have a pretty good hierarchy in my head of what I think the most important is, and I can value that appropriately while getting the rest done to some degree—whether or not it's in a healthy way is beside the point.
- **John:** Depends on what I'm optimizing for. For my happiness and my sanity, yes.

16.3.3 *Decision Process*

When you're deciding what to work on, what factors usually matter the most to you?

- **Ashton:** Quantity of assignments and then which class it is, because my social psych class has a very chill teacher. Her assignments are easier—it's often just summarizing a concept, which I've already done in my note-taking phase. The more difficult the task, the more time I will allot to it.
- **Austin:** Ideally I would like to knock out smaller tasks or tasks that take less effort early; that's been important in the past.
- **Annalise:** Mostly when things are due, but familiarity is a factor. I find I'm less likely to want to start a brand new thing and will likely start with things I know.
- **Gabriel:** The number 1 is comfort. Normally the things I can do comfortably would be things I can do from home or that I can do with my headphones in. If I have to call someone or work with additional people, that's less comfortable so I prioritize it last. It's a very ineffective and not good system that Mom is ashamed of.
- **Callum:** Whether it is an assignment for a grade or not, but for doing reading assignments for the same day maybe when in the day it is, then lastly the level of engagement I need for the material for the class at hand—I'd prioritize a class where my professor does cold calls over one where the professor doesn't.
- **John:** Urgency; sometimes that's imposed deadlines or something intrinsic to the task at hand. For example, we had cost overrun on the Google Cloud service that if not resolved immediately we'd rack up charges. That would override an internal deadline. That applies to household chores as well; there's a point at which emptying a litterbox is not optional if you want to live in a dwelling that is habitable. I think that there's definitely part of it that is on enjoyment; I'll prioritize tasks that I enjoy doing. I don't always want to be doing cognitively intensive work, even if it is what I enjoy doing.

16.3.4 *Tradeoffs & Conflict*

What happens when two tasks feel equally urgent or important?

- **Ashton:** I would probably go off of what's due first.
- **Austin:** I typically would pick the one that's easier to knock out, so I can spend the lion's share of my time on what I would have more difficulty with, and that gives me time to reach out to the professor if I need more help.
- **Annalise:** I honestly work on them concurrently and trade off. Or I choose the more familiar task first as a default.
- **Gabriel:** I do the one that I want to do more.
- **Callum:** Whichever one is sooner comes first, and if they're the same, then that's when the lower factors come into play. When things are longer or shorter, more or less stressful—if all else is equal.
- **John:** I'll typically engage with one of my managers on which one to prioritize if it's at work. Outside of that, I'll look at time to complete; if it can be done quickly I'll push through that one.

How do you break that tie?

- **Ashton:**
- **Austin:** Choose the easier one first.
- **Annalise:** Go by due date.
- **Gabriel:** I would default to the one I can do with earbuds on and with a video on in the background. That is a winner. If I have to move from my current location to do it, that is a lower chance.
- **Callum:**
- **John:** Whatever I'm more engaged or productive on. If there's something I've been working on more recently so it's top of mind, I can be more productive on it—momentum.

Do you ever feel stuck deciding?

- **Ashton:** No, because of my 1-2-3 system.
- **Austin:** Absolutely.
- **Annalise:** All the time.
- **Gabriel:** No, because it's usually very urgent.
- **Callum:** Yeah, I think so.
- **John:** Not typically, I don't spend that much time thinking about it. A lot of

this is internalized and subconscious. I don't actively think through what each item's priority is. It's intuitive into how I plan my day.

Have you ever chosen something and later felt like it wasn't the right choice?

- **Ashton:** Oh, yeah—social psych is due Wednesday, developmental is due Sunday, and stats is due Monday. Sometimes I get my wires crossed and wish I had given more time to stats earlier.
- **Austin:** As far as prioritization, no. Maybe choice of major though.
- **Annalise:** Yes, I sometimes spend way too much time studying and not enough time on deliverables.
- **Gabriel:** I don't think so. I usually gaslight my teachers into thinking I'm ahead in a class when really I'm behind.
- **Callum:** Yes, I think so. Sometimes something that feels more urgent you find out later is maybe less important than something else.
- **John:** Maybe on a micro scale, like "oh I spent those two hours doing the wrong thing," but I typically just change my allocation over the next few days.

16.3.5 *Emotion & Avoidance*

Are there tasks you tend to avoid or put off? Why?

- **Ashton:** I think it's probably big projects and stuff that are due a month out. I should work on them a little bit but I never do. I start a few days before it's due even though it's suggested to start earlier. But that's because I don't have time.
- **Austin:** I wouldn't say so. I don't really have an aversion to most of my work. Any kind of putting off is purely procrastination.
- **Annalise:** All the time. I'm a big-time procrastinator and have a hard time starting things, especially when they are unfamiliar.
- **Gabriel:** Of course. Yes, overdue work I hate to do. Because it's past the deadline, so why would I do it? I hate to do it. I hate to clean my room. If you want me to clean my room, you are asking a monumental task of me.
- **Callum:** I think there are certain classes I put off more because they're drier or more annoying. I think they start to fall down the priority list, so they happen at the latest date.
- **John:** Laundry is very tedious. It's probably good for me to do something tedious and low stimulation, but my little ADHD brain likes to be stimulated. I have a growing stack of internal memos that I'm supposed to write—I don't

really want to spend the time, and neither does my manager want me to spend the time, so it's kind of intentionally put off. Trivial internal things.

What makes a task feel avoidable?

- **Ashton:** If it's not due that week.
- **Austin:** If it seems like I may be able to do it quickly, and I don't have to think about it as much.
- **Annalise:** Being due later, being unfamiliar.
- **Gabriel:** Oh, how easily I can forget about it. If I can forget about it by doing something else, that's scoring high points for me for not doing it.
- **Callum:** The level of immediate knowledge that I need to have on the topic. If I know I have a chance to be called on, that makes the task a higher priority. If there's a low chance of being called on, that makes it more avoidable.
- **John:** Low priority, no clear deliverable or stakeholder, no consequence of it not getting done (that is defined, at least). Things that I think can be delegated if it becomes important.

What do you usually do instead?

- **Ashton:** What is due the soonest.
- **Austin:** Either a different project or general weekly work like readings and class videos.
- **Annalise:** Easy tasks that are more rewarding.
- **Gabriel:** Either other tasks that I want to do more or I take my guilty conscience and I jump into social media, because it's good for dissolving your identity. Don't tell Mom.
- **Callum:** Either do a different task that maybe at the time feels more important (whether or not it is) or I end up procrastinating by doing a chore or something I would put below schoolwork.
- **John:** Something higher priority. I know something is really silly if I choose to clock out an hour early or go to play pool.

How does your mood or energy affect what you choose to work on?

- **Ashton:** No, because I know that there's no other option, I have to power through it. I will say when I messed up the deadline last week—if I really don't have it, I will not give it my all. But that's if I'm really down bad.
- **Austin:** Yeah for sure, especially if I'm tired and feel like I'm not getting the

best work out of something, I'll take some time off. But I try not to put it off for more than a day or a sleep. But the stress is constant.

- **Annalise:** Yes, all the time. I'm usually running on empty and have to switch out of high-level cognitive tasks and over to easier things because I get tired.
- **Gabriel:** Oh, baby. Overwhelmed, okay, I can do work. But if I'm tired, that's way overestimating what I can do. Asking me to work when I'm tired is like asking me to stop a train with my bare hands. That being said, I am doing very well in all of my classes and jobs.
- **Callum:** I would say that it does. It definitely makes me want to take on tasks less, start a new task less. Increases procrastination.
- **John:** Yes, it's hard to say systematically. It makes my decision-making and time allocation generally worse. I had bad writer's block last week when I was working on a proposal and felt like I wasn't making much headway. To get out of that, I talked to my boss who suggested doing a low-cognition task for a few hours and then get back to it. It's hard when you're in the weeds to prioritize. If there's one task that is stressing me out I feel like I need to focus on that task, but at the same time I know I'm not being productive because I'm weeded. So I need to do something else so I can reset.

16.3.6 *Reflection & Gaps*

Do you feel like your current way of deciding what to work on works well for you?

- **Ashton:** Not really.
- **Austin:** Absolutely not. We're always searching for the best way to decide and never really finding it.
- **Annalise:** Oh, not at all.
- **Gabriel:** No, no I don't. But I also don't think I'm going to change, so I'm at peace with that.
- **Callum:** I would say generally it does, but it's not perfect and there are certain things that definitely fall through the cracks that should probably be higher up on the priority list, but they don't seem imminently important so I prioritize them less.
- **John:** I think so; I think I've had one of my more productive quarters, and I'm on a good trajectory. I've had a better work-life balance.

What's the hardest part about deciding what to work on?

- **Ashton:** I think knowing that I have to give more to stats even though I don't like it. Me and liking my classes is a big thing—you have to do requirements even if you don't like it. It takes emotional energy to motivate myself to do it. I get frustrated because I'm savvy academically and stats doesn't come as naturally.
- **Austin:** Being sure I have the prerequisites ready for it. If I have a problem set, do I have enough information to make a sizable dent in it, or am I just doing it one question at a time from prior material?
- **Annalise:** Not having enough time to do everything perfectly. Managing burnout in longer study sessions.
- **Gabriel:** Balancing how much I want to do it with how much I need to do it. I do have work that needs to get done that I really don't want to do, and there's some work that needs to get done later that I'm okay with doing.
- **Callum:** Choice paralysis. There's a lot of things to work on and a lot of them do end up being relatively equal. So hard to know what things to do first, and hard to decide what to work on vs. things to do—all of the other things in existence.
- **John:** Misalignment between what I believe is important and what I am told is important.

16.3.7 *Summarized Interview Results*

Table 3—Summarized insights from student interviews on task prioritization behavior.

Insight	Interpretation	Design Implication
Prioritization is reactive, not planned	Participants generally do not follow stable prioritization strategies. Instead, they respond to what feels most urgent or salient in the moment, often describing their behavior as "putting out fires."	Systems should scaffold forward-looking decisions without requiring heavy upfront planning.
Deadlines dominate, but only at the short-term level	Imminent deadlines are the primary driver of action, but longer-term tasks are systematically ignored until they become urgent, creating cycles of cramming and stress.	Tools should surface long-term consequences and encourage earlier engagement with distant deadlines.
Perceived effort drives decisions more than time	Participants consistently prioritize based on how difficult or mentally taxing a task feels, rather than how long it will take. This introduces bias toward avoiding cognitively demanding work.	Systems should account for subjective effort (cognitive load), not just time estimates.
Avoidance and comfort shape task selection	When given flexibility, users default to tasks that are easier, more familiar, or more enjoyable, especially when tasks feel equally urgent.	Systems should intervene in tie-breaking moments and support initiation of avoided tasks.
Decision-making is highly context- and state-dependent	Task choices shift based on energy, mood, environment, and available time. Users dynamically adjust their priorities depending on what feels doable in the moment.	Systems should incorporate context-awareness (e.g., energy level, time constraints) into recommendations.
Tracking systems are fragmented and unreliable	Participants use a mix of tools, memory, and ad hoc strategies, but none are consistently maintained. Many revert to mental tracking, which increases cognitive load and risk of oversight.	Tools should reduce maintenance overhead and passively integrate into existing workflows.
Users experience uncertainty and cognitive overload when prioritizing	Participants frequently report feeling stuck, overwhelmed, or unsure when multiple tasks compete for attention, often defaulting to heuristics or avoidance.	There is a clear opportunity for decision-support systems that reduce cognitive burden and provide guidance.

16.4 Appendix 4: Needfinding Survey Data

21 survey responses were collected, below is a compact representative summary.

16.4.1 *Participant Demographics (Answer: Frequency)*

- High School: 1
- Undergraduate: 1
- Graduate: 19

16.4.2 *Applications Used for Scheduling and Prioritization (Answer: Frequency)*

- Google Calendar: 16
- Notes app (e.g., Apple Notes, Google Keep): 7
- Notion: 4
- Physical Planner/Journal: 4
- Other tools (e.g., Discord, Proton Drive documents, Proton Calendar + Claude): 3

16.4.3 *Frequency of Primary Tool Use (Answer: Frequency)*

- Daily (multiple times a day): 14
- Weekly (several times a week): 5
- Occasionally (a few times a month): 1
- Rarely (less than once a month): 1

16.4.4 *How Often Participants Follow Their Plans (Answer: Frequency)*

- Daily (stick to it most of the time): 5
- Weekly (try to follow it, but often make changes): 9
- Occasionally (follow it when tasks become overwhelming): 5
- Rarely (often abandon the schedule quickly): 2

16.4.5 *Primary Factors Driving Prioritization (Answer: Frequency)*

- Deadline (most urgent tasks first): 16
- Importance (tasks with the biggest impact): 3
- Time/Energy: 1
- No explicit prioritization strategy: 1
- Difficulty: 0

16.4.6 *Ease of Prioritizing Tasks Effectively*

Responses were given on a 1-5 scale, where 1 indicates very hard and 5 indicates very easy.

- Mean rating: 3.05
- Distribution: 1 (1 response), 2 (5 responses), 3 (7 responses), 4 (8 responses), 5 (0 responses)

16.4.7 *Procrastination Due to Uncertainty About What to Start Next (Answer: Frequency)*

- Yes, frequently: 11
- Sometimes: 9
- No, never: 1

16.4.8 *Most Desired Features in a Future Tool (Answer: Frequency)*

Participants could select multiple options.

- Automatic scheduling based on estimated time and availability: 15
- Visual timelines / Gantt charts: 15
- Smart reminders and alerts: 14
- Task prioritization suggestions: 12
- AI suggestions for breaking down large projects: 12
- Integration with academic platforms (e.g., Canvas, Blackboard): 10
- Progress tracking and completion analytics: 10

16.4.9 *Appeal of an App Providing Personalized Recommendations (Answer: Frequency)*

- Very appealing: 10
- Slightly appealing: 9
- Neutral: 1
- Not appealing: 1

16.4.10 *Time Estimation Patterns (Answer: Frequency)*

Small Tasks (less than 1 hour)—

- Accurate estimate: 13
- Slightly underestimate: 4

- Slightly overestimate: 3
- Underestimate significantly: 1
- Overestimate significantly: 0

Medium Tasks (1-3 hours)—

- Slightly underestimate: 9
- Accurate estimate: 5
- Slightly overestimate: 5
- Underestimate significantly: 2
- Overestimate significantly: 0

Large Projects (3+ hours or multi-stage)—

- Slightly underestimate: 6
- Underestimate significantly: 6
- Slightly overestimate: 5
- Accurate estimate: 2
- Overestimate significantly: 2

16.4.11 *Effectiveness of Current System in Reducing Stress*

Responses were given on a 1-5 scale.

- Mean rating: 3.00

16.4.12 *Preparedness at the Start of a Typical Week*

Responses were given on a 1-5 scale, where 1 indicates not prepared at all and 5 indicates fully prepared.

- Mean rating: 3.19

16.4.13 *Representative Open Ended Challenges from Participants*

Recurring themes in the open ended responses included:

- Too many overlapping deadlines
- Difficulty deciding what to prioritize first
- Distraction and loss of focus
- Overcommitment and exhaustion
- Unplanned tasks disrupting schedules
- Difficulty tracking multiple responsibilities across tools

16.4.14 *Representative Open Ended Suggestions from Participants*

Common suggestions for a future tool included:

- Better prioritization guidance
- Automatic scheduling support
- Visual deadline timelines
- Integration with academic systems
- AI-based project breakdown and next-step recommendations
- Progress tracking and clearer completion indicators

16.5 Appendix 5: Needfinding Heuristic Evaluation Data

16.5.1 *Results Table*

Table 4—Heuristic Evaluation Results for Google Calendar

Task	Heuristic Violated	Description	UI Location	Severity
1 – Multi-stage project	Match between system and real world	Calendar scrolls horizontally rather than vertically, conflicting with expected calendar navigation	Month view	2
1 – Multi-stage project	User control and freedom	No way to set checkpoint dates or interim reminders within a single task, making multi-stage planning impossible without workarounds	Event creation screen	4
1 – Multi-stage project	Flexibility and efficiency of use	Three separate entry types (birthday, task, event) add unnecessary complexity for a simple scheduling action	Add item menu	1
1 – Multi-stage project	Recognition rather than recall	Deadline field defaults to the selected day, giving the illusion of functionality without supporting actual deadline planning	Event creation screen	2
2 – Equal deadlines	Flexibility and efficiency of use	Tasks with identical deadlines are sorted alphabetically with no prioritization cues, leaving no decision support	Day view	4
3 – Rescheduling	Match between system and real world	Pencil icon implies editing content rather than rescheduling; users must discover functionality through trial and error	Event detail view	3
4 – Time estimation	Flexibility and efficiency of use	No mechanism to estimate, log, or track time spent on tasks; requires separate app (Google Tasks)	Event/task creation	4

4 – Time estimation	User control and freedom	Core functionality is offloaded to an external app, fragmenting workflow and increasing cognitive load	App-wide	3
5 – Weekly overview	Visibility of system status	Week view truncates task names, hiding meaningful information unless users open each item	Week view	3
5 – Weekly overview	Visibility of system status	No visual indication of task importance, urgency, or workload density; all tasks appear identical	Week view	4
5 – Weekly overview	Recognition rather than recall	"Current day" button is visually ambiguous and requires trial-and-error to understand	Top navigation bar	2

16.5.2 *Walkthrough*

when i go to the date i want to in the next month, i cannot scroll down like i would a normal calendar. i have to swipe sideways which is problematic (match between system and real world). click on the day, it opens up the daylike normal. plus button on the bottom right which indicates to add something. 3 buttons, birthday, task or event. i feel like this may be overkill. there is one option to have a deadline which is kind of useless because i clicked on the deadline day to set the event. there is also no way to set any reminders or checkpoint dates before that so i cannot make this multistage. for task 2, i have set a new event and set the deadline as the same day. it organizes my tasks alphabetically but there is no way to decide between which of the two to start. for task 3 i am now going to move my second event and place it later on in the week. there is no intuitive way to do this. i had to press the pencil edit button to change the date. i was initially going to delete it before i remembered that the pencil edit means i can also edit the date. i usually assume the pencil means edit the event itself and not its scheduling aspect. the app has no way of estimating or log how long a project will take and in order to view more about a task, i have to download another app called google tasks. for the overview of the week's work, i press on the hamburger icon on the top left to get out of the month view, and i press on schedule which is the top button, this gives me every task for every month in a scrollable format. This is not for a week though, i press hamburger icon again and press on week which is 4 buttons down, and that shows me the week but it scrunches up the tasks so i can only read the first word of the task. i have no idea how important anything is or anything about the task unless i click on it. there is also a button that looks like a rectangle with a corner cut off and the current day number of the month. i had no idea that it was even that until i clicked on it and it brought me to my current day in the current format i clicked (week format).

16.6 Appendix 6: Initial Prototypes

16.6.1 *Prototype C: Gamification*

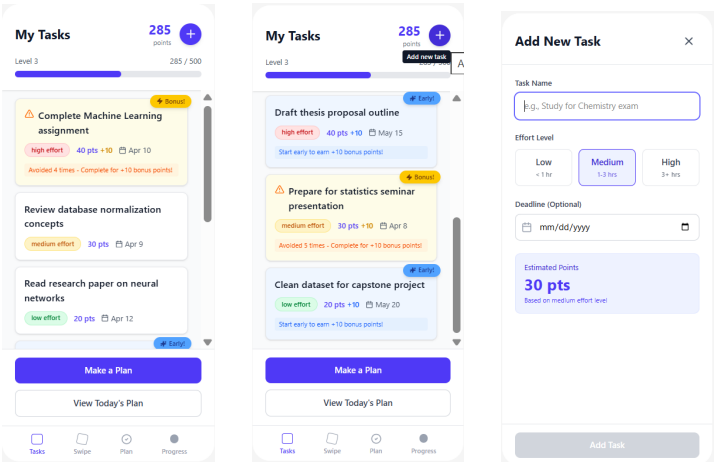


Figure 13—Task management interface showing the full list of tasks alongside the task creation flow. Tasks are assigned effort-based point values and may surface either avoidance-based or early-start bonuses. This screen establishes the underlying reward structure while keeping task entry and organization straightforward. Images created with Figma.ai.

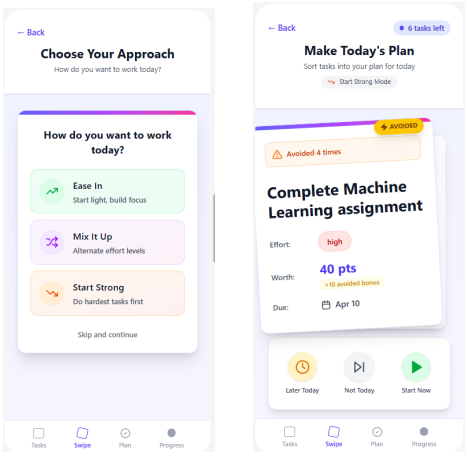


Figure 14—Planning interface combining mode selection and swipe-based prioritization. Users first choose a pacing strategy (ease in, mix it up, or start strong), then sort tasks into "start now," "later today," or "not today." This separates how work is structured from which tasks are selected, while keeping the decision process lightweight. Images created by Figma.ai.

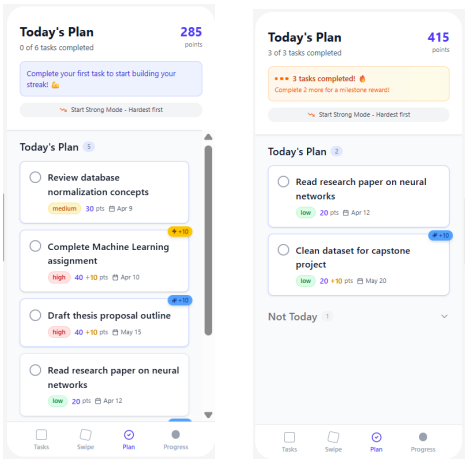


Figure 15—Today's plan view, where prioritized tasks are organized into a concrete, actionable list. Tasks can be completed directly from this screen, contributing to a session-based streak and point accumulation. This serves as the main bridge between deciding what to do and actually doing it. Images created by Figma.ai.

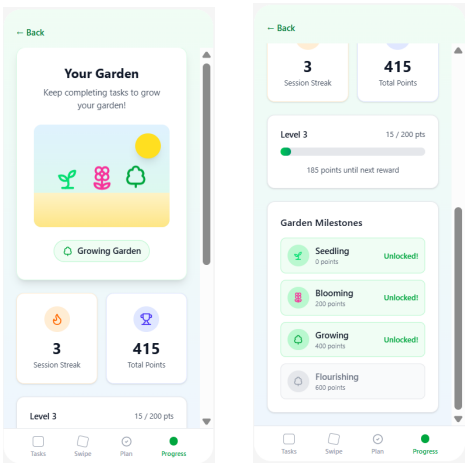


Figure 16—Progress and gamification view representing accumulated points and streaks through a growing garden. Visual milestones provide lightweight feedback on long-term progress, while remaining secondary to the core planning and execution workflow. Images created by Figma.ai.

Figma.ai prompts for Prototype C

This appendix documents the prompts used to generate and iteratively refine the gamified swipe-based prioritization prototype in Figma.ai.

Initial Prompt

Design a medium-fidelity mobile app prototype for a student task prioritization tool that uses gamification to motivate task completion.

The app should help students decide what to work on by combining task management with a points-based reward system and light game mechanics.

Include the following screens:

Home / Task List Screen

- Clean list of tasks
- Each task shows:
  - Title
  - Effort level (low, medium, high)
  - Point value
  - Deadline (if present)
- Visual indicators for:
  - Tasks that have been avoided multiple times (e.g., warning icon)
  - Tasks eligible for bonus points
- A progress bar or score indicator at the top
- Button to add a new task

Add Task Screen

- Input fields:
  - Task name
  - Effort level selector (low / medium / high)
  - Optional deadline
- Show estimated points earned based on effort
- Simple, minimal UI

Swipe / Prioritization Screen (Tinder-style)

- Show one task at a time as a card
- Swipe right = "Do now"
- Swipe left = "Later"
- Swipe up = "Not today"
- Include subtle feedback (color or animation)
- Show point reward for selecting a task
- Highlight bonus if task has been avoided multiple times

Gamified Progress Screen

- Visual theme-based progress (example: garden growing, bakery filling with items, or dungeon progress)
- Points contribute to unlocking visual elements
- Show streak count (e.g., "3 tasks completed in a row")

- Show next reward milestone

Bonus / Notification State

- Modal or notification when:
  - A task has been avoided multiple times
  - A bonus is offered for completing it
- Example: "Complete this task now for +20 bonus points"

Design style:

- Medium-fidelity wireframe (not high-polish UI)
- Clear layout and hierarchy
- Light use of color to indicate states (bonus, avoided, completed)
- Playful but not childish (designed for college students)

Interaction emphasis:

- Make it easy to quickly decide what to work on
- Reduce cognitive load in prioritization
- Reinforce motivation through visible progress and rewards

Iteration Notes

- I wish the tasks/swipe/progress bottom menu was available on the swipe menu as well.
- For the streak, I wish it were by number of items completed rather than number of days. So if in your study session you have already completed 2 things, maybe there's a small bonus for the third, etc.
- I think the point spread is too high. Maybe instead of 10/25/50 it should be 20/30/40.
- The interface shows 3 things with bonuses for avoidance. Let's just do 2 so it isn't cluttered.
- Can we change the example assignments to be more college/grad school oriented? Bonus points if some of the assignments (not all) are computer science or data related.
- On the swipe menu, can we have some sort of title for the page (something like "Make today's plan" or "Let's prioritize," or something, I'm open to suggestions).
- I like that everything is clean in terms of UI, but I wish the swipe menu were a bit more engaging.
- I love the "bonus opportunity" screen that popped up during swiping.
- Maybe as a counterpart to bonuses for completing avoided tasks, we also have a bonus for starting something early. E.g., if it's the first step of a project due in May, we should have a flag to start early so we aren't inadvertently rewarding avoidance.
- The swipe left/swipe up/swipe right icons should work as buttons in addition to the swipe capability.
- On the "My Tasks" page, can you show the early start bonus on one of the tasks?
- Selecting tasks on the swipe page shouldn't increase points. That would be done when a student checks off that they're completing a task (not shown in this prototype).
- Maybe the tiles that show a bonus (either an avoidance or a start early) should have a slight color to them to help them stand out. Not bright, but noticeable.
- I think the bonuses should be lower—maybe just 10 additional points.

Additional Screen Prompt

Add this screen:

"Today's Plan" Screen

Design a screen that represents a structured daily plan generated from task prioritization.

Include:

Header

- Title: "Today's Plan"
- Small progress indicator (e.g., tasks completed today, streak count)

Sections

- Do Now
  - Tasks the user selected as "Do Now" in swipe mode
  - Show:
    - Task name
    - Points (+ bonus if applicable)
    - Effort level
  - Include checkbox or "Complete" button
- Do Later
  - Tasks marked as "Later"
  - Visually de-emphasized
- Not Today (collapsed section)
  - Hidden by default
  - Expandable list

Interactions

- Completing a task:
  - Updates streak (based on consecutive tasks completed)
  - Triggers point reward
  - Show subtle feedback (animation or highlight)

Motivation layer

- Show:
  - Current task streak (e.g., "2-task streak ")
  - Hint for next reward ("Complete one more for +5 bonus")

Navigation

- Accessible from bottom navigation
- Serves as the primary screen users return to after swiping

Navigation and Interaction Updates

- I think on tasks the options should be to add new task, swipe (make a plan), or view today's plan.

- Can we nest those options somehow or organize them so they don't take up so much of the screen?
- Maybe add a new task is just a plus at the top where if you hover over it it says add a new task (and clicking takes you to new task), and then we have make a plan/view today's plan at the bottom.

Add the following improvements:

Swipe interaction updates

- Swipe right:
  - Label: "Start Now"
  - Icon: play button ()
- Swipe left:
  - Label: "Later Today"
  - Icon: clock ()
- Swipe up:
  - Label: "Not Today"
  - Icon: skip or forward icon ()

- Also include these as tap buttons, not just gestures.
- Add a short instruction at the top: "Sort tasks into your plan for today."

Planning mode selection (first card in swipe flow)

- Title: "How do you want to work today?"
- Include 3 selectable options:
  - Ease In — "Start light, build focus"
  - Mix It Up — "Alternate effort levels"
  - Start Strong — "Do hardest tasks first"

Logic and Ordering Updates

- So when we go to the swipe screen, I want to be able to swipe through all of my possible tasks, not just a subset of them.
- That will sort them into do now / do later / not today.
- Then to convert them to today's plan, we will use two data points: the approach + the swipe factors.
- If we selected Start Strong, the order of the tasks on the plan page would be:
  - Start now tasks: difficult to easy
  - Later today tasks: difficult to easy
  - Not today (collapsed): by due date
- In the today's plan view, we don't need to break up the start now/later today sections, as they'll just be part of the plan.
- For some reason, it's only having me swipe on 3 items instead of all 6 (or however many we have). Can we fix that so I swipe on every item?
- In the priority list, selecting "Start Now" should trump "Later Today" regardless of which mode you select.
- So it would show in this order:
  - All start now tasks: organized by selected mode

    - Later today tasks: organized by selected mode
    - Not today (collapsed): by due date
- Can we add a third nested layer or organization?
- It should be swipe order > selected mode > due date.
- So the same as before, but if two or more items were in the same swipe group and have the same difficulty, let's put earlier due date first.

16.7 Appendix 7: Evaluation Plan

16.7.1 *Interview instrument*

Warm-up questions

- Can you tell me a bit about your current semester and workload?
- How do you usually decide what to work on when you have multiple tasks?
- On a scale from 1 to 5, how confident do you usually feel in those decisions?
- Do you ever feel stuck deciding what to do next? Can you describe a recent example?

Prototype task prompt For each prototype, participants will be given the same short scenario:

> Imagine you have three academic tasks: one assignment due tomorrow, one large project due next week, and one reading you have been avoiding. Use this interface to decide what you would work on next.

Think-aloud prompts during interaction

- What are you thinking as you look at this?
- What stands out to you first?
- How are you deciding what to choose?

Questions after each prototype

Quantitative questions—

- On a scale from 1 to 5, how easy was it to decide what to work on using this prototype?
- On a scale from 1 to 5, how confident are you in the decision you made here?
- On a scale from 1 to 5, how mentally effortful did this feel?
- On a scale from 1 to 5, how likely would you be to use a tool like this?

Qualitative questions—

- What helped you make a decision here?
- What felt confusing, frustrating, or missing?
- Did this change how you would normally choose what to work on?
- How did this compare to how you usually make these decisions?

Cross-prototype comparison questions

- Which of the three prototypes felt easiest for deciding what to do next? Why?

- Which prototype felt the most overwhelming or difficult to use? Why?
- Which prototype would you trust the most to help you make a good decision?
- Did any prototype seem especially helpful for tasks you would normally avoid?

Comparative quantitative questions

- Rank the three prototypes from most helpful to least helpful for deciding what to work on next.
- Which prototype reduced your decision stress the most?

Closing reflection questions

- What is still missing across all three prototypes?
- If you could combine parts of these designs, what would you keep?
- Do you think any of these would realistically change how you manage your work? Why or why not?

16.8 Appendix 8: Evaluation Data

16.8.1 *Raw Data from Interviews*

Participant Overview—

Participant	Status	Prototype Order
Carline 1 (Emre)	PhD Graduate, Computer Science	A → B → C
Carline 2 (Esin)	Master's Student, Business Analytics	A → C → B
Sophia 1 (Gabriel)	Undergraduate, Art Student	B → A → C
Sophia 2 (Serena)	Master's, International Studies	B → C → A
Sophia 3 (Ashleigh)	Master's, Strategic PR	C → A → B
Shivani 1 (Anu)	Undergraduate CS Student	C → B → A
Shivani 2 (Priya)	Master's Cybersecurity Student	A → C → B
Hasan 1 (Samantha)	Graduate, HCI	A → B → C
Hasan 2 (Joyce)	Graduate, HCI	B → C → A

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Interviewer Script—

> "Thanks for doing this! This should take about 15–20 minutes. We're testing a few early design concepts for helping students decide what to work on.
>
> There are no right or wrong answers—we're really interested in how you think through things.
>
> As you go through the prototypes, I'll ask you to think out loud and talk through your thought process as much as possible."

—

Warm-Up/Context Responses—

Can you tell me a bit about your current semester and workload?

Carline 1: I'm working and no longer a student

Carline 2: I'm taking 3 classes while working part time.

Sophia 1: I work 2 part time jobs as well as being a full time graphic design student at the Columbus College of Art and Design.

Sophia 2: I have 15-16 credit hours, including a grant based class twice a week (including going into the city in Cleveland), a history of Africa class (once a week at night, 3 hours), a sociology class (three times a week), and Tai Chi for half a credit hour. I also have a capstone project that I meet twice a week for to cover my graduating thesis.

Sophia 3: 2 classes in the evening, 1 on audiovisual media and website design and 1 on research and development- so surveys, focus groups, putting together a full campaign.

Shivani 1: I'm currently a CS student taking classes like data structures, encryption, nlp, etc. The overall workload is project based and some theory papers. Mostly readings, problem sets, watching lectures, and labs. The projects take a long time because I need to understand the topic first based on the lectures and then apply the concepts. Other assignments are just busy work mostly.

Shivani 2: this semester is pretty intense as a masters student, i'm taking a mix of application and systems focused classes. There are a couple of hands-on assignments(code, testing, iteration) and design thinking assignments. I mostly prepare for lectures, especially for conceptual material. I'm also a dancer out of school which takes up a lot of time, so balance is the key for day to day work.

Hasan 1: Im in 2 classes, pretty manageable. The classes have a suggested weekly schedule. The TAs send this out for this week and next week. It's been pretty manageable so far so long as you follow that.

Hasan 2: Im currently taking two classes, Digital Health Equity and a pass/fail data structures and algos. For the first, it's varied, some weeks I spend at most 1 hour and other weeks I'm spending 8-10 hours to wrap it up. The other class is a pretty consistent 6-7 hours a week.

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How do you decide what to work on when you have multiple tasks?

Carline 1: Whichever is closer, deadlines

Carline 2: The first priority is by deadlines, then by whatever is easier.

Sophia 1: Famously and embarrassingly, I go by order of how much I want to do things, and the level of urgency follows that.

Sophia 2: I go by due date first, so whatever is most pressing I tend to focus on first. If there are multiple things due at the same time, I do the easiest thing first, so not type of work but what is nearest.

Sophia 3: Whichever one is due first.

Shivani 1: Priority based on deadlines, complexity, and effort.

Shivani 2: Mostly based on priority, effort and importance. I like to pick up things in the middle of my breaks, before my classes, while commuting, etc. I like to also factor in reducing my load by chunking my assignments to focus on small parts one at a time.

Hasan 1: [this one was partially answered in the first question] I'll usually work on one task then rotate and swap back to the other one and keep doing this until they are both done.

Hasan 2: Whichever one is due sooner. I operate on deadlines.

—

On a scale from 1 to 5, how confident do you usually feel in those decisions?

Carline 1: [4]

Carline 2: [3]

Sophia 1: [2] because I'm getting good work done, but I know that there is more work that needs to be done that I'm not doing.

Sophia 2: [5] because I usually wait until very little time is left, so I'm certain it's the right choice to prioritize it that way.

Sophia 3: [4]

Shivani 1: 3

Shivani 2: 4, it takes a bit of planning but I eventually figure things out as I'm used to.

Hasan 1: [4]

Hasan 2: [4] Sometimes I forget smaller deadlines like discussion posts because I don't check Canvas often enough.

—

Do you ever feel stuck deciding what to do next? Can you walk me through a recent example?

Carline 1: I don't.

Carline 2: Yes, especially when multiple assignments have similar deadlines. I can go back and forth sometimes or procrastinate.

Sophia 1: No, because I just do what I want to do. My soul guides me.

Sophia 2: No, I never really work ahead, it's truly deadline-based.

Sophia 3: Yes- design aspects needing different photos or pieces that I don't necessarily have- is it better to start all at once or start with what I do have and do it piecemeal? What part am I at, what can I complete in its entirety?

Shivani 1: Sometimes because I often procrastinate but the deadlines keep me going.

Shivani 2: yes, definitely. It happens a lot more than I like to admit. Especially if I have a lot of things due at the same time and are equally important. classes these days seem to have deadlines on the same date. It is often difficult to select which assignment i should prioritize especially if they fall on the same deadline.

Hasan 1: Sometimes when I have too many assignments for one week, I definitely feel overwhelmed, I don't have a specific example. But if I have a very busy week, this was more of an issue in undergrad, but when I get overwhelmed I don't do anything.

Hasan 2: If I have two important things due at once– I don't have an example, I guess when I have schoolwork and work work I don't know which to prioritize. I guess I just didn't sleep for 1 day because I don't have a plan. It depends on my mood, if I'm burnt out I'll do the easier thing but if I'm not, then I'll do the harder one.

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Prototype A – AI Breakdown—

Pause! Make sure you're showing these (and writing in notes) in the correct order! Refer to the table at the top.

Script (read before showing)

"Now I'm going to show you a prototype. For this one, imagine you have:

- one assignment due tomorrow
- one large project due next week
- and one reading you've been avoiding

Using this interface, decide what you would work on next. As you're doing that, please talk through your thinking out loud."

Think-Aloud Prompts (use as needed)

(Only prompt if they go quiet)

- "What are you thinking right now?"
- "What stands out to you first?"
- "How are you deciding what to choose?"

Notes (Think-aloud + behavior)

Carline 1: I create tasks and see them on the calendar. You can write the assignment description. I would just type, because I'm not sure what to upload. It's a little bit confusing reading the breakdown. I don't know if it will add them to the calendar or if I need to add them manually. Why can't I assign subtasks to different days or times? Subtasks may have different deadlines that would help me to choose. It should be flexible to assign them different times.

Carline 2: I like the idea of automatically breaking things down, because I usually don't do that myself. Most likely I would type in the description over uploading anything. The subtasks are helpful, but I can't immediately tell how they connect to my schedule. I would

still likely start with the assignment due tomorrow, but I would use the subtasks to help decide which part to do first.

Sophia 1: Well, my first glaring and major thing that I'm noticing is that I am vehemently opposed to the use of AI in constant work. I know that makes me behind the times. If I were to not have that viewpoint, I would think this is cool, but I am admittedly a little woke about AI so that may change my answers. The idea of breaking things into subtasks really cool. Today I wrote down the tasks for the rest of the project I'm working on, and I just went onto essentially google docs and made a bullet point list, and made an indent bullet point for the subtasks I need to do. Maybe that's too analog for kids these days, and I don't know how AI would know how to do it. I love breaking things down into small tasks, because that is more manageable. I would love for there to be a tool where I can add events and add my own subtasks that I can check off. On the smart task breakdown, the word 'task' is an orphan, so from a graphic design standpoint I would change the text somehow to not have an orphan. Everything else is perfect and I love it all. I like the pretty colors, I like that you can break things down into tasks, I wish I could do it manually rather than with AI, I like how easy things are to see, I like that it looks like you can move things around. It looks like there should be a time breakdown on the Y-axis, but maybe that's not necessary for normal people. I would decide what to choose by doing the next subtask- I would look at the subtasks and see what I can manage within my timeframe, breaking it down makes it easy to see what I can do right now. But yet again, I'm getting woke about AI. I think this is contributing to my cognitive decline because I need to break things down too far. I think I need help doing menial tasks and I find that concerning, but I don't think that's related to this prototype.

Sophia 2: I really don't love generative AI, but that's not what it's about. It seems unnecessary for me personally to break down a task into further subtasks, but I know that is not the case for everyone. This isn't really helping me make a choice per se, it feels more like describing the task rather than making the choice. I would revert to what is due most urgently.

Sophia 3: I think what stands out first is that having to put in the subtasks myself feels like more work that I would want to. Being able to upload the syllabus tasks or description and having the AI work on it makes it more appealing than having to self-do that. But I think if I were in a busy time workload wise, this could be a good tool to see the pieces involved in the project and seeing the completion. I like the completion bar part of the concept. I'd choose what to work on next based on what task is most immediate- either by deadline or in order to further along other steps in a task. I think this would be good in a group project concept in terms of seeing what tasks other people in the group have done and how that contributes to the work overall.

Shivani 1: I would add my tasks and information to the calendar. Use the smart task breakdown to create actionable steps. This AI feature is really interesting since it can take a complex task and break it into smaller actionable steps. I would be able to easily identify which subtable to pick up this way.

Shivani 2: This is a clean concept, I will simply go to the left and click on the task button. Once this brings up the page, I will enter information about my task either by uploading

an assignment details document or use the box and AI feature to process the information. I will then use this feature to further break down the tasks. I will pick up whichever is shown as the first subtask. But, i would probably need some additional information on why these tasks were split a certain way.

Hasan 1: If there is an assignment due tomorrow and depending how big it is I would do a chunk of it then swap to the project or the reading. But the assignment is the first priority. The AI smart task breakdown is really helpful for a really big assignment or the project. This would actually help with the class I have right now with the sections of the assignment. I would actually use this to split up my tasks and do the swapping method I do currently. If the assignment was just one math problem for example, I'm not going to break that up. But if it was our assignments that have specific sections, then I would use this [only use the breakdown when it makes sense]. I like the subtasks as well as how much time they would take. If I want to prioritize the easy ones, I can but if I want to tackle the hard subtask, I can go for it right now. Super easy and clean to read. I like [screen 4] I like todo lists so I like how this is designed. The greyed out buttons also look clickable so I can see what's been done and what hasn't. This is nice.

Hasan 2: Well the assignment due tomorrow is critical so I would not use the task breakdown, I need to finish it now. After Monday, I might break down the second and third task. I might break the longer assignment into 3 parts and then break the reading into 3 parts too. I'll click on the breakdown task just to see what it does but normally, I just do all this in my head. So what I would do at this point after seeing the Smart breakdown, I would go to Canvas and Ctrl C and Ctrl V the entire assignment into the textbox and generate breakdown. [Sees next screen] Interesting. My first thought for this screen is I am a slow worker so for the estimated time, I am definitely going to surpass it for all these options. But yeah, I guess, 4 tasks is good. I don't think I'll click on anything else here besides accept all. Okay I see this screen and I acknowledge it, I won't interact with it any further. I just start working. Maybe if I finish a subtask I'll click on them one by one but I'm not very good at checking off to-do lists I make, not going to lie.

—

Quantitative Questions

On a scale from 1 to 5, how easy was it to decide what to work on using this?

Carline 1: [3]

Carline 2: [3]

Sophia 1: [5]

Sophia 2: [3] because I think it would just have me do whatever is due next; it doesn't feel like it's helping me make a choice, just that it's encouraging me to work on whatever is due closest. Breaking it down into smaller bites doesn't really help me make a choice.

Sophia 3: [3] Fairly easy, looking at it directly as it's laid out, it's not as easy to look at priorities, but on its own if there was a way to show what task needed to be prioritized

(and where that would fall overall) that would be helpful.

Shivani 1: 3

Shivani 2: 3

Hasan 1: [5] reluctant to give 5 because nothing is perfect.

Hasan 2: [4] I think it was easy to visualize everything and I think it's good to see everything in one place but once it's broken into subtasks I lose the understanding of how long each subtask is going to take me. For example, I know some of these subtasks are harder than other ones but they all look the same on this interface. Like one subtask was 3 hours but another was 30 minutes. I think there's something visually satisfying to know, let's say you finished the 3 hour task, that the bar goes up way higher than if you finished the 30 minute task. I have a tendency when I use to-do lists that I do the smaller tasks first because it is equally checking things off and making me feel just as accomplished as if I did something actually important or larger. The interface here should be more accurate to the level of the task I just did [so it doesn't mimic my already existing to-do lists].

—

On a scale from 1 to 5, how confident are you in the decision you made?

Carline 1: [4]

Carline 2: [4]

Sophia 1: [5]

Sophia 2: [5] because I don't think terribly hard about it.

Sophia 3: [3]

Shivani 1: 4

Shivani 2: 3

Hasan 1: [4] because it is an AI generated. AI can also give too much information or wrong info.

Hasan 2: [3] I feel pretty confident but I feel hesitant about using AI to do this because I don't want to accidentally miss anything that may be crucial on the assignment rubric. If I could review the subtasks again against the assignment description, that would be ideal.

—

On a scale from 1 to 5, how mentally effortful did this feel? One being low effort, 5 being high effort

Carline 1: [2]

Carline 2: [3]

Sophia 1: [1]

Sophia 2: [4] It feels more effortful than the others, I personally think breaking it down into tasks helps me decide. This model feels like it would be good as an additional step or feature rather than something on its own. It feels like extra effort to me that I wouldn't need to put in, personally.

Sophia 3: [4]

Shivani 1: 3

Shivani 2: 2

Hasan 1: [2]

Hasan 2: [2] The AI tasks do help, but I don't think I'll use it for smaller assignments because it would be extra effort from my side to do that. For the larger design assignments that I do, it would help way more.

—

On a scale from 1 to 5, how likely would you be to use something like this?

Carline 1: [4]

Carline 2: [3]

Sophia 1: [1] If it were not AI, it would be a 3 or 4. But with just AI it would be a 1.

Sophia 2: [1] I don't think I'm very likely to use this.

Sophia 3: [3]

Shivani 1: 2

Shivani 2: 3, it's nice to see the subtasks broken down from an assignment. makes it easier to understand what to start with.

Hasan 1: [2] because I am a hand written task person. Online schedulers do not work for me. But for someone else I would say [4] or [5]. This is a me problem.

Hasan 2: [3] just because I prefer to keep tasks in my head, but if this was integrated into Canvas where I click on a calendar and this pops up, this would be very helpful. I don't think I would go out of my way to look for or use this.

—

Reflection Questions

What, if anything, helped you make a decision here?

Carline 1: The calendar

Carline 2: The subtasks made approaching the larger project more manageable.

Sophia 1: The layout. I think that this is a hit, I just happen to be adverse to AI.

Sophia 2: Probably just the calendar view, which would have me work on whatever is closest to me (time-wise)

Sophia 3: I think probably the progress bar in terms of how far I've made it into completion and how many more steps there are, and how much more there needs to be done.

Shivani 1: I like the AI breakdown of the assignment to actionable steps.

Shivani 2: The subtasks are easy to see what to pick up next.

Hasan 1: The faded out buttons told me those things were done and no longer relevant. The level of effort and time estimates would also help me prioritize. The buttons look like buttons.

Hasan 2: Whenever I see a CTA [button] I just interact with it so that's why I interacted with that.

—

What felt confusing, frustrating, or missing?

Carline 1: Breaking tasks in the same time spot

Carline 2: It was unclear how the subtasks fit into the calendar or how they're prioritized compared to other tasks.

Sophia 1: AI is terrible for the environment and threatens creators and designers across the world. Is that woke? Is that crazy? I don't think so, I think that's normal. Sorry!

Sophia 2: It doesn't really help me decide what to work on, I'm not sure where that comes in. Confusing because I don't think it's personally helpful for me to break down a task; this also doesn't necessarily help me with ranking. The descriptors that you have per subtask maybe help you decide but it seems a little clunky- you have to load in all of the information but it doesn't do the work to rank anything for you. It feels like a Google Calendar with extra steps.

Sophia 3: I think the urgency or prioritization is missing, or that functional optional aspect. It showing that you've completed however many pieces is great, especially on a weekly or monthly schedule, but an aspect of 'hey you need to revisit this' could be helpful.

Shivani 1: Layout is clean, but the calendar view was a bit confusing.

Shivani 2: How would happen if I don't pick up a certain subtask and miss in my calendar? Will this be moved else where? Not clear, what would happen.

Hasan 1: The placeholders were not clear at first but once you told me then I understood.

Hasan 2: I didn't realize this before but the graphs are all the same width, I'm not sure what the width entails, I guess maybe I'll work on this from Monday to Tuesday? [context: the assignment bubbles overlap with certain days so it seems like an assignment is meant for multiple days rather than just one]. I also didn't realize that the Smart tool was AI at first until I read it and implied to myself its AI. Actually, I still don't know if it's AI, I'm just assuming. [I told her she is correct about it]. Well I don't mind that it's AI but as a user,

I appreciate transparency, I want to know that it is AI that I am using. That way I can be more careful to double check things against the actual assignment itself.

—

Did this change how you would normally decide what to work on?

Carline 1: No

Carline 2: A little. It may help me start larger assignments sooner, but deadlines would still be the main way.

Sophia 1: Yeah, I would probably do more tasks at the same time knowing that I can do one subtask from 1 event and then 1 from another, rather than needing to knock out a whole project or larger task at a time. That's nice because I'm taking 4 classes, and it would be nice to be able to break out the work in the day.

Sophia 2: I don't think so, no. Because if I looked at this, I would just choose what is nearest to me.

Sophia 3: I don't think so.

Shivani 1: Maybe, needs a couple of changes in features.

Shivani 2: Yes, I like the idea of subtasks.

Hasan 1: No, I don't use online tools and I just go on vibes and what I feel like working on. But if someone does use online tools, then it would help them for sure.

Hasan 2: No, this interface doesn't change how I work, it just breaks things down.

—

How did this compare to how you usually make these decisions?

Carline 1: Similar, I also used a calendar. The AI helps with breakdown.

Carline 2: Similar, with more structure

Sophia 1: This focuses on small, actionable tasks, which is what my therapist says I need to work on when I feel powerless or hopeless, which is often how I feel when I make decisions currently (which is vibes based), and that makes me feel paralyzed. So this would be helpful for me.

Sophia 2: Honestly, it is the same but with extra steps. I had a planner in college- pen and paper- and I similarly decided based on what is due most imminently. This feels the same but on a computer.

Sophia 3: I think in the reflection of how much progress I've made vs how much I need to make, in relation to the due date, it's nice to see where things fall. If I've only completed 2 parts of a 5 part thing and another task due around the same time I've completed 4 parts, that helps me decide.

Shivani 1: Seems like a nice layout, definitely similar to what I use regularly.

Shivani 2: This layout is similar to what I use on a daily basis, it includes a calendar which I generally use to mark deadlines and tasks that are approaching.

Hasan 1: It is the same in the sense it's not useful for me but it could be useful for someone. It's not any better or worse.

Hasan 2: N/A

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Prototype B – Heatmap—

Pause! Make sure you're showing these (and writing in notes) in the correct order! Refer to the table at the top.

Script

"Now we'll look at a different version. Same scenario:

- one assignment due tomorrow
- one large project due next week
- and one reading you've been avoiding

Again, decide what you would work on next and talk through your thinking."

Think-Aloud Prompts (use as needed)

(Only prompt if they go quiet)

- "What are you thinking right now?"
- "What stands out to you first?"
- "How are you deciding what to choose?"

Notes (Think-Aloud + behavior)

Carline 1: Deadline comes first, then the large project, then the reading. It looks nice, but the heatmap looks kind of confusing. I would choose based on what's coming up. The "what's coming up" section stands out, because it shows things in order it's easy to track. "How are you feeling" is interesting too. It reorders things based on your mood, so it's helpful. The heatmaps are not clear. The reminders are helpful and that you can press don't remind me again which is helpful.

Carline 2: The heatmap stands out right away. It makes it easier to see which days are busier, I would still start with the assignment due tomorrow, but some of the work can be moved earlier if there are less busy days. The "how are you feeling today" section is interesting.

Sophia 1: I would be grateful that I can schedule things multiple days in advance, my current app doesn't do this because there's no monthly overmap. The heatmap looks cool, but makes me worried that I wouldn't keep up with it because I can't spend 5 minutes creating a new event. Adding a heatmap asking me how intense something is is another step that would add a few seconds, which makes me want to use it less. The blue-to-purple

gradient stands out first, the month-long calendar stands out to me, the orange bar (heat map, level of effort for the day) stands out as well. I would be just low energy all of the time. I like that the add event button is easy to see, the procrastination thing is really clever- I've never seen or heard of that before- I think that's really cool. I like that all of the action is so easy. I would love if the 'how are you feeling today' was a widget on the home screen that opens the app. That's what I would do if it were my project. I would also change the color palette for low-moderate-high energy. I like the pastel colors (behind team meeting, HCI project, etc) would be nice- maybe a 50% opacity over all of these of the strong overhead color could bring unity and synergy. But overall everything is very professional from a design standpoint. [[interviewer note- apparently he's going to give us a color palette using an overlay. I cannot control my hectic younger brother.]] To decide what to choose, I would probably still go by how much I want to do the thing, but I'd also weigh in the importance. It would be good to see any overdue work and have those noted in- and having those noted as an 'emergency'. If you're able to not do something and forget about it, it turns into invisible work.

Sophia 2: The task that I've been skipping stands out to me- reminding me that i should be doing that even if it's not due immediately. That stands out for me as someone that procrastinates a lot. Especially for larger things like the project, I could spread out tasks over days. I like that I can skip things if I don't have the energy or don't feel it that day. I think this would help me decide which to prioritize. I'd decide how to choose based on the "day" view, especially using the high/low/moderate energy. That would help me reorganize it if I feel stuck- especially in my job rather than in school where tasks may not have a specific deadline.

Sophia 3: I think what stood out to me first is what type of day I'm having, and being able to select that and having that reorganize tasks (as much as I can within reason). Especially prioritizing something that's due even if I'm low energy because it has to get done. I like that aspect of having it re-auto generate. For the week view, I like being able to see what that week looks like, just in a stress and anxiety balancing format of what else I need to reorganize and mapping that around- being able to move things around if it's going to be a stressful day. The month concept, I like that the dots only show for the week and what's coming up- I think I'd still like to see the other dots, especially if there are any linking concepts. Some of my projects have a lot of pieces related to that, they're multi-week things that relate to each other. Having a side-task bar could be helpful so I can make progress on an overall project when a project may have smaller tasks with loose deadlines that need to get done.

Shivani 1: This interface has a nice layout, but seems like there is a lot of content as I'm looking at it my first time. Its very compact, but offers a lot of great suggestions, details and features. The buttons are placed properly and work as expected. It is easy to navigate.

Shivani 2: Overall its a good design i can tell what to do next from what I see. There are a bunch of options here. I like the interface layout and the colors used its very nice. This section [plant] is something that i've used before while I study. Its an app that i use to stop phone usage while studying. It helps with procrastination. I like this integration actually.

Hasan 1:

Hasan 2: [Reads procrastination script] Sure let's schedule it [I told her the screen is not available but it would normally take her to a scheduling screen and that the prompt goes away with more tasks showing up behind it after completing it]. I don't know if I'm right or wrong, but I'm assuming that the Week at a Glance and the Scheduling button [on the procrastination prompt] is pulling events from your backlog from other weeks.

—

Quantitative Questions

On a scale from 1 to 5, how easy was it to decide what to work on using this?

Carline 1: [5]

Carline 2: [4]

Sophia 1: [5] It's so easy, it lays everything out in front of you and you can just decide what to do. The heatmap and calendar view are nice and helpful, but the real tea is that you can see everything in front of you and decide what to do. I don't know the difference between 'add event' and 'schedule'. So maybe you can take off schedule unless its a fully different thing.

Sophia 2: [5] It accounts for my procrastination and decision paralysis (with ADHD!) pretty well. It would help me prioritize if I can't envision my tasks without a tool.

Sophia 3: [5]

Shivani 1: [4]

Shivani 2: [4]

Hasan 1:

Hasan 2: [3] I like how it puts things forward to me, but I was a little bit confused at where this was pulling tasks from, especially for me, I'm always busy so I don't know if I would pull tasks from another day.

—

On a scale from 1 to 5, how confident are you in the decision you made?

Carline 1: [4]

Carline 2: [4]

Sophia 1: [5] I wonder if this would make you second guess- but I suppose you can always come back to it if you need to.

Sophia 2: [5] Maybe I'm not a very serious student or worker, but it doesn't seem that risky to choose one task over the other if they're both getting done.

Sophia 3: [4]

Shivani 1: [3]

Shivani 2: [4]

Hasan 1:

Hasan 2: [4]

—

On a scale from 1 to 5, how mentally effortful did this feel? One meaning low effort, 5 being high effort

Carline 1: [2]

Carline 2: [2]

Sophia 1: [3] It looks like the designer took a lot of mental effort to use it. For me as a user, it would take too much effort to create an event, between due date, due time, title, effort, description, urgency- it's too much effort. If it were my app, which it's not, it would have to be the final option when you've filled out the event description- when you hit finalize/add event to calendar, you'd swipe up and the higher the swipe the higher the urgency.

Sophia 2: [1.5] It feels like it's making the decisions for me, I just have to present the information. It's very visually pleasing and easy to look at and comprehend.

Sophia 3: [3] I guess I have a follow up- how do I get all of my information in here? Do I physically type it all in? Is there a scanning concept? Do I have to assign priority? Does within the app does it have a way of doing that- like if I put things in over multiple weeks, does it build in priority?

Shivani 1: [3]

Shivani 2: [3]

Hasan 1:

Hasan 2: [3, 2 with more clarification of the interactions (clickability)]

—

On a scale from 1 to 5, how likely would you be to use something like this?

Carline 1: [4]

Carline 2: [4]

Sophia 1: [2.5] I don't think it's better than my current tool. But I think it's beautiful, and I like it, but it's also not pink so it can't really compete.

Sophia 2: [4] If I was able to put in the front-loading work of entering all of the tasks that I need to do, it would be helpful. As a student I honestly might not put in that work.

Sophia 3: [4] I think if I could easily get the information in, it could be a good tool for prioritizing assignments, especially in longer-term or semester-long projects.

Shivani 1: [3]

Shivani 2: [3]

Hasan 1:

Hasan 2: [3]

—

Reflection Questions

What, if anything, helped you make a decision here?

Carline 1: "What's coming up" section

Carline 2: The weekly view and seeing which days are busier at a glance

Sophia 1: I like the month calendar, I like the procrastination question, and I like the how are you feeling today question. Those make it easy to make a decision. It's easy and inventive.

Sophia 2: Both the week screen and the day screen- I like the skipping/blocking time feature as well as the energy prompt.

Sophia 3: I think being able to see the mental capacity that a task would take- but also seeing its priority of when it lands or when its due, and being able to decide on how I'm feeling, is very helpful. Especially with the mental capacity and a deadline concept.

Shivani 1: I like the options in the interface but at times there were too many and couldn't decide what to select next.

Shivani 2: Its a good interface overall but would've preferred something thats easy to navigate and simple with less features.

Hasan 1:

Hasan 2: I liked everything in order, I like a listview and its very conversational language. This is asking me questions that I can easily answer. I think that kind of helps me pick the CTA options, the language for sure.

—

What felt confusing, frustrating, or missing?

Carline 1: Heatmaps

Carline 2: Heatmap color definitions were not immediately obvious

Sophia 1: I don't think anything is too frustrating or missing. What I'm struggling is when you boil a day down to one color of urgency; I can have five not very urgent things on a day that makes it a high-pressure day; ranking by the most urgent thing is maybe not the most accurate. Also, things change in urgency or cognitive load closer to the due date. I personally would find it helpful to organize by how much I want to do it, because then I

can see that the thing that I want to do that's due in 3 weeks needs to be done after the thing that I don't want to do that's due today.

Sophia 2: I don't think it's confusing at all, it's pretty straightforward. I like that you can skip the mood survey in case I already know what to work on. The colors didn't immediately make sense to me at first glance, but with explanation they make sense. Maybe if it was discussed in an intro it would click faster.

Sophia 3: The inputting of the information (automatically), being able to see a longer, big-picture overview. And maybe a reminder or an alert concept, even just something to check in, like a "hey, this is your day".

Shivani 1: It was hard to focus on my goal because of the endless features available in this app.

Shivani 2: Need to include more user control options and changes in layout to be more simple.

Hasan 1:

Hasan 2: Unclear about where something is populated from. Is it backlogged? More emphasis on deadlines. As someone who is oriented on deadlines mostly, I have to ask why is it recommending me to do something? This needs to get explained better as I usually just look for the next deadline and work toward that. Like are these tasks really big and I should be prioritizing it earlier? [I explained to her how the color coding of tasks worked]. Okay so labeling needs to be more clear [on the Cognitive heatmapping of tasks], less relying on color coding and more usage of language such as easy or hard along with the colors to make it more clear. I think the function actually does help where it reorganizes tasks based on energy level, but it's more how I am oriented. This would probably work in my workflow with more clear labeling. LIke for example, if you told me you've been skipping your long, hard, daunting task of something, please schedule time for it [she was not speaking about the procrastination prompt (though it is applicable there too), this was something she was imagining looking at the color coded tasks and what they mean to her if they had proper labeling]. That would help me so this needs better labeling.

—

Did this change how you would normally decide what to work on?

Carline 1: No

Carline 2: A little. It would make me more likely to plan ahead more and not only focus on deadlines.

Sophia 1: No, I would still do the thing that I want to do.

Sophia 2: Yeah, it would, because I would normally just work on what was due next. If I used it more than once it would help me prioritize in a different way, not wait until the last minute maybe.

Sophia 3: I think so.

Shivani 1: No

Shivani 2: Yes it provides an incentive to complete my tasks. It has a lot of features i would be interested in using.

Hasan 1:

Hasan 2: It would

—

How did this compare to how you usually make these decisions?

Carline 1: Similar, I use calendar too and this is a calendar

Carline 2: It is more structured than what I currently do and considers the workload among several days.

Sophia 1: It wouldn't- I would just do the things I want to do. Although seeing things that are past due would be helpful. It seems like something that would have AI in it, that's the vibe. No offense to your day job, but AI is not my favorite guy.

Sophia 2: It feels better. Not procrastination based; it feels very logical and organized in a way that I have never been naturally. But, for the better!

Sophia 3: Currently, with my planner and my calendar and inputting into three places- this being consolidated and maybe being able to link to my calendar, and being able to see all of the things i need to see at once, makes it easier. I'm someone who writes everything down, but I like having it in a digital format as well.

Shivani 1: Its a bit complicated that what I usually use (a simple to do list).

Shivani 2: I decide to work on assignments based on priorities and deadlines so its pretty similar to what i do regularly.

Hasan 1:

Hasan 2: I like how it takes into account my energy level which I normally do in my head but this puts it on paper.

—

Prototype C – Gamification—

Pause! Make sure you're showing these (and writing in notes) in the correct order! Refer to the table at the top.

Script

"Now we'll look at a different version. Same scenario:

- one assignment due tomorrow
- one large project due next week
- and one reading you've been avoiding

Again, decide what you would work on next and talk through your thinking."

Think-Aloud Prompts (use as needed)

(Only prompt if they go quiet)

- "What are you thinking right now?"
- "What stands out to you first?"
- "How are you deciding what to choose?"

Notes (Think-Aloud + behavior)

Carline 1: I would do the project tomorrow first again, then the project next week, then the reading. It looks confusing now, because it's not easy to see the dates and they are mixed. I also don't see a sorting feature. Right, left swipe, I don't know what it means exactly. I don't know if you can edit the tasks, but clicking them is straightforward. The garden growing is a good idea. It can help you do the streaks and get points, which may motivate you to do stuff. The gamification and swipes stand out the most. I would still choose based on the dates.

Carline 2: This feels easy to use. It doesn't show all of the tasks at the same time. The swipe feature is cool and feels quicker to decide. I would still start with the assignment due tomorrow, but this is fun to use. It seems like there is less thinking involved when organizing.

Sophia 1: I have SO many thoughts, I think this is so cool. To gamify your tasks is neat. I like that it's brief, I can't spend too much time adding tasks. I like the sorting system of what needs to be done first/second/last. I like that there are multiple things to do. I like that there are points you can earn and you can buy little things. I'm imagining a mascot where you can buy it cute little outfits. I think this would be helpful. I think that the first screen [task list] is super functional and could maybe do with some beautification, maybe finding a different way to color them to make them a little more beautiful. That sounds so shady but it's not. I guess I'm wondering how I can make it pink. But this is probably the ugliest page because the others are so beautiful. I would delight in seeing some beautiful vector imagery or some more tantalizing colors and maybe some other stuff, I don't know. But it's very utilitarian. The garden page is so much more beautiful. I'm deciding what task to choose through the tinder thing, through the swipe. And so easily too.

Sophia 2: What stands out to me first is that I would be more likely to use this model over prior things that I've used. I love to-do lists and checking them off, I love the reward element. I also think that looking ahead, just the way that I'm only presented a few things at a time is easier for me to manage (as opposed to typical scheduling apps where I see a month or week at a time). It feels friendly to my brain. I'm deciding what to work on by choosing the 'ease your way in' option, and I would work in the order that the tasks are presented. If I was feeling ambitious I would choose 'mix it up' though.

Sophia 3: I like the aspect of the estimated duration for certain tasks, and that it highlights what is due sooner and/or that I've skipped already that needs to be prioritized because I've neglected it. I'd probably still choose what's due soonest- if I have a day where I have

an hour here and two hours there, having an estimated time of how long it would take to complete tasks might change the way that I prioritize it, but I would probably default to 'start strong' and prioritize things that I've avoided in the past.

Shivani 1: [Prototype A] its easy to determine what to work on next since it analyzes details like energy level.

Shivani 2: [Prototype B] overall the interface is simple but efficient. It also offers an AI assistant which would be useful to quickly pick up tasks instead of understanding what is required to complete assigned work.

Hasan 1:

Hasan 2: [interviewer let the user use it herself]. There is a lot on this page, I'm being very alerted right now. [Tried clicking on assignments, wasn't clickable] [clicked on make a plan] Oh my god this is like a workout [app]. Let me see, start light build focus. Suuuure. [dragged all tasks up] oh interesting. Okay so I guess I just clicked on the first clickable big thing which was this big blue button. The directions after were clear enough.

—

Quantitative Questions

On a scale from 1 to 5, how easy was it to decide what to work on using this?

Carline 1: [4]

Carline 2: [5]

Sophia 1: [5] It's so easy.

Sophia 2: [5] It does take the decision paralysis out of it, it gives you a question and based on your question and response it decides for you, which I kind of like.

Sophia 3: [4]

Shivani 1: [4]

Shivani 2: [4]

Hasan 1:

Hasan 2: [4] I kinda liked that. It's very prescriptive to what I need to do and it helps but almost too much. Like what if I don't want to do this assignment? [seems that the user did not understand that she could deny tasks too in the tinder format]

—

On a scale from 1 to 5, how confident are you in the decision you made?

Carline 1: [4]

Carline 2: [4]

Sophia 1: [5]. Can you reorganize it if you made a mistake? Like in the plan stage?

Sophia 2: [5]

Sophia 3: [5]

Shivani 1: [3]

Shivani 2: [4]

Hasan 1:

Hasan 2: [3]

—

On a scale from 1 to 5, how mentally effortful did this feel? One being low effort, 5 being high effort.

Carline 1: [2]

Carline 2: [2]

Sophia 1: [1], this feels relaxing and enjoyable.

Sophia 2: [1], it feels very easy. It also feels very rewarding, I love checking off tasks.

Sophia 3: [3]

Shivani 1: [3]

Shivani 2: [3]

Hasan 1:

Hasan 2: [3] I think it clicked to me that this sort of functioned like a workout app so it was intuitive. I think there are too many badges going on, too much important stuff in my face.

—

On a scale from 1 to 5, how likely would you be to use something like this?

Carline 1: [4]

Carline 2: [4]

Sophia 1: [3.5]. I'm really not the type of guy to put all of my assignments into something, but if I had a specific day where I need to put everything in because I have a lot going on, this would be a good way to do it. I prefer this over my current app.

Sophia 2: [5] I would absolutely use this.

Sophia 3: [3.5] with a gamification, a 3-4. I like the plants.

Shivani 1: [3]

Shivani 2: [3]

Hasan 1:

Hasan 2: [2]

—

Reflection Questions

What, if anything, helped you make a decision here?

Carline 1: Dates

Carline 2: The simplified view and the swipe interaction

Sophia 1: everything about it. And how beautiful it is. Scratch beautiful from the record.

Sophia 2: I really like checking off tasks, that's really it. I like that. To have it presented as a check-able list helps me decide, and the question on how I want to start (easiest to hardest) is nice. That would make the decision real easy.

Sophia 3: I'd probably say time, and urgency. That's always going to be a thing. The research paper due in a month is not going to take priority. Even if it should! Even if there's bonus points! Probably not going to take priority.

Shivani 1: I like the options in the interface but at times there were too many and couldn't decide what to select next.

Shivani 2: Its a good interface overall but would've preferred something thats easy to navigate and simple with less features.

Hasan 1:

Hasan 2: I liked the step by step aspect of this. I didn't need to think about what was next.

—

What felt confusing, frustrating, or missing?

Carline 1: Swipes and after creating the tasks, I don't know if there is an edit

Carline 2: I didn't know how to go back and edit after swiping

Sophia 1: Nothing felt confusing, nothing felt bad. I felt that the other two had some great radial gradients in a way that feels like a flat gradient, and this could use that too, but that's maybe less important.

Sophia 2: Nothing really felt confusing or missing, it all feels very well presented in this model. I didn't have any questions about this one.

Sophia 3: I think a reminder aspect- I'm not the best at using a reminder app on my phone (I usually use a calendar)- maybe having an aspect where it would would link with your calendar.

Shivani 1: It was hard to focus on my goal because of the endless features available in this app.

Shivani 2: Need to include more user control options and changes in layout to be more simple.

Hasan 1:

Hasan 2: I think I need some ability to customize my list of stuff. As much as it's helpful it has these 3 different modes, I want to be able to do it myself, though it's a cool idea.

—

Did this change how you would normally decide what to work on?

Carline 1: No

Carline 2: Yes, it appeared faster and I didn't overthink much

Sophia 1: Yeah, this is totally different from how I would decide to work on things.

Sophia 2: Yeah, again it's the same as other prototypes I've seen. It would probably make me do things before they're due, which is a good thing.

Sophia 3: Yes, I think so. I think being able to see it in that mapped out format with timing, priority, and when things are due- being able to see that with when things are due. Having it in a digital format where you can take it with you and see it in multiple ways is definitely helpful.

Shivani 1: No

Shivani 2: Yes it provides an incentive to complete my tasks. It has a lot of features i would be interested in using.

Hasan 1:

Hasan 2: Yes in that it prescribes it to me.

—

How did this compare to how you usually make these decisions?

Carline 1: It's similar

Carline 2: This is very different. Typically, I look at everything at once, this is simpler.

Sophia 1: It does what I like, which is that it doesn't judge me or make me feel bad for not doing something. It simply presents a clear list of objectives and tells me the order to do it.

Sophia 2: It's definitely better organized than my current system, and gets me to stop procrastinating. So it's not just what is next, but what I can do with my capacity today.

Sophia 3: I think it's very nice to be able to see things all at once; I typically like having the days in a week separated out, color coding by the day that they're due or the week that they're due. This makes it easier to access when I'm on the go- especially working and being a student.

Shivani 1: Its a bit complicated that what I usually use (a simple to do list).

Shivani 2: I decide to work on assignments based on priorities and deadlines so its pretty similar to what i do regularly.

Hasan 1:

Hasan 2: [already answered]

—

Cross-Prototype Comparison—

Looking across all three, which one felt easiest for deciding what to do next? Why?

Carline 1: [Prototype B] because it's easy to see what's coming up

Carline 2: [Prototype C] because I overthought less

Sophia 1: [Prototype C] because I get to swipe and figure out which order I want to do things in. It's closest to how I work. I like the heatmap, but it's just not how I work.

Sophia 2: [Prototype C] felt easier to me, but I think it's because it's similar to how I already think. No, that's not right. I felt like [Prototype C] was only giving me the most relevant things to do in the near future, and then with a choice to work ahead. I liked that because it appealed to both my procrastinator and wanting to do better. And I just love making lists and checking stuff off! So it felt good and exciting.

Sophia 3: I think [Prototype C] because the breakdown pieces of that were easier. The gamification part that breaks it down, and being able to see priority, what needs to get done, have I skipped it more than once, and that reminder aspect. I like that.

Shivani 1: [Prototype A] its easy to determine what to work on next since it analyzes details like energy level.

Shivani 2: [Prototype B] overall the interface is simple but efficient. It also offers an AI assistant which would be useful to quickly pick up tasks instead of understanding what is required to complete assigned work.

Hasan 1:

Hasan 2: [Prototype A] operates more along my mental model of how I operate. Its a very linear calendar.

—

Which one felt the most overwhelming or difficult to use? Why?

Carline 1: [Prototype C] because it has a lot of features

Carline 2: [Prototype A] because there was too much information at once

Sophia 1: None of them felt overwhelming or difficult, they were all fab and slay. But if I had to choose, the heatmap [Prototype B]. I think there were too many steps [when creating a new task], I need fewer steps.

Sophia 2: None of them felt terribly overwhelming. [Prototype A] seems the most difficult to use because I'm still doing all of the work prioritizing-wise, so not overwhelming but clunky, I guess.

Sophia 3: I don't know that any of them feel difficult to use, depending on what type of person you are. I would say [Prototype B] feels like it would take the most front-end effort but would have the best back-end result, comparatively. As opposed to [Prototype A] which does that front-end task for you.

Shivani 1: Prototype [B] less user control and options.

Shivani 2: Prototype [A] I like the idea of a heat map, but I would prefer something more simple instead of spending time to create a task.

Hasan 1:

Hasan 2: [Prototype C]

—

Which one would you trust the most to help you make a good decision?

Carline 1: [Prototype B] because it's easier to see

Carline 2: [Prototype B] because it felt more realistic and structured

Sophia 1: [Prototype C] I think.

Sophia 2: [Prototype B] and [Prototype C] are on similar footing, I think, but if I had to choose just one I would say [Prototype C]. Why? It seemed easier to choose or delay a task than in [Prototype B] even though it was an option in both.

Sophia 3: Probably [Prototype B].

Shivani 1: [Prototype B] AI assistance provides clarity on assignment understanding.

Shivani 2: [Prototype B] it is simple and realistic to navigate.

Hasan 1:

Hasan 2: [Prototype A] because it's more oriented to how I work.

—

Did any of these feel especially helpful for tasks you would normally avoid?

Carline 1: No

Carline 2: [Prototype B] because seeing the workload helped me better think ahead

Sophia 1: All of them would be helpful, I don't think any of them specifically.

Sophia 2: [Prototype B] and [Prototype C] both had features that incentivized or prompted me to consider tasks that I have been procrastinating on. I like that [Prototype C] incentivized it rather than just reminding, but I think they were both helpful.

Sophia 3: I think [Prototype C], it showing the deadline, what capacity level it requires, if it has been skipped, if it had a sooner deadline than something else. And being able to grow a plant helps push it over the edge in terms of completing a task that you may have otherwise pushed off.

Shivani 1: Yes prototype B has AI assistance which would be useful in determining what to work on next.

Shivani 2: No

Hasan 1:

Hasan 2: I do like the energy level ideas [Prototype B] but it's just that I know I would misuse it by pressing low energy low energy everyday. It just doesn't work for me personally. I think [Prototype A] helps most, because it helps me visualize everything better and I can just suck it up and work on the hard tasks.

—

Rankings

If you had to rank them from most helpful to least helpful for deciding what to work on, how would you rank them?

Carline 1: [B, A, C]

Carline 2: [C, B, A]

Sophia 1: [Prototype C][Prototype A][Prototype B] They all have their own merit. I love the energy meter in the heatmap, I liked the subtasks.

Sophia 2: [Prototype C][Prototype B][Prototype A]

Sophia 3: [Prototype C] [Prototype B] [Prototype A], although I like the taskbar concept.

Shivani 1: [C,B,A]

Shivani 2: [A,C,B]

Hasan 1:

Hasan 2: [A, B, C]

—

Which one reduced your decision stress the most?

Carline 1: [Prototype B] because you can reorder things based on how you feel

Carline 2: [Prototype C]

Sophia 1: [Prototype C], I like the idea of there being a list of things that I've ranked to do first or in an order. The subtasks helped as well.

Sophia 2: I would say [Prototype C], but [Prototype B] is not far behind. If I had a particularly stressful week, [Prototype B] may stress me out to see the whole month or the whole week, but I liked that [Prototype C] only showed you a few things.

Sophia 3: I think [Prototype B], being able to put in the kind I'm having energy-wise and having it reorganize the tasks eliminates the feeling of being stressed when you're not at your best.

Shivani 1: All prototypes have great features but I would prefer prototype B or C since it has AI assistance and features as an incentive.

Shivani 2: [Prototype C] I love the idea of gamification since it would help me understand and pick up work items quicker.

Hasan 1:

Hasan 2: [Prototype B] it's pretty unintrusive and helps me make decisions faster, though I don't know if I'm trying to go faster or not.

—

Final Reflection—

What still feels missing across all three designs?

Carline 1: Updating tasks or making them repetitive (recurring tasks)

Carline 2: Something that prioritizes the tasks for me

Sophia 1: No pink! Where is all of the pink? You guys all like blue or purple- if you must think, think pink! On a real note, these could use some ornamentation. The gamification one needs outfits that I can buy.

Sophia 2: I think across the three of them, I don't know what isn't there. Between the three of them you seem to capture a lot of elements. Even if I personally wouldn't use the task breakdown, I know a lot of people would. It seems like together they cover a lot of ground.

Sophia 3: Uploading the tasks front end more easily, linking to your own calendar, reminders/updates to get you to check the app and check what your day looks like.

Shivani 1: Overall the designs seem too compact. I like to interact with interfaces that have a good layout in terms of aesthetics and also core functionalities. Try to pick 2-3 functionalities instead of doing everything.

Shivani 2: Simplicity and basic layout. Oftentimes students like me tend to pick up tasks quickly if we give more time to organization then there is less time for actually doing the work. I would prefer something simple to handle and navigate.

Hasan 1:

Hasan 2: A lot of them operate on the notion that you do one assignment at a time. I like to be able to swap between assignments midway so there needs to be a way to account for that working style.

—

If you could combine parts of these, what would you keep?

Carline 1: "What's coming up" summarizing upcoming tasks

Carline 2: The breakdown from A, the heatmap from B, and the simple interactions from C

Sophia 1: I would keep the energy meter from the heatmap- I don't know what the energy meter would do, but I like that. I would keep the month/week/day calendar from the heatmap. I like the ability to break things down into subtask- being able to add subtasks, check mark, progress bar. I like everything about the gamification one.

Sophia 2: I really like the task list part of [Prototype C], I like the energy check-in of [Prototype B]- it helps me determine what I can do based off of how capable I feel. I know the task breakdown would be helpful to others even if it isn't appealing to me. I think the week view of [Prototype B] is helpful, even if it's sometimes stressful. I liked the color-coded days in the heatmap, but if I'm feeling overwhelmed I maybe don't want to see that.

Sophia 3: I like different aspects of all of them! I think I like it reorganizing and moving things around of [Prototype C], I like being able to decide the level of heatmap energy and the month/day/week of [Prototype B] (being able to see the bigger picture as well as day-to-day). A lot of my projects build on each other so being able to have both aspects of what I need to do today- even if its a smaller task thats a chunk of a bigger one. And that for the week as well- can something be pushed later or earlier? But I really like the progress bar of [Prototype A] to see how far you are into a project. It's nice to be able to see that aspect as well. And the onboarding of information- I like being able to directly import the information within [Prototype A], it's definitely more user friendly than needing to put everything in.

Shivani 1: The heatmap, Ai assistance, and gamification from the prototypes could be integrated to create the final one. These are features I would probably use the most and probably would be the most helpful for a student.

Shivani 2: The energy level section from prototype A is very useful especially when I want to pick up a task quickly based on my mood. The calendar format and the AI assistant is another useful feature that can help further clarify an assignment/instructions. I would love to see something similar to this prototype.

Hasan 1:

Hasan 2: I think [Prototype A] has the best general structure. I like the calendar view. [Prototype B] visualized tasks well but it just needed better labeling. A color coded label for me to see clearly. [Prototype C] I notice the points, I don't know if I'll work with it but I guess fake points are nice.

—

Do you think any of these would realistically change how you manage your work? Why or why not?

Carline 1: No, I wouldn't trust AI to organize my stuff based on my mood so I would use them all as normal calendars.

Carline 2: Yes, particularly the heatmap and gamification, may motivate me to start tasks earlier

Sophia 1: No, maybe [Prototype C]. But I don't know that you can make me want to do something that I don't want to do.

Sophia 2: Realistically, probably yeah. I think [Prototype C] could change the way that I do work. Having a tiny, fake reward maybe would help me do something that I've been avoiding, and it gives me the satisfaction of checking off a list. I think [Prototype C] could potentially realistically be helpful and change my approach.

Sophia 3: Yes, I think so. I think that being a person that takes my planner and puts it all in three different sections- this is just a digital version of being able to do that. I like that it takes the mental part of what I have to do out of it. I like the idea of it helping me find time and space to prioritize and to move things around when things pop up. I like being able to visually see it rather than doing the mental gymnastics.

Shivani 1: Yes these are all great prototypes and have amazing features. I definitely see myself using it if some of the ideas were combined and integrated to create an application.

Shivani 2: No, these prototypes have too many features I would not use. I prefer working with something simple and something that can ensure that I spend less time planning and more time working.

Hasan 1:

Hasan 2: If they were integrated into something already existing that I use such as Canvas then I would probably play around with it.

16.8.2 *Interview Summary*

Table 5—Key insights from prototype evaluation interviews across design alternatives.

Insight	Evidence from Evaluation	Design Implication
Swipe-based prioritization reduces decision friction	Participants were able to quickly choose tasks when presented one at a time and reported feeling less overwhelmed compared to viewing a full task list. Some described the interaction as "easier to start" because it removed the need to compare multiple tasks simultaneously.	Interfaces should reduce simultaneous comparisons and instead support sequential, lightweight decision-making to lower cognitive load.
Users want guidance, but not full automation	While participants appreciated suggestions (e.g., what to work on next), several expressed hesitation about fully automated scheduling or decisions being made for them. Users preferred retaining final control.	Design systems that provide recommendations with clear rationale while preserving user agency and override capability.
Visual workload representations improve awareness but require clarity	Participants responded positively to the heatmap concept for understanding busy vs. lighter days, but some reported confusion about how to interpret color intensity or what actions to take.	Visualizations should be paired with clear legends, explanations, and actionable suggestions to translate awareness into decisions.
Task breakdown reduces overwhelm but must remain editable	Participants found AI-assisted task breakdown helpful for getting started on large tasks, but wanted the ability to modify or reject generated subtasks.	Support flexible scaffolding that users can adjust, rather than rigid system-generated structures.
Motivational features increase engagement but risk distraction	Gamification elements (e.g., points, streaks) were seen as motivating for task completion, but some participants noted they could become secondary to actual planning or feel superficial.	Use lightweight gamification to reinforce progress without overshadowing core decision-making functionality.
Context-sensitive recommendations are perceived as valuable	Participants indicated that recommendations based on energy level, time availability, or workload would be helpful, especially during periods of fatigue or overload.	Incorporate context-aware inputs (e.g., time, energy) into prioritization support to better match real-world decision conditions.

16.9 Appendix 9: Final Prototype

16.9.1 *Main weekly calendar view*

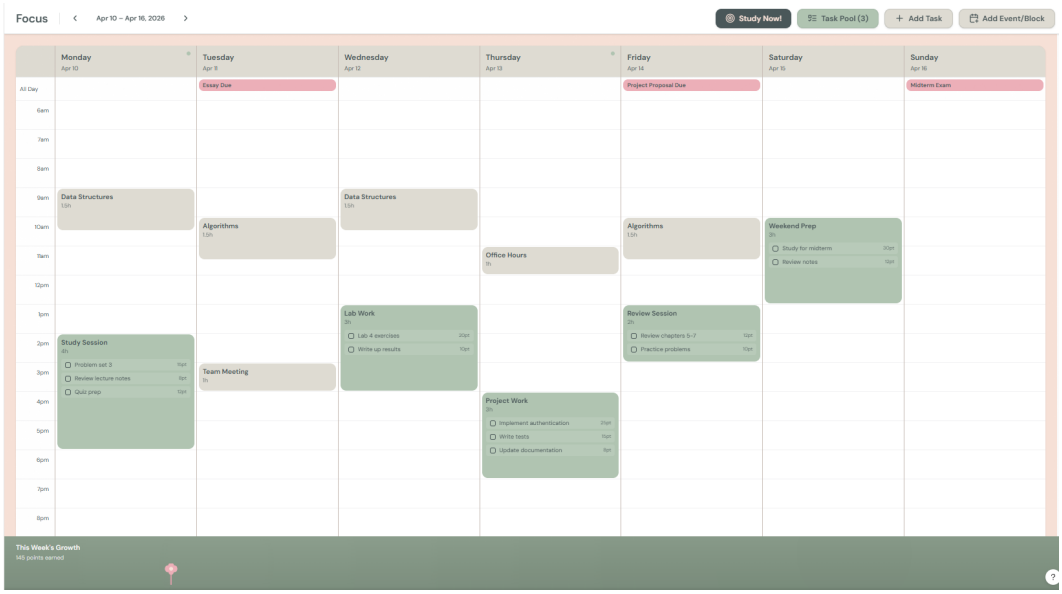


Figure 17—Main weekly calendar view of the final prototype. This screen integrates fixed-time events, deadlines, flexible study blocks, and top-level entry points to key workflows including task pool access, task creation, event/block creation, and immediate entry into Focus Mode. Image created with Figma.ai.

16.9.2 *Event and study block creation*

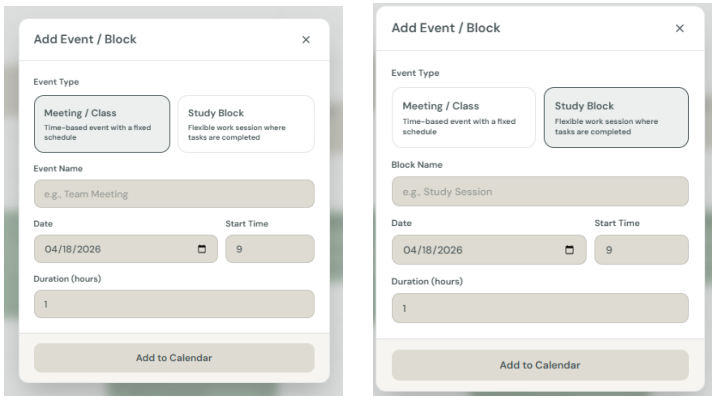


Figure 18—Event and study block creation modal. Users can add either a fixed-time meeting/class event or a flexible study block. The interface adapts based on the selected event type so that the same entry point supports both scheduling structures. Images created with Figma.ai.

16.9.3 *Task pool creation flow*

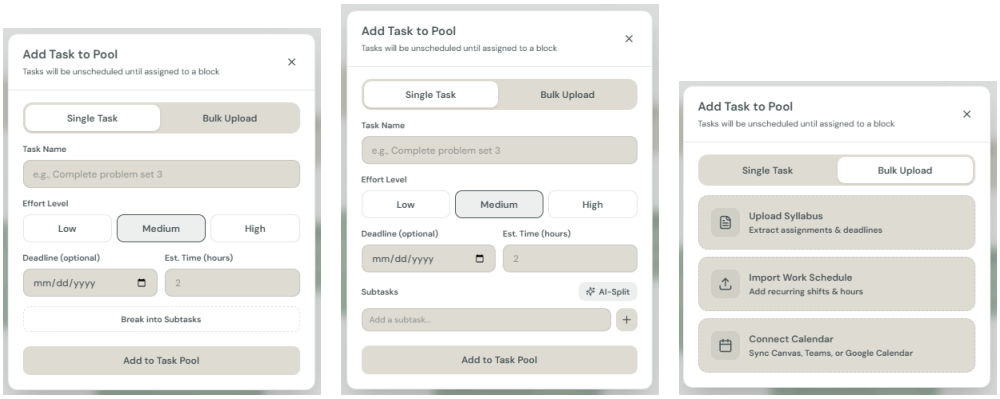


Figure 19—Task pool creation flow. Users can add an individual task with effort, deadline, and estimated time; optionally define subtasks or use AI-supported task splitting; or switch to bulk upload pathways such as syllabus upload, work schedule import, or external calendar connection. Images created with Figma.ai.

16.9.4 *Task pool views*

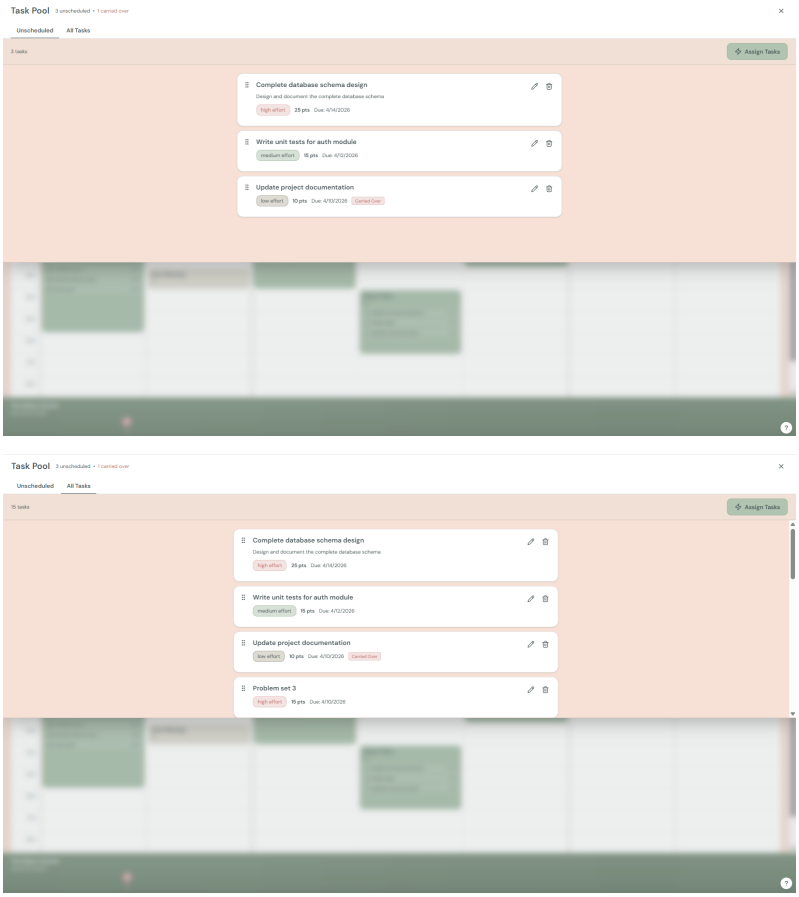
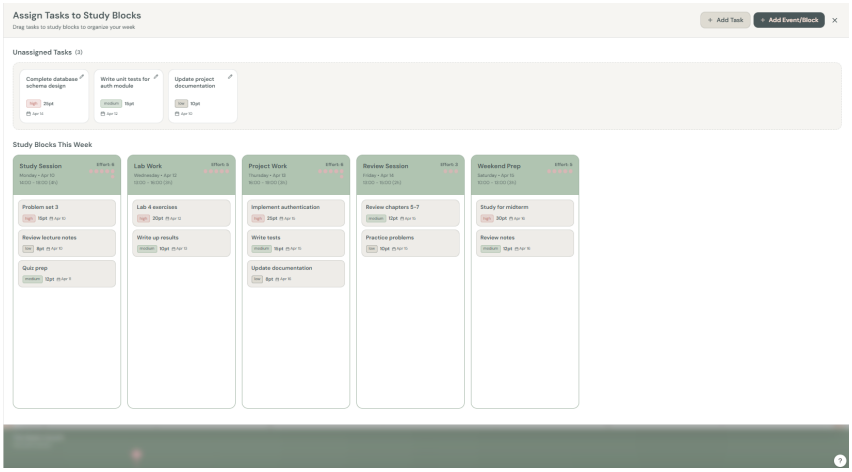


Figure 20—Task pool views for unscheduled tasks and the full task list. These screens support backlog management, highlight carried-over work, and provide a staging area for tasks before they are assigned to study blocks or surfaced in Focus Mode. Images created with Figma.ai.

16.9.5 *Assigning tasks to study blocks*



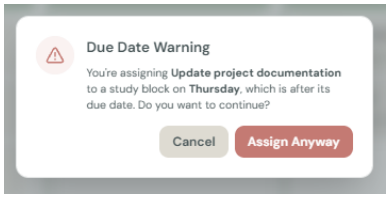


Figure 21—Task assignment workflow for study blocks. Users drag unscheduled tasks into flexible study blocks across the week, while the warning modal appears when a task is assigned to a block occurring after the task's due date. Estimated effort level heatmap allows users to better plan study blocks based around combined effort of tasks. Images created with Figma.ai.

16.9.6 *Study block detail view*



Figure 22—Study block detail view. The sidebar shows tasks assigned to a study block, total and earned points, editing controls, and the ability to add tasks directly to the block or launch Focus Mode from within that session context. Image created with Figma.ai.

16.9.7 *Focus Mode entry*

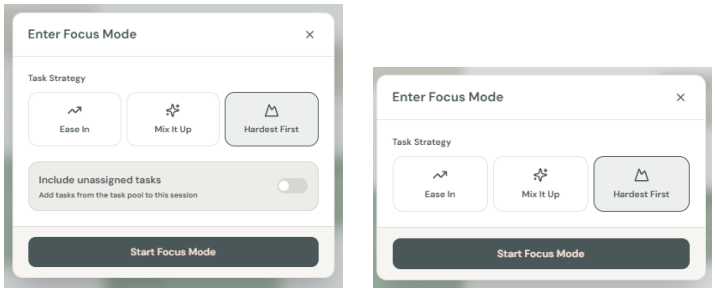


Figure 23—Focus Mode entry from two contexts. Users can launch Focus Mode either from within a study block or from the main weekly view. Entering from a study block prioritizes tasks already assigned to that session with the option to include unassigned tasks, while entering from the main view limits prioritization to unassigned task pool items. Images created with Figma.ai.

16.9.8 *Focus Mode prioritization interaction*

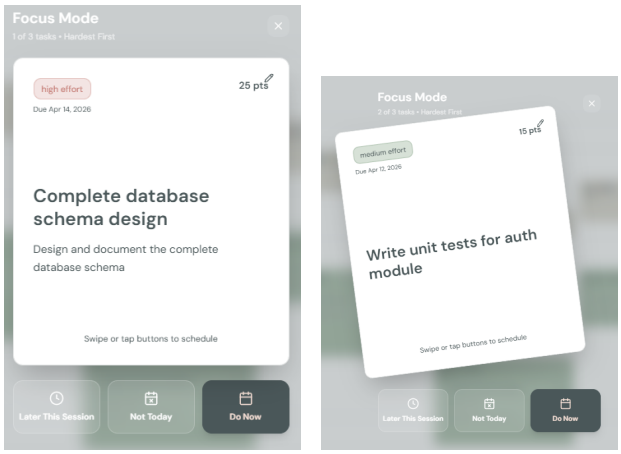


Figure 24—Focus Mode prioritization interaction. Tasks are presented one at a time as cards, and users assign them to categories such as do now, later this session, or not today using either swipe gestures or button-based controls. This sequential interaction reduces comparison burden and supports rapid decision-making. Images created with Figma.ai.

16.9.9 *Task completion feedback*

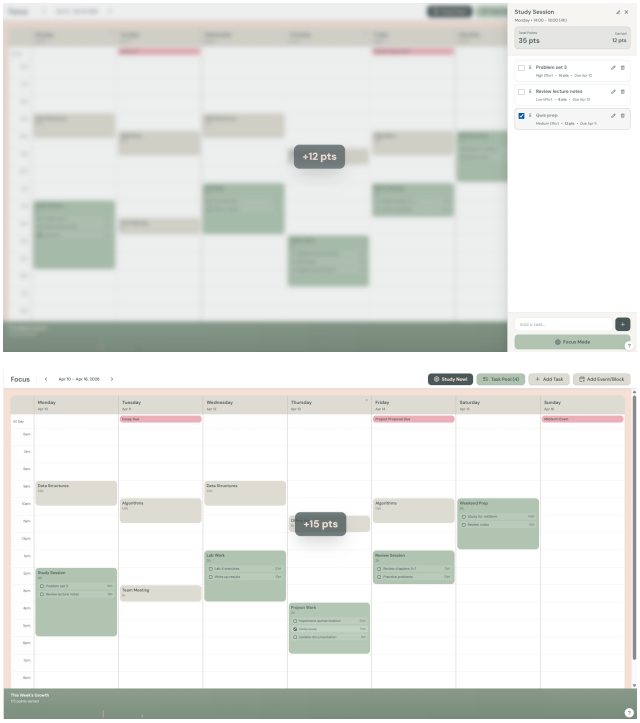


Figure 25—Task completion feedback from two locations in the interface. Tasks can be completed either within the study block detail panel or directly from the weekly calendar view, with lightweight point feedback reinforcing progress without interrupting the scheduling workflow. Images created with Figma.ai.

16.9.10 *Garden and progress view*

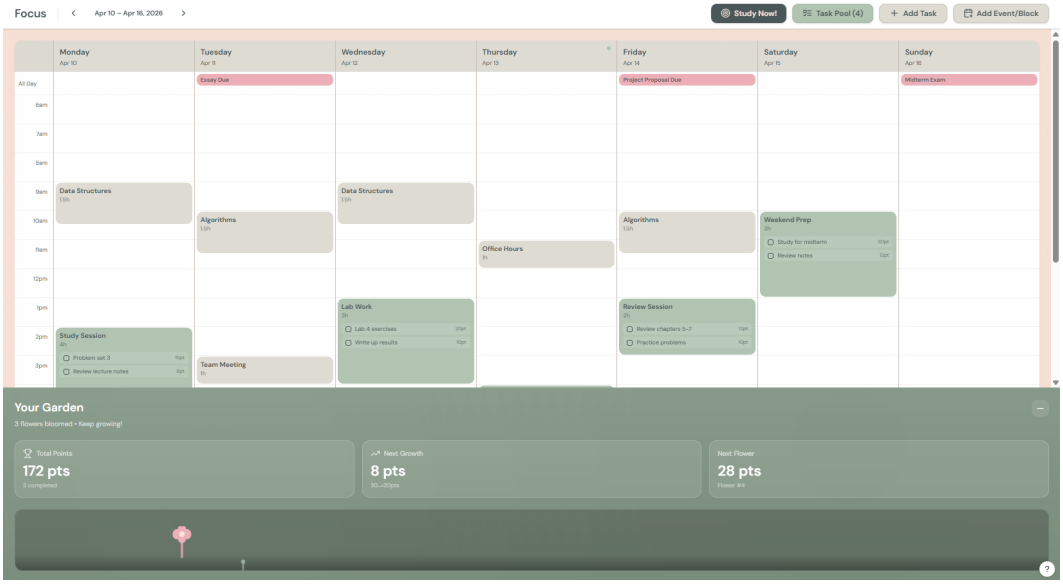


Figure 26—Garden and progress view within the weekly interface. This lower panel visualizes accumulated points and weekly growth using a lightweight garden metaphor, providing motivational feedback while remaining secondary to the main planning surface above. Image created with Figma.ai.

16.9.11 *Figma.ai prompts for final prototype*

Prompt 1

Design a medium-fidelity desktop web app (NOT mobile) for a student scheduling + prioritization tool focused on decision-making, not just task tracking

- main screen = weekly calendar view (like google calendar but cleaner + more decision-support focused). show a full week (mon–sun) with time blocks, including a mix of classes / meetings (fixed events) and study blocks (more flexible). make it feel realistic (not empty calendar).
- top of screen: simple nav bar (maybe week switcher, current week label, etc.), with a + button in the top right corner to add a task. when clicking +, it opens a modal / pop-up menu (centered, not full page).
- inside the pop-up: include a toggle/slider at the top with 2 modes — left = "add single task / engagement", right = "bulk upload / connect".
- if LEFT (single task): include fields for task name, effort level (low / medium / high), optional deadline, maybe estimated time. keep it clean + fast to enter (low friction).
- if RIGHT (bulk / connect): include options like upload syllabus, import work schedule, connect to Canvas / MS Teams / Google Calendar. this should feel visually distinct from single task mode (more "system-level").

- bottom of main screen: persistent gamification "garden" strip across the bottom. shows "plants" growing based on points earned this week, where each plant represents progress from completed tasks. should feel light + motivating, not childish, with subtle animation or growth stages (seed → sprout → plant).
- overall vibe: clean, slightly academic but warm, not overwhelming (reduce cognitive load), and should feel like it helps you decide what to do rather than just list things.
- extra small details if space: maybe subtle indicators on the calendar for heavy vs light days, or avoided tasks / high-effort tasks — but don't over-clutter.

Prompt 2

- calendar updates (main screen): keep the weekly calendar layout (mon–sun, time grid), but introduce 3 distinct color-coded event types so the structure is immediately clear. time-bound events (classes, meetings, office hours) use one consistent color, deadlines / due dates are shown as all-day banners at the top of each day column (like MS Teams / Google Calendar), and study / work blocks use a third color. the goal is a visual hierarchy where it's immediately obvious what is fixed vs flexible vs due soon.
- study/work block behavior: each study/work block should display multiple smaller task items inside the block as a vertical list. if there are too many tasks, the block should scroll internally rather than expanding the whole calendar. each task shows the task name and maybe a small indicator (effort or points). for example, a block like "Study Session (2h)" might include problem set 3 (partial), review lecture notes, and quiz prep.
- interaction: clicking a study block opens a centered modal popup where the user can edit the block's time/day and also drag to reschedule directly on the calendar (with direct manipulation also supported outside the modal). inside the popup, the user sees the list of tasks in that block and can click into a task to edit, drag to reorder, or remove/add tasks — this should feel like a lightweight task manager. this will later connect to swipe prioritization.
- NEW: "schedule tasks" flow — add a "Schedule Tasks" button on the main screen (top right or near calendar controls). clicking it opens a Tinder-style swipe interface that shows one task at a time. users swipe to assign: now / today, later, or not today, and the outcome is that tasks get automatically placed into available study blocks.
- fix existing UI issue: in the "add to schedule" popup, fix the slider toggle (single task vs bulk upload) so it fully moves left right (currently broken / partial movement).
- task-level features: when viewing a task (either from the add task screen or by clicking a task inside a block), include a "split into subtasks" button. subtasks behave like nested checklist items.
- points + gamification integration: each task / subtask has points assigned based on difficulty (low/med/high), completing early, or completing after avoiding (bonus). points should be visible in the task UI (small but present) and feed into the garden at the bottom of the screen.
- work session "play mode": when clicking a study/work block, include a "play" button. clicking play opens a right-side panel (slides in from the right) that shows a checklist of all tasks in that block, including task list, completion checkboxes, and points earned per task. this becomes the "execution mode" (focus + doing, not planning).
- overall vibe: still clean + minimal, but now clearly distinguishes types of work, reduces decision-making overhead, and bridges planning → prioritization → execution → reward.

Prompt 3

- focus of this iteration = interaction flow between calendar → tasks → execution → prioritization. currently clicking a study block only lets you add a task → change this so that clicking a study/work block opens a right-side panel (slides in from right, NOT centered modal). this panel replaces the need for a "play" button (remove play button entirely).
- right-side panel (study block view): this becomes the main "work session / execution + editing" space. it should include the block title + time at the top and total points in the block, followed by a task list in checklist format (checkboxes). each task shows name and points, and supports interactions like checking off tasks, dragging to reorder, clicking to edit, and dragging tasks out of the block into another block/day (cross-block movement). also include buttons for "+ add task" and "prioritize tasks" (opens swipe flow).
- tinder / swipe flow: currently goes straight into swiping → add a pre-step modal. when the user clicks "prioritize tasks", first open a small centered popup asking "how do you want to approach today?". include two selection sections: mode (ease in, mix it up, hardest first) and energy / mood (low energy, moderate, focused, high energy). once selected → THEN go into swipe interface, and swipe logic should adapt ordering based on these selections.
- task creation improvements: when adding a task (either from add task button OR inside study block panel), include a "break into subtasks" button. when clicked, expand the popup downward (progressive disclosure). the expanded view includes a list where the user can manually add subtasks, and also a button "AI-Split" to automatically generate subtasks.
- task editing behavior: clicking a task (anywhere: block or panel) opens a task detail modal. inside, the user can edit name, effort, deadline, etc., see subtasks, and edit subtasks individually.
- calendar interaction upgrade: allow dragging of study blocks (time/day) AND individual tasks between blocks (important).
- growth / gamification panel (bottom): currently static → make interactive. when the user clicks the bottom growth bar, it expands upward to take bottom 1/3 of the screen. the expanded panel includes total points earned (this week), points to next goal, points to next level, and a larger view of the garden growth. top right of the expanded panel includes a "–" button to collapse back down.
- overall interaction goals: remove friction between deciding what to do, organizing tasks, and actually doing them. unify planning (calendar), prioritization (swipe), execution (right panel), and reward (garden).

Prompt 4

- focus = fix interaction gaps + make everything consistent + connect actions to rewards. currently when clicking a study session → panel opens but no tasks show → fix this so that when the panel opens, it shows ALL tasks assigned to that block in a checklist format.
- each task should include a checkbox (for completion), task name, points value, and a pencil icon (edit). interactions should support checking off tasks, drag & drop to reorder, and clicking the pencil icon to open the task edit modal (same UI as task creation, including ability to split into subtasks).
- task completion → gamification: checking off a task should immediately add points to the total and visually update both the total points and the garden (more growth / plants appear or grow). this should feel like instant feedback and a slightly satisfying micro-reward moment.
- task editing behavior: editing a task should reopen the same add/edit task modal and allow name

edit, effort edit, deadline edit, subtask editing, and AI-split option.

- prioritization / swipe flow: when clicking "prioritize tasks", an optional modal appears (mode + energy), where the user can select options or skip entirely and go straight to swipe.
- swipe actions rename: swipe right → "Do Now", swipe left → "Later This Session", swipe up → "Not Today". (optional alt vibe: right → "Start Now", left → "Later", up → "Skip Today").
- calendar interaction consistency: clicking ANY calendar item (study block OR meeting) should open an edit modal. meetings / time-bound events should be editable (name, time, etc.).
- drag & drop: allow dragging of study blocks (to move time/day) and meetings/time-bound events (to reschedule the same way). this should feel like direct manipulation and be smooth + intuitive.
- task movement across blocks: from the right-side panel, allow dragging a task out of the panel and dropping it into another study block (same day or different day).
- overall system consistency: make interactions predictable — click = edit, drag = move, check = complete + reward.
- gamification feedback loop: action = check off task → result = points increase, garden grows (visibly), maybe small animation / highlight. goal = reinforce connection between doing work, earning points, and visual progress.
- overall vibe: still clean / minimal, but now highly interactive, responsive, and clearly supports the decision → action → reward loop.

Prompt 5

- Design an updated version of the desktop web app for student scheduling + prioritization.
- focus = clearly separate tasks vs calendar, introduce task pool, and improve how users view + prioritize tasks.
- core system structure: separate system into tasks (things to decide), calendar blocks (time containers), and task pool (unassigned work).
- task creation vs scheduling: introduce two distinct actions. Add Task creates a task in the task pool (unassigned), does NOT place it on the calendar, and is used for prioritization and later assignment into study blocks. Add Event / Block creates meetings, classes, and study/work blocks, and is placed directly on the calendar.
- UI: split top-right button into "Add Task" and "Add Event / Block".
- task pool: create a task pool / backlog view. this is where all unassigned tasks live, and it should be accessible via the "Schedule Tasks" button.
- Schedule Tasks screen: clicking "Schedule Tasks" should open a dedicated task management + prioritization view.
- this screen includes top filter / toggle options where the user can choose what tasks to view/prioritize: unscheduled tasks (task pool), tasks for a specific day, tasks within a specific study block, tasks for the full week, and all tasks combined.
- within this screen, the user can view tasks as a list, drag tasks into study blocks or specific days, OR click "prioritize" → go into swipe mode.
- prioritization behavior: swipe mode operates on whichever set is selected — unscheduled tasks, day-specific tasks, week tasks, etc. — and the output assigns tasks into study blocks.
- task lifecycle: tasks move through states — unassigned → task pool, assigned → inside study block, completed → checked off (points awarded).

- unfinished task behavior: if a task is NOT completed, it is removed from the study block and returned to the task pool.
- visually marked as "carried over" or "not completed".
- goal: reinforces re-prioritization and supports decision-making loop.
- study block panel: shows tasks assigned to that block, checkboxes, and points. when tasks are completed, points are awarded immediately and the garden updates.
- task movement: tasks can be dragged from pool → into study block, dragged between blocks, or removed back into pool.
- calendar remains clean: calendar ONLY shows time-bound events and study blocks, NOT raw tasks (unless inside blocks).
- make it obvious: tasks = what needs to be done, task pool = what hasn't been scheduled yet, study blocks = when work happens, swipe = how decisions are made, garden = reward for completion.

Prompt 6

- task pool: task pool should open as a top-down panel that slides down from the top, covers top 60% of the screen, and still leaves the calendar partially visible underneath.
- task pool tabs: include tabs at the top for This Week and All Tasks. This Week shows ALL tasks relevant to the current week, including assigned + unassigned tasks and completed + incomplete tasks (optional visual distinction). All Tasks shows ALL incomplete tasks across all time and acts as a long-term backlog.
- task pool interactions: within task pool, tasks are shown as a list. users can drag tasks into study blocks (visible behind the panel or via drop zones), click to edit, or select a subset → prioritize. include a "Prioritize Tasks" button.
- prioritization scope: prioritization works based on the current tab or selection — "This Week", "All Tasks", or a selected subset.
- add event / block: replace the day-of-week dropdown with a calendar + time picker. default date = today.
- event type selection: when selecting event type, Meeting / Class should include the subtext "time-based event with a fixed schedule", and Study Block should include the subtext "flexible work session where tasks are completed". remove or deprioritize standalone "deadline" event type — deadlines should come from tasks themselves.
- calendar interaction fixes: fix broken behaviors so that clicking a meeting / time-bound event opens an edit modal, and dragging meetings or study blocks allows rescheduling. make this consistent with all calendar items.
- study block panel: right-side panel still opens and shows all tasks in the block, checklist, drag reorder, and edit via pencil icon.
- task lifecycle: unfinished tasks are removed from the study block, returned to the task pool, and marked as "carried over".
- gamification system: garden MUST be tied directly to points (currently broken).
- growth model: within a plant (next goal), growth goes seed → sprout → stem → leaves → flower (petals appear in color variation). next level = once a flower is complete → new plant starts.
- points behavior: each completed task adds points immediately, and plant growth updates in real time based on accumulated points.

- scaling: thresholds for next growth stage and next plant are based on the total available points for the week. if new tasks are added, thresholds should adjust smoothly (no jarring jumps).
- garden interaction: clicking the garden expands it to the bottom 1/3 of the screen. the expanded view shows total points, progress to next growth stage, and progress to next plant (level).
- minimize behavior: include a "–" button in the top right, OR clicking anywhere on the calendar collapses the garden.
- overall: task pool = what hasn't been scheduled, study blocks = when work happens, swipe = how decisions are made, calendar = structure, garden = reward.

Prompt 7

- the garden is no longer in view. can we bring it back?
- i like that the task pool defaults to showing the unscheduled tasks, but when clicking into it, we should be able to flip between unscheduled tasks and all tasks. From there, we should be able to either schedule individual tasks into study blocks, schedule all unscheduled tasks, or reschedule all tasks for the week.
- the current version does not allow me to edit the meetings by clicking on them. Can you make it so that clicking on a meeting brings up the edit window (the SAME window as add event/block, prefilled to the event's information?)
- when i click into a study block, i should be able to make edits to the block itself- change the name, change the time, etc
- when i click into a study block, i see that there is a pencil icon to edit the individual items. clicking this should allow the user to edit the task.

Prompt 8

- The "All Tasks" tab of the task pool should show every task, assigned or not. Every task that is currently on the screen should be included in that, whether or not they have been assigned.
- Clicking on the pencil/edit icon of a task should allow you to edit the task, whether in this menu or when you open it via study block. These changes should stay consistent throughout.
- If you are in the task pool menu and you are looking only at the unassigned tasks, clicking prioritize should prompt you with the option to either select the study block(s) you'd like to assign to, create a new study block, or start now (e.g. automatically create a new study block right now).

Prompt 9

- Can you make it so all of the items within each study block (study session, lab work, review session, weekend prep, etc) are counted as 'tasks'? they should be characterized in the same way as the unscheduled tasks in the task pool.

Prompt 10

- Great, I like that the tasks that are assigned to individual study sessions are now shown in 'all tasks'.
- can we show them as checkboxes within the study block in the calendar as well?
- can we make the edit functionality for the blocks actually functional? e.g. if I change the start time, name, or duration, it shows up on the calendar as such?
- can we make it so that the calendar can scroll so you can see past 4pm if needed?

Prompt 11

- The calendar's hours are no longer even; can we make it spaced evenly again and have the lines even? I think the issue is that the items are 'stuck' to staying within their starting time rather than filling up time the way a calendar does.
- Can we change the font and checkbox style within the study blocks to better match the theme? they kinda stick out weird.
- When you click into a study block and have the checkboxes and info to the right, the edit buttons should be functional and should allow you to edit the individual tasks. These edits should change the task itself internally rather than just visually on this view.

Prompt 12

- I'd like to change the way that we do the swipe/prioritizing.
- Swiping should exist "as-is" in 2 places only: within a study block when opened, and under a "study now!" button. This is what we will call "focus mode".
- When swiping/prioritizing from a study block, in focus mode, there should be a toggle for "include unassigned task pool items". If the toggle is turned OFF (default), the tasks the user is swiping between will ONLY include tasks already assigned to that block. If the toggle is turned ON, the user will swipe between both items already in the block and items that are unassigned in the task pool.
- If studying from a "study now!" button, also in focus mode, it should pull from the unassigned task pool items.
- I like the three current swipe options. They should function as such:
  - right- do now- task is prioritized to the top of the block
  - left- later this session- stays in block, lower priority
  - up- not today- gets sent back to the unassigned task pool.
- To convert them to the study block's plan, we will use two data points- the approach + the swipe factors. So if we selected 'hardest first', the order of the tasks on the plan page would be [do now tasks: difficult to easy], [later this session tasks: difficult to easy]. anything selected as 'not today' goes back into the unassigned task pool.

Prompt 13

- This is looking pretty great! When you click "study now" and go into swiping for focus mode, the output should be a sidebar- the exact same as the study session- with the tasks ordered and able to be checked off for points. The title can just be "Instant Study Block" and it should function the same as the normal study blocks.
- The points earned should go towards the flowers. Let's recalibrate so each flower "matures" at 50 points, with every 10 points giving more detail. So 0 points is just a seed, 10 points is a seedling, 20 points is a taller seedling, 30 points is a taller seedling with a leaf, 40 points is a taller seedling with a leaf and a colored flower bud, 50 points is a full flower with petals. After one flower is created, the points continue on to the next flower, which can "grow" anywhere within the garden (doesn't have to be left to right). Let's have 5 possible colors for the flowers to keep it exciting.
- For example, if a user is at 180 points, they would have 3 full flowers (50 points each) and a 4th flower in the tall seedling with leaf phase. If they do 30 points of work, that flower would complete and the next flower would be started.

Prompt 14

- I'd now like to build in another feature: "assign tasks mode". In this mode, the user assigns tasks from their task pool to either existing or new study blocks. They can access this in two places: the unassigned task pool and the 'all tasks' task pool.
- This menu should be integrated into the task pool itself and should function as a large pop up that pulls down from the top of the screen, covering nearly all of the calendar.
- This function should 'pull up' all of the study blocks for the week, keeping them in order and keeping all of the dates and times clear. The blocks should show all of the tasks currently assigned to them.
- Any unassigned tasks should be above the study blocks as a stack of cards, and the user can drag the task 'card' to the study block they would like to add it to. The card should show the assignment name, effort level, points, and due date. Dragging the card to a study block should add it to the bottom of that study block's task list. Tasks can be rearranged within the study block by dragging, if the user chooses.
- The "Add Event/Block" button should be visible from here, so the user should be able to add a new study block without leaving "assign tasks mode." If the user makes a new study block within this mode, it should be accessible both within the mode (added to the view) but also persist on the calendar.
- In this mode, users should be able to drag tasks between study blocks, as well as drag them back to the pool of unassigned task cards to reorganize.
- A 'heatmap' of small dots should be shown at the top of each study block in this mode indicating the total effort level of the block. This would be the sum of the effort for each task within the block (easy task = 1 dot, medium task = 2 dots, hard task = 3 dots). This is so the user can easily see the difficulty of the block and can make decisions to even it out should they so choose. Let's also color the dots to be more saturated the more there are in the row so that the user can see difficulty either by number of dots or by color saturation.
- The system should flag and prevent the user from assigning a task to a date after the due date. The flag should warn them but allow them to accept anyway should they so choose.

Prompt 15

- Can you ensure that all of the example tasks within the system have assigned due date tags?

Prompt 16

- Can you ensure that the due dates for each task is visible when opening a study block, as well as within the swipe/focus mode?
- within assign tasks mode, can you make it so that the study blocks look more visually similar to the study blocks in the calendar? I'm envisioning them left-to-right across the space rather than top-to-bottom so the user does not need to scroll.
- within assign tasks mode, Can the cards on top (the unassigned tasks) look a little more fun- like maybe more of a tile shape than just a wide rectangle?
- within assign tasks mode, can you make the add event/block button functional? (right now it presses but does not bring up the popup)
- within assign tasks mode, can you make every task editable?
- within assign tasks mode, can you have the add task button at the top? this should be functional and

bring up the add task popup.

Prompt 17c

- I see that pressing add task or add event/block buttons within assign tasks mode brings up the popup menus, but they are ordered underneath the assign tasks mode interface. can we make them pop up on top?

Prompt 18

- When in the assign tasks mode, can you make it so we see not just the day of the week and time of each study block, but also the date?

Prompt 19

- Can you make it so that any time you click on an edit button near a task, the task opens up into an edit popup (prepopulated with the tasks info)? this should be literally anywhere you see a task- within a study block, in the task pool, in the swipe/focus mode, in assign task mode. This should be functional and actually edit the task itself.
- In addition, when the edit popup comes up, I would like for it to have a button option to break the task into subtasks. Pressing this button would allow the user to add rows of subtasks underneath the task. There should be a sparkle button revealed only within the subtask feature that says "AI-split" which would automatically split the task into subtasks by effort level (more for higher effort tasks)

Prompt 20

- Can you have the same task/subtask/ai split option available in the 'add task' menu as well?

Prompt 21

- Can we update the color scheme across the board? I found a pallete we like:

Earthy Tones

- Earthy mauve-pinks, creamy white, sandy taupe, gentle sage, and organic charcoal harmonize.

Colors

- Cherry Blossom #edafb8
- Powder Petal #f7e1d7
- Dust Grey #dedbd2
- Ash Grey #b0c4b1
- Iron Grey #4a5759

- Lively, fresh, and charming, its playful pastel spirit conjures spring petals fluttering in a gentle breeze.
- Like airy chiffon curtains filtering sunbeams, creamy, subtle warmth breathes tranquility and a peaceful lightness.
- Subtle, soft tone mirroring dusted stone, ideal for inviting warmth and understated sophistication.
- Muted blend balancing green and grey for an earthy effect; subtly sophisticated and understatedly modern.

- Slate grey tinted with a blue-green cast suggests durability and seriousness, echoing iron's stoic and unwavering resolve.

Prompt 22

- Can we reshuffle these? the light green and light pink are a little too low-contrast to the white text

Prompt 23

- can we maybe flip the colors for the study blocks and the events?

Prompt 24

- Can we change the background color for the garden as well to be a bit brighter?

Prompt 25

- Maybe not brown for the garden background? my team didnt like the brown

Prompt 26

- can we swap every instance of powder petal and dust grey?

Prompt 27

Deadlines (all-day banners at top) → Cherry Blossom (EDAFB8)

Study/work blocks → Ash Grey (B0C4B1)

Meetings/classes (time-bound events) → Dust Grey (DEDBD2)

Use Iron Grey (4A5759) for text, icons, outlines

Use Powder Petal (F7E1D7) for main backgrounds / containers

Garden (bottom panel):

- background → slightly darker / richer version of Ash Grey (earthy, muted green tone)
- should feel grounded / "soil-like" but still soft (not dark brown or harsh)
- plants (growth stages) sit on top and use Cherry Blossom / Ash Grey variations

Inside study blocks:

- keep tasks readable (slightly lighter cards if needed)

Prompt 28

- Can we make the text within the meetings/classes and study blocks darker? i cant really read them.

Prompt 29

- Can we slightly increase font weight across the board for readability, and maybe stick with a san serif font throughout

Prompt 30

- Can we make the effort dots within the assign tasks mode bigger and the effort label/number slightly more pronounced?

16.10 Appendix 10: Final Evaluation Survey Questions

Instructions
Participants are first asked to watch the video prototype, then complete the questions below.

16.10.1 *Demographic Questions*

What is your current level of study?

- High School (or lower)
- Undergraduate
- Graduate
- Other

How often do you plan your academic work schedule?

- Daily
- Weekly
- Monthly
- Rarely (less often than monthly)

16.10.2 *Quantitative Questions*

Instructions:
Please rate each statement on a scale of 1 to 5.
1 = Strongly Disagree
2 = Disagree
3 = Neutral
4 = Agree
5 = Strongly Agree

1. While using this prototype, I was able to easily decide what to work on next.
2. When choosing a task, I felt confident in my decision.
3. The prototype reduced the amount of thinking needed to choose a task.
4. The prototype clearly explained why tasks are recommended.
5. The prototype helped me understand my workload.
6. I would use this prototype in my daily academic workflow.
7. The prototype felt intuitive and easy to follow.
8. I did not feel overwhelmed while using this prototype.
9. This prototype helped me get started on tasks that I would normally avoid.
10. This prototype would reduce procrastination for me.

16.10.3 *Qualitative Questions*

1. Which feature did you find the most helpful?
2. Which feature was the least helpful or the most confusing?
3. What did you like the most about the prototype?
4. What did you find the most difficult to understand?
5. Would this prototype change how you decide what to work on? Why or why not?
6. What features would you add or improve?
7. In what situation would you use this prototype the most?